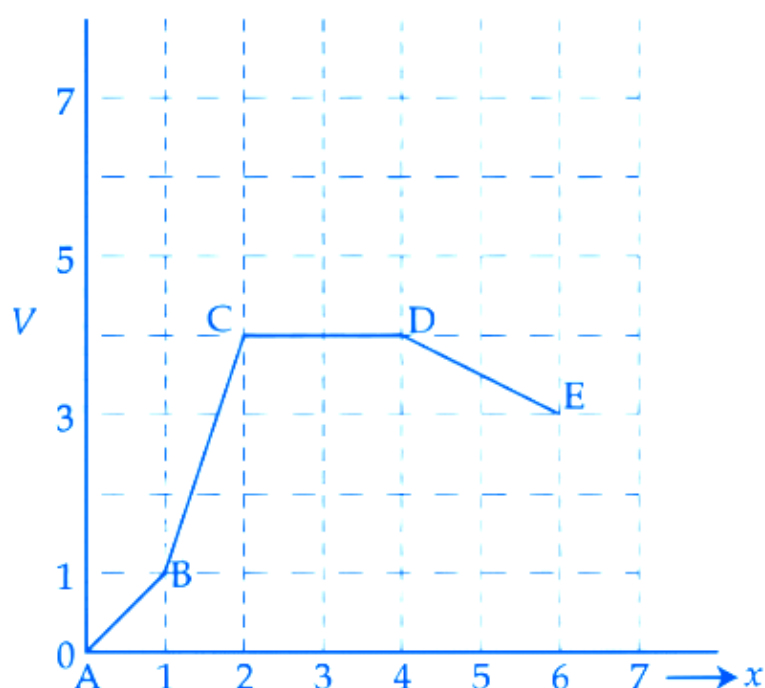


Work, Energy and Power

Question1

Potential energy (V) versus distance (x) is given by the graph. Rank various regions as per the magnitudes of the force (F) acting on a particle from high to low.



Options:

A.

$$F_{BC} > F_{AB} > F_{DE} > F_{CD}$$

B.

$$F_{BC} > F_{CD} > F_{DE} > F_{AB}$$

C.

$$F_{CD} > F_{AB} > F_{BC} > F_{DE}$$

D.

$$F_{CD} > F_{DE} > F_{AB} > F_{BC}$$

Answer: A

Solution:

The force (F) acting on a particle is defined as the negative gradient of its potential energy (V) with respect to position (x) :

$$F = -\frac{dV}{dx}$$

Here, $\frac{dV}{dx}$ is the slope of the V versus x graph. So, the magnitude of the force ($|F|$) is the absolute value of the slope in each region :

$$|F| = | \text{slope of the } V - x \text{ graph} |$$

Using the formula of slope

$$\text{Slope} = m = \frac{V_2 - V_1}{x_2 - x_1}$$

For region AB (from $x = 0$ to $x = 1$) :

Points are (0,0) and (1,1)

$$|F_{AB}| = \left| \frac{1-0}{1-0} \right| = 1 \text{ unit}$$

For region BC (from $x = 1$ to $x = 2$) :

Points are (1,1) and (2,4)

$$|F_{BC}| = \left| \frac{4-1}{2-1} \right| = 3 \text{ unit}$$

For region CD (from $x = 2$ to $x = 4$) :

Points are (2,4) and (4,4)

$$|F_{CD}| = \left| \frac{4-4}{4-2} \right| = 0 \text{ unit}$$

For region DE (from $x = 4$ to $x = 6$) :

Points are (4,4) and (6,3)

$$|F_{DE}| = \left| \frac{3-4}{6-4} \right| = 0.5 \text{ unit}$$

The descending order of the magnitude of forces is,

$$3 > 1 > 0.5 > 0 \Rightarrow F_{BC} > F_{AB} > F_{DE} > F_{CD}$$

Therefore, region BC has the greatest slope and thus the strongest force, while region CD has no slope and zero force.

Hence, the correct option is (A).

Question2

A body of mass 2 kg begins to move under the influence of time dependent force $\vec{F} = (2t\hat{i} + 6t^2\hat{j})\text{N}$, where $\hat{i}$ and $\hat{j}$ are unit vectors

along x and y -axis respectively. The power produced by the force at $t = 2$ s is _____ W.

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Answer: 200

Solution:

Power (P) is defined as the rate at which work is done. It can be calculated as the dot product of the force vector ($\vec{F}$) and the velocity vector ($\vec{V}$).

According to Newton's Second Law ($\vec{F} = m\vec{a}$), the acceleration is :

$$\vec{a} = \frac{\vec{F}}{m} = \frac{2t\hat{i} + 6t^2\hat{j}}{2}$$

$$\vec{a} = (t\hat{i} + 3t^2\hat{j})\text{m/s}^2$$

Velocity is the integral of acceleration with respect to time. Since the body begins to move, we assume the initial velocity is zero at $t = 0$:

$$\vec{v} = \int \vec{a} dt = \int_0^t (t\hat{i} + 3t^2\hat{j}) dt$$

$$\vec{v} = \left(\frac{t^2}{2}\right)\hat{i} + \left(\frac{3t^3}{3}\right)\hat{j} = \left(\frac{t^2}{2}\right)\hat{i} + t^3\hat{j}$$

Force at $t = 2$:

$$\vec{F}(2) = [2(2)]\hat{i} + [6(2)^2]\hat{j} = (4\hat{i} + 24\hat{j})\text{N}$$

Velocity at $t = 2$:

$$\vec{v}(2) = \left[\frac{(2)^2}{2}\right]\hat{i} + (2)^3\hat{j} = (2\hat{i} + 8\hat{j})\text{m/s}$$

So, the power at $t = 2$ s is,

$$P = \vec{F} \cdot \vec{v}$$

$$P = (4\hat{i} + 24\hat{j}) \cdot (2\hat{i} + 8\hat{j})$$

$$P = (4 \times 2) + (24 \times 8)$$

$$P = 8 + 192 = 200 \text{ W}$$

Therefore, the correct answer is 200 .

Question3

A 1 kg block subjected to two simultaneous forces $(2\hat{i} + 3\hat{j} + 4\hat{k})\text{N}$ and $(3\hat{i} - \hat{j} - 2\hat{k})\text{N}$ is moved a distance of 25 m along $(3\hat{i} - 4\hat{j})$ direction. The work done in this process is ____ J.

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Answer: 35

Solution:

When multiple forces act on a body simultaneously, the net force is the vector sum of all individual forces :

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2$$

$$\Rightarrow \vec{F}_{\text{net}} = (2\hat{i} + 3\hat{j} + 4\hat{k}) + (3\hat{i} - \hat{j} - 2\hat{k})$$

$$\Rightarrow \vec{F}_{\text{net}} = (2 + 3)\hat{i} + (3 - 1)\hat{j} + (4 - 2)\hat{k}$$

$$\Rightarrow \vec{F}_{\text{net}} = (5\hat{i} + 2\hat{j} + 2\hat{k})\text{N}$$

The block moves a distance of 25 m along the direction of the vector $\vec{A} = 3\hat{i} - 4\hat{j}$. The unit vector in this direction is :

$$\hat{u} = \frac{\vec{A}}{|\vec{A}|} = \frac{3\hat{i} - 4\hat{j}}{\sqrt{3^2 + (-4)^2}} = \frac{3\hat{i} - 4\hat{j}}{\sqrt{9+16}} = \frac{3\hat{i} - 4\hat{j}}{5}$$

The displacement vector $\vec{d}$ is

$$\vec{d} = 25 \times \left(\frac{3\hat{i} - 4\hat{j}}{5} \right) = 5(3\hat{i} - 4\hat{j}) = (15\hat{i} - 20\hat{j})\text{m}$$

Work done is defined as the dot product of the force vector and the displacement vector :

$$W = \vec{F}_{\text{net}} \cdot \vec{d}$$

$$\Rightarrow W = (5\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (15\hat{i} - 20\hat{j} + 0\hat{k})$$

$$\Rightarrow W = (5 \times 15) + (2 \times (-20)) + (2 \times 0)$$

$$\Rightarrow W = 75 - 40 + 0 = 35 \text{ J}$$

Therefore, the correct answer is 35 .

Question4

A body of mass 2 kg is moving along x -direction such that its displacement as function of time is given by $x(t) = \alpha t^2 + \beta t + \gamma$ m, where $\alpha = 1 \text{ m/s}^2$, $\beta = 1 \text{ m/s}$ and $\gamma = 1 \text{ m}$. The work done on the body during the time interval $t = 2 \text{ s}$ to $t = 3 \text{ s}$, is _____ J.

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Options:

A.

42

B.

24

C.

49

D.

12

Answer: B

Solution:

Velocity (v) is the rate of change of displacement (x) with respect to time (t) :

$$v(t) = \frac{dx}{dt}$$

The displacement function is given as:

$$x(t) = \alpha t^2 + \beta t + \gamma$$

Differentiating with respect to t :

$$v(t) = \frac{d}{dt} (\alpha t^2 + \beta t + \gamma)$$

$$\Rightarrow v(t) = 2\alpha t + \beta$$

Substituting the given values ($\alpha = 1 \text{ m/s}^2, \beta = 1 \text{ m/s}$) :

$$v(t) = 2(1)t + 1$$

$$\Rightarrow v(t) = 2t + 1$$

Initial velocity (v_i) at $t = 2 \text{ s}$:

$$v_i = 2(2) + 1 = 5 \text{ m/s}$$

Final velocity (v_f) at $t = 3 \text{ s}$:

$$v_f = 2(3) + 1 = 7 \text{ m/s}$$

Using Work-Energy Theorem formula,

$$W = \Delta K \cdot E = \frac{1}{2} mv_f^2 - \frac{1}{2} mv_i^2$$

$$\Rightarrow W = \frac{1}{2} m (v_f^2 - v_i^2)$$

$$\Rightarrow W = \frac{1}{2} \times 2 \times (7^2 - 5^2)$$

$$\Rightarrow W = 1 \times (49 - 25)$$

$$\Rightarrow W = 24 \text{ J}$$

Therefore, the work done on the body during the given time interval is 24 J .

Hence, the correct option is (B).

Question5

Given below are two statements :

Statement I : An object moves from position r_1 to position r_2 under a conservative force field $\vec{F}$. The work done by the force is

$$W = - \int_{r_1}^{r_2} \vec{F} \cdot \vec{dr}.$$

Statement II : Any object moving from one location to another location can follow infinite number of paths. Therefore, the amount of work done by the object changes with the path it follows for a conservative force.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Statement I is true but Statement II is false

B.

Statement I is false but Statement II is true

C.

Both Statement I and Statement II are false

D.

Both Statement I and Statement II are true

Answer: C

Solution:

Let a conservative force $\vec{F}$ is acting on an object. For infinitesimally small displacement $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$, the work done is,

$$dW = \vec{F} \cdot d\vec{r}$$

The work done (W) by a variable force $\vec{F}$ when displacing an object from an initial position vector $\vec{r}_1$ to a final position vector $\vec{r}_2$ is the total work done in moving the object,

$$\text{The work done is, } W = \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}$$

So, this formula is valid for all forces whether it is conservative or non-conservative force. But the statement I states that the work done is $W = - \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}$. Hence, the statement I is false.

Work done by a conservative force is path independent. And the work done by the conservative force is negative of the change in potential energy, i.e., the work done by the conservative force is stored in object as potential energy.

The work done depends only on the initial and final positions.

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r} = -(U_f - U_i)$$

For a particle going from point A to point B, there are infinitely many possible paths. So, it is a geometrical fact and is correct.

A force is called conservative if $\oint \vec{F} \cdot d\vec{r} = 0$

That means work done in a closed path is zero.

Hence for motion from A to B :

$$W_{A \rightarrow B} = U_A - U_B$$

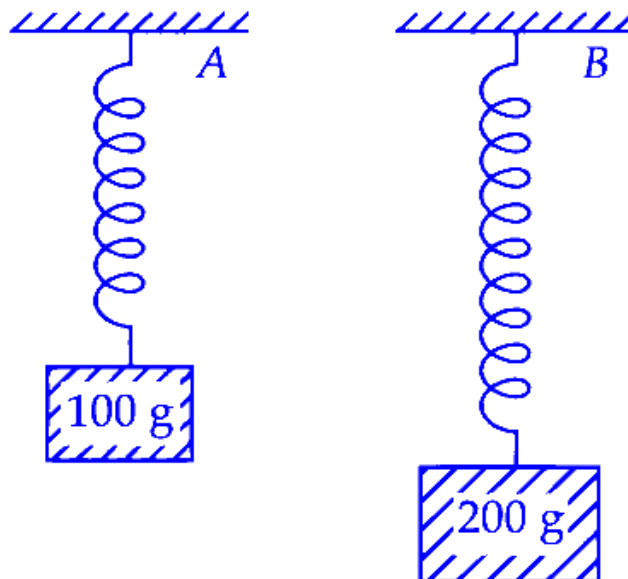
So work done depends only on initial position and final position and not on the path followed.

But the statement says, "work done changes with path for a conservative force", so the statement is incorrect because for conservative forces, straight path, curved path, zig-zag path all give the same work between the same two points. Hence, the statement II is incorrect.

Therefore, both Statement I and Statement II are false. Hence, the correct option is (C).

Question6

Two blocks with masses 100 g and 200 g are attached to the ends of springs A and B as shown in figure. The energy stored in A is E . The energy stored in B, when spring constants k_A, k_B of A and B, respectively satisfy the relation $4k_A = 3k_B$, is :



JEE Main 2026 (Online) 23rd January Morning Shift

Options:

A.

$$4E$$

B.

$$2E$$

C.

$$\frac{4}{3}E$$

D.

$$3E$$

Answer: D

Solution:

When a mass m is suspended from a spring of constant k , the spring stretches.

If the stretch in spring is x , $kx = mg \Rightarrow x = \frac{mg}{k}$

The potential energy (U) stored in the spring is given by :

$$U = \frac{1}{2}kx^2$$

Substituting $x = \frac{mg}{k}$:

$$U = \frac{1}{2}k\left(\frac{mg}{k}\right)^2 = \frac{(mg)^2}{2k}$$

Mass of block A is $m_A = 100 \text{ g}$.

The spring constant is k_A .

So, the stored energy in spring A is $E_A = E$.

Using the formula :

$$E = \frac{(m_A g)^2}{2k_A}$$

Mass of block B is $m_B = 200 \text{ g} = 2 \times 100 \text{ g} = 2 m_A$

The spring constant of spring B is k_B .

It is given that $4k_A = 3k_B \Rightarrow k_B = \frac{4}{3}k_A$.

So, the stored energy in spring B is

$$E_B = \frac{(m_B g)^2}{2k_B}$$

Substituting $m_B = 2 m_A$ and $k_B = \frac{4}{3}k_A$:

$$E_B = \frac{(2m_A g)^2}{2(\frac{4}{3}k_A)}$$

$$\Rightarrow E_B = \frac{4 \times \frac{3}{8} \times (m_A g)^2}{k_A}$$

$$\Rightarrow E_B = \frac{12}{8} \times \frac{(m_A g)^2}{k_A}$$

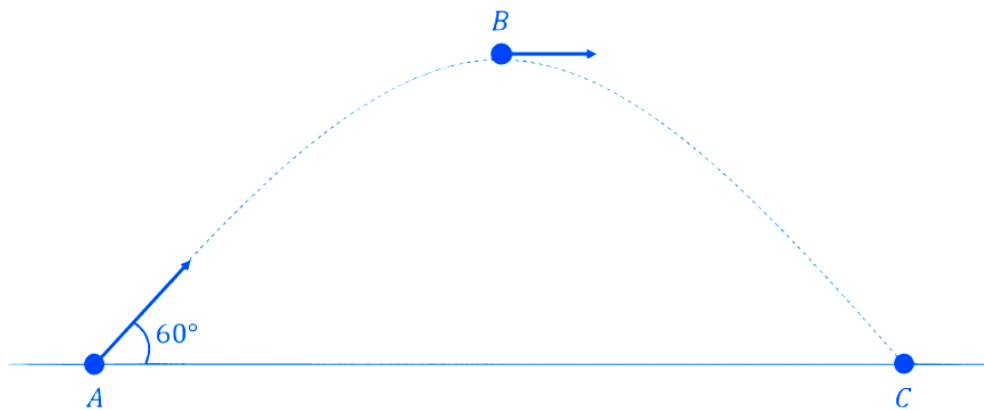
$$\Rightarrow E_B = \frac{3}{2} \times \frac{(m_A g)^2}{k_A}$$

$$\Rightarrow E_B = 3 \times \left(\frac{(m_A g)^2}{2k_A} \right)$$

$$\Rightarrow E_B = 3E$$

Question7

An object is projected with kinetic energy K from a point A at an angle 60° with the horizontal. The ratio of the difference in kinetic energies at points B and C to that at point A (see figure), in the absence of air friction is :



JEE Main 2026 (Online) 23rd January Morning Shift

Options:

A.

3 : 4

B.

1 : 4

C.

2 : 3

D.

1 : 2

Answer: A

Solution:

Let the mass of the object be m and the initial speed be u .

The angle of projection is $\theta = 60^\circ$.

At point A

The kinetic energy at point A is given as K .

$$K_A = \frac{1}{2}mu^2 = K$$

At Point B (Maximum Height)

At the highest point of a projectile's path, the vertical component of velocity becomes zero ($v_y = 0$). The object only possesses the horizontal component of velocity (v_x).

The horizontal velocity remains constant throughout the motion (assuming no air resistance):

$$v_B = u_x = u \cos \theta$$

$$\Rightarrow v_B = u \cos 60^\circ = u \left(\frac{1}{2} \right) = \frac{u}{2}$$

So, the kinetic energy at B is,

$$K_B = \frac{1}{2} m(v_B)^2$$

$$\Rightarrow K_B = \frac{1}{2} m \left(\frac{u}{2} \right)^2$$

$$\Rightarrow K_B = \frac{1}{2} m \left(\frac{u^2}{4} \right)$$

$$\Rightarrow K_B = \frac{1}{4} \left(\frac{1}{2}mu^2 \right) = \frac{1}{4} K_A = \frac{K}{4}$$

$$\Rightarrow K_B = \frac{K}{4}$$

At Point C

Point C is at the same horizontal level as point A. By the law of conservation of energy, the speed of the projectile when it hits the ground at the same level is equal to the speed of projection.

$$v_C = u$$

Therefore, the kinetic energy at C is equal to the kinetic energy at A :

$$K_C = K_A = K$$

The difference in kinetic energies at B and C :

$$\Delta K_{BC} = |K_C - K_B| = \left| K - \frac{K}{4} \right| = \frac{3K}{4}$$

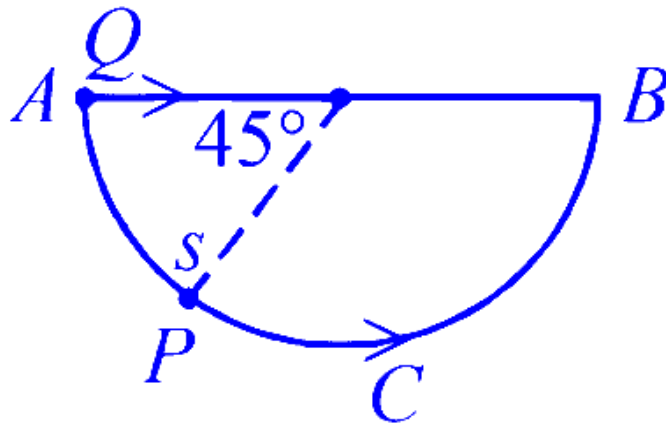
So, the ratio of the difference in kinetic energies at points B and C to that at point A is

$$\text{Ratio} = \frac{\Delta K_{BC}}{K_A} = \frac{(3/4)K}{K} = \frac{3}{4}$$

Hence, the correct option is (A).

Question8

A bead P sliding on a frictionless semi-circular string (ACB) and it is at point S at $t = 0$ and at this instant the horizontal component of its velocity is v . Another bead Q of the same mass as P is ejected from point A at $t = 0$ along the horizontal string AB , with the speed v , friction between the beads and the respective strings may be neglected in both cases. Let t_P and t_Q be the respective times taken by beads P and Q to reach the point B , then the relation between t_P and t_Q is



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Options:

A.

$$t_P < t_Q$$

B.

$$t_P = t_Q$$

C.

$$t_P > 1.25t_Q$$

D.

$$t_P > t_Q$$

Answer: A

Solution:

For bead Q

Bead Q moves along the horizontal straight string AB. The length AB is $2R$ where R is the radius of semicircular path.

Since the string is frictionless and horizontal, there is no component of gravity acting along the direction of motion.

Bead Q maintains a constant horizontal velocity v throughout the motion.

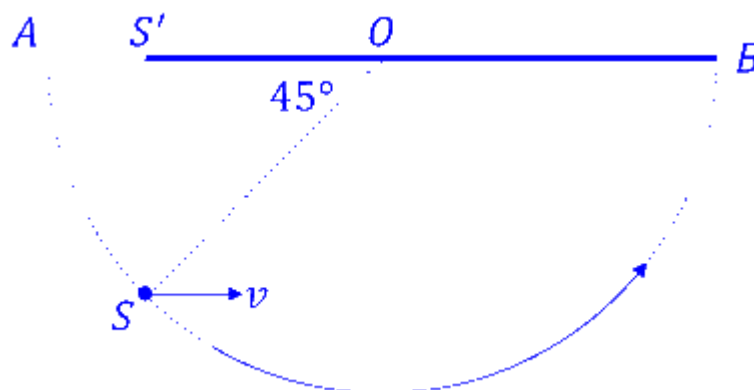
$$t_Q = \frac{AB}{v} = 2 \left[\frac{R}{v} \right]$$

For bead P

Bead P moves along the semi-circular path ACB under the influence of gravity. And along horizontal the net force is zero, i.e., the acceleration along vertical will be non-zero and the acceleration along horizontal will be zero.

At point S, the horizontal component of its velocity is given as v .

As bead P moves from S toward the bottom-most point C, the vertical component of velocity changes but the horizontal component remains same.



When bead P reaches point B then the horizontal displacement is $S'O + OB$

$$S'O = R \cos 45^\circ$$

$$\text{Then the time taken to reach point B is } t_P = \frac{S'B}{v} = \frac{R \cos 45^\circ + R}{v} = \frac{R}{v} \left[\frac{1}{\sqrt{2}} + 1 \right]$$

So, the ratio of time taken by Q and P to reach B is,

$$\frac{t_Q}{t_P} = \frac{2[R/v]}{\left[\frac{1}{\sqrt{2}}+1\right][R/v]} = \frac{2\sqrt{2}}{\sqrt{2}+1} = [8 - 2\sqrt{2}] > 1$$

$$\Rightarrow \frac{t_Q}{t_P} > 1 \Rightarrow t_Q > t_P$$

Question9

A spherical ball of mass 2 kg falls from a height of 10 m and is brought to rest after penetrating 10 cm into sand.

The average force exerted by sand on the ball is _____ N.

(Take $g = 10 \text{ m/s}^2$)

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

1980

B.

2020

C.

2000

D.

1000

Answer: B

Solution:

Using the work-energy theorem, the work done by all forces acting on a body is equal to the change in its kinetic energy.

The ball falls from height $h = 10$ m under gravity. And the ball enters the sand and travels a distance $d = 10$ cm $= 0.1$ m before coming to rest.

Let m be the mass of the ball, h be the initial height, d be the penetration depth, and F_{avg} be the average resistive force exerted by the sand.

The total work done on the ball from the moment it is dropped until it stops is:

$$W_{\text{total}} = W_{\text{gravity}} + W_{\text{sand}}$$

Since the ball starts from rest at height h and ends at rest after depth d , the total change in kinetic energy is zero ($\Delta K = 0$).

$$W_{\text{gravity}} + W_{\text{sand}} = 0$$

Gravity acts over the entire vertical displacement ($h + d$).

So, the work done by gravitational force is $W_{\text{gravity}} = mg(h + d)$

The resistive force F_{avg} acts upwards, opposite to the displacement d .

So, the work done by the sand is $W_{\text{sand}} = -F_{\text{avg}} \cdot d$

Substituting these into the energy equation :

$$mg(h + d) - F_{\text{avg}} \cdot d = 0$$

$$\Rightarrow F_{\text{avg}} = \frac{mg(h + d)}{d}$$

The mass of ball is $m = 2$ kg, the height of fall is $h = 10$ m, the distance of penetration in sand is $d = 0.1$ m, and $g = 10$ m/s².

$$F_{\text{avg}} = \frac{2 \times 10 \times (10 + 0.1)}{0.1}$$

$$\Rightarrow F_{\text{avg}} = \frac{20 \times 10.1}{0.1} = 20 \times 101$$

$$\Rightarrow F_{\text{avg}} = 2020 \text{ N}$$

Therefore, the average force exerted by sand on the ball is 2020 N .

Hence, the correct option is (B).

Question10

A mass of 1 kg is kept on a inclined plane with 30° inclination with respect to horizontal plane and it is at rest initially. Then the whole assembly is moved up with constant velocity of 4 m/s. The work done by the frictional force in time 2 s is ____ J. (Take $g = 10$ m/s²)

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

20

B.

25

C.

30

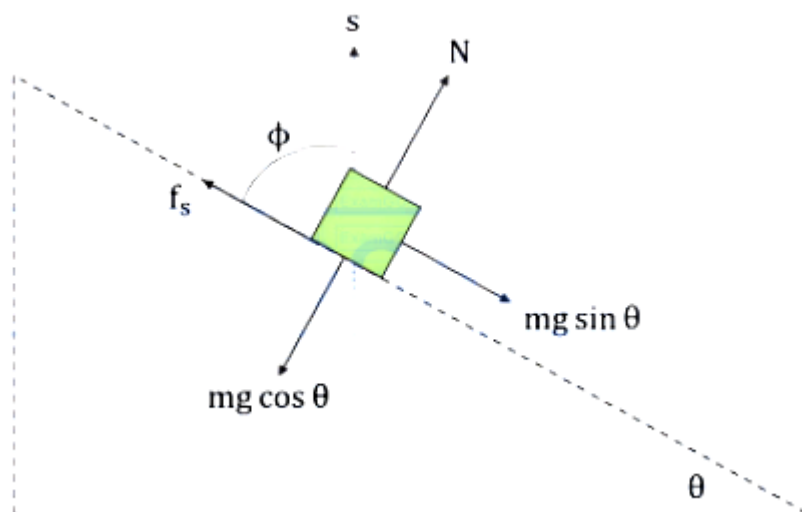
D.

10

Answer: A

Solution:

When the block is at rest on the incline, the forces acting on it are :



$F_g = mg$ is weight acting vertically downward.

N = Normal force acting perpendicular to the incline.

f_s = Static friction acting up the incline, preventing the block from sliding down.

Since the block is at rest relative to the incline, the frictional force must balance the component of gravity acting down the slope :

$$f_s = mg \sin \theta$$

The mass of block is $m = 1 \text{ kg}$, acceleration due to gravity is $g = 10 \text{ m/s}^2$ and $\theta = 30^\circ$

So, value of static friction is

$$f_s = 1 \times 10 \times \sin 30^\circ = 10 \times 0.5 = 5 \text{ N}$$

The friction force is 5 N directed up along the incline.

As the entire assembly moves vertically upward with a constant velocity $v = 4 \text{ m/s}$, so in time $t = 2 \text{ s}$, the vertical displacement S is:

$$S = v \times t = 4 \times 2 = 8 \text{ m}$$

Work done is defined as $W = \vec{F} \cdot \vec{d} = Fd \cos \phi$, where ϕ is the angle between the force and the displacement.

The angle between the friction force and the vertical is $90^\circ - 30^\circ = 60^\circ = \phi$.

So, work done by frictional force is,

$$W = f_s \times S \times \cos 60^\circ$$

$$\Rightarrow W = 5 \times 8 \times 0.5 \text{ J}$$

$$\Rightarrow W = 5 \times 4 = 20 \text{ J}$$

Therefore, the work done by the frictional force is 20 J .

Hence, the correct option is (A).

Question11

The rain drop of mass 1 g , starts with zero velocity from a height of 1 km . It hits the ground with a speed of 5 m/s. The work done by the unknown resistive force is _____ J.

(take $g = 10 \text{ m/s}^2$)

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

-8.75

B.

-8.35

C.

-9.55

D.

-9.98

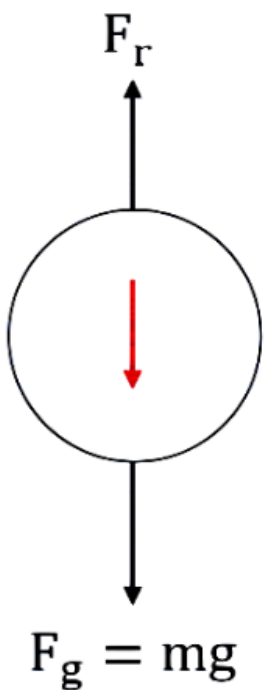
Answer: D

Solution:

The Work-Energy Theorem states that the total work done by all forces (conservative and non-conservative) on a particle is equal to the change in its kinetic energy.

$$W_{\text{total}} = \Delta K$$

When rain drop is falling, it experiences gravitational force (vertically downward) and resistive force which opposes motion of rain drop (vertically upward).



$$W_{\text{gravity}} + W_{\text{resistive}} = K_{\text{final}} - K_{\text{initial}}$$

Work done by gravity when an object falls a height h is:

$$W_g = F_g h = mgh$$

The mass of the rain drop is $m = 1 \text{ g} = 1 \times 10^{-3} \text{ kg}$

The height of drop is $h = 1 \text{ km} = 1000 \text{ m}$

And the acceleration due to gravity is $g = 10 \text{ m/s}^2$

Putting the values,

$$W_g = (10^{-3}) \times (10) \times (1000) = 10 \text{ J}$$

The initial velocity is $u = 0$ as it started from rest, so the initial kinetic energy is $K_{\text{initial}} = \frac{1}{2}mu^2 = 0$.

The final velocity when rain drop hits the ground is $v = 5 \text{ m/s}$.

Thus, the final kinetic energy of the rain drop is

$$K_{\text{final}} = \frac{1}{2}mv^2$$

$$\Rightarrow K_{\text{final}} = \frac{1}{2} \times (10^{-3}) \times (5)^2$$

$$\Rightarrow K_{\text{final}} = \frac{1}{2} \times 10^{-3} \times 25 = 12.5 \times 10^{-3} \text{ J} = 0.0125 \text{ J}$$

Using the work-energy theorem equation,

$$W_g + W_r = K_{\text{final}} - K_{\text{initial}}$$

$$\Rightarrow 10 \text{ J} + W_r = 0.0125 \text{ J}$$

$$\Rightarrow W_r = 0.0125 - 10$$

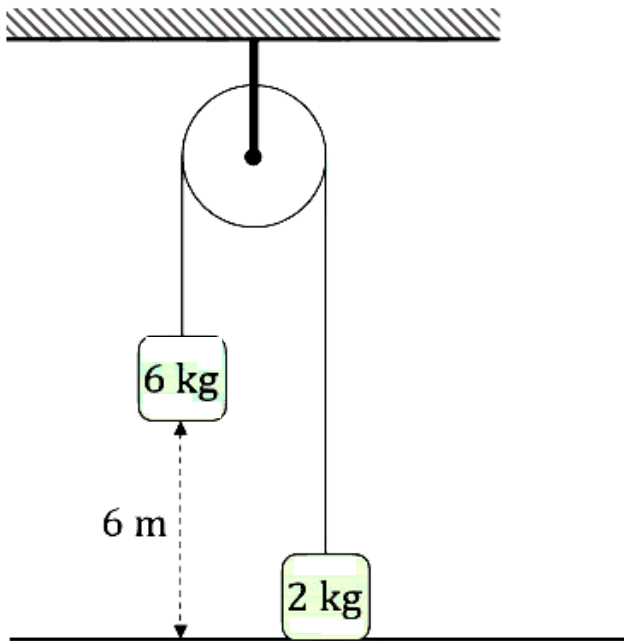
$$\Rightarrow W_r = -9.9875 \text{ J}$$

Therefore, the magnitude of the work done by the resistive force is approximately 9.98 J.

Thus, the correct option is (D).

Question12

The velocity at which 6 kg mass (shown in figure) strikes the ground when it is released from a height of 6 m above the ground is _____ m/s. Assume pulley is massless and string is light and inextensible. (Take $g = 10 \text{ m/s}^2$)



JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

7.74

B.

7.20

C.

6.55

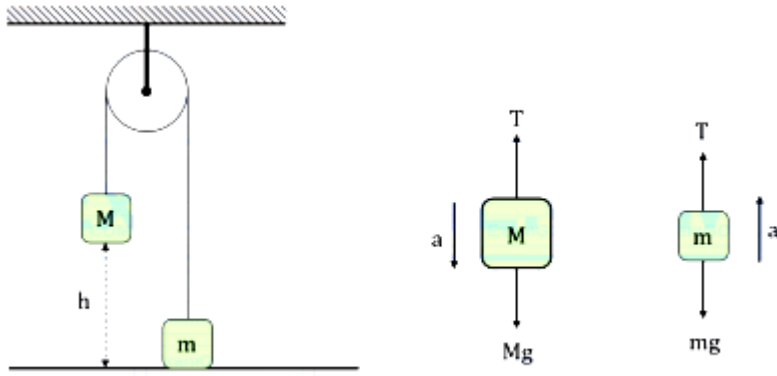
D.

4.50

Answer: A

Solution:

For the 6 kg mass (M) moving down and the 2 kg mass (m) moving up, let the acceleration of system is a and tension in the string is T .



Using Newton's second law ($F = ma$) of motion for M

$$Mg - T = Ma \dots (i)$$

For m ,

$$T - mg = ma \dots (ii)$$

Adding the two equations :

$$(M - m)g = (M + m)a$$

$$\Rightarrow a = \frac{(M - m)}{M + m} g$$

Substituting $M = 6$, $m = 2$, and $g = 10$:

$$a = \frac{6 - 2}{6 + 2} \times 10 = \frac{4}{8} \times 10 = 5 \text{ m/s}^2$$

Using the kinematic equation for constant acceleration, starting from rest ($u = 0$) :

$$v^2 = u^2 + 2as$$

$$\Rightarrow v = \sqrt{2as}$$

Where s is the height $h = 6 \text{ m}$.

$$v = \sqrt{2 \times 5 \times 6}$$

$$\Rightarrow v = \sqrt{60} \approx 7.745 \text{ m/s}$$

Therefore, the velocity is approximately 7.74 m/s .

The correct option is (A).

Question13

A body of mass 1 kg moves along a straight line with a velocity $v = 2x^2$. The work done by the body during displacement from $x = 0$ to 5 m is _____ J.

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Options:

A.

0

B.

250

C.

1250

D.

1000

Answer: C

Solution:

Work done (W) is defined as the integral of force (F) over a displacement (dx):

$$W = \int_{x_i}^{x_f} F dx$$

From Newton's Second Law, $F = ma$, and acceleration $a = \frac{dv}{dt}$.

Using the chain rule, $a = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \frac{dv}{dx}$.

$$W = \int_{x_i}^{x_f} \left[m \left(v \frac{dv}{dx} \right) \right] dx$$

$$\Rightarrow W = \int_{v_i}^{v_f} m v dv = m \int_{v_i}^{v_f} v dv = m \left[\frac{v^2}{2} \right]_{v_i}^{v_f}$$

$$\Rightarrow W = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$\Rightarrow W = \Delta K. E.$$

The mass of the body is $m = 1$ kg

The velocity of body is given as function of position, $v = 2x^2$

Initial position is $x_i = 0$ m

Final position is $x_f = 5$ m

At $x_i = 0$:

$$v_i = 2(0)^2 = 0 \text{ m/s}$$

At $x_f = 5$:

$$v_f = 2(5)^2 = 2(25) = 50 \text{ m/s}$$

On substituting these values into the Work-Energy equation :

$$W = \frac{1}{2} m (v_f^2 - v_i^2)$$

$$\Rightarrow W = \frac{1}{2} \times 1 \times (50^2 - 0^2)$$

$$\Rightarrow W = \frac{1}{2} \times (2500)$$

$$\Rightarrow W = 1250 \text{ J}$$

Therefore, the work done by the body during displacement from $x = 0$ to $x = 5$ m is 1250 J .

Hence, the correct option is (C).

Question14

A force $\mathbf{f} = x^2y\hat{i} + y^2\hat{j}$ acts on a particle in a plane $x + y = 10$. The work done by this force during a displacement from $(0, 0)$ to $(4 \text{ m}, 2 \text{ m})$ is _____ Joule (round off to the nearest integer)

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Answer: 152

Solution:

To calculate the work done by the force $\mathbf{f} = x^2y\hat{i} + y^2\hat{j}$ during the displacement from $(0, 0)$ to $(4 \text{ m}, 2 \text{ m})$, we proceed as follows:

Given the constraint that the particle moves in the plane $x + y = 10$, we can express y in terms of x :

$$y = 10 - x$$

The work done, W , is the integral of the force along the path of the displacement:

$$W = \int_0^4 x^2(10 - x) dx + \int_0^2 y^2 dy$$

Calculate the integral of the $\hat{i}$ component:

$$\begin{aligned}\int_0^4 x^2(10 - x) dx &= \int_0^4 (10x^2 - x^3) dx \\&= \left[\frac{10x^3}{3} - \frac{x^4}{4} \right]_0^4 \\&= \left(\frac{10(4)^3}{3} - \frac{(4)^4}{4} \right) - \left(\frac{10(0)^3}{3} - \frac{(0)^4}{4} \right) \\&= \frac{640}{3} - \frac{256}{4}\end{aligned}$$

Calculate the integral of the $\hat{j}$ component:

$$\begin{aligned}\int_0^2 y^2 dy &= \left[\frac{y^3}{3} \right]_0^2 \\&= \frac{(2)^3}{3} - \frac{(0)^3}{3} \\&= \frac{8}{3}\end{aligned}$$

Combine the results:

$$W = \left(\frac{640}{3} - 64 \right) + \frac{8}{3}$$

Simplify:

$$\begin{aligned}W &= \frac{640}{3} - 64 + \frac{8}{3} \\&= \frac{640 + 8}{3} - 64 \\&= \frac{648}{3} - 64 \\&= 216 - 64 \\&= 152 \text{ J}\end{aligned}$$

Thus, the work done by the force is 152 Joules.

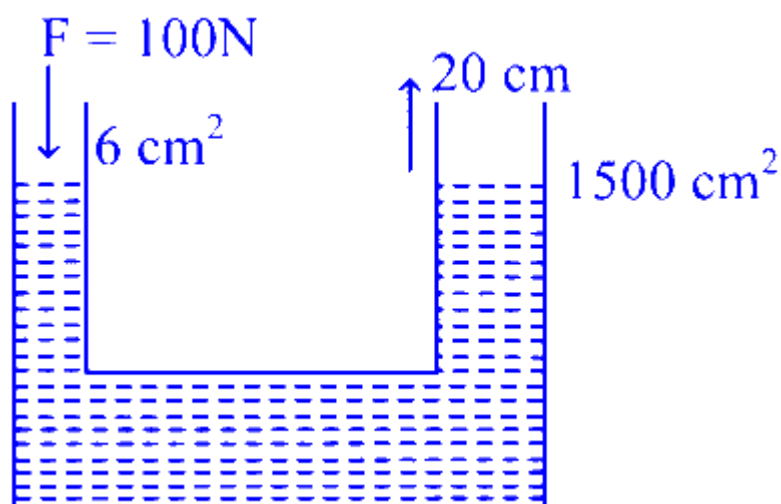
Question15

In a hydraulic lift, the surface area of the input piston is 6 cm^2 and that of the output piston is 1500 cm^2 . If 100 N force is applied to the input piston to raise the output piston by 20 cm, then the work done is _____ kJ.

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Answer: 5

Solution:



$$\frac{F_1}{A_1} = \frac{F_2}{A_2}, \frac{100}{6} = \frac{F}{1500}, F = \frac{50}{3} \times 1500$$

$$F = 50 \times 500 = 25 \times 10^3 \text{ N}$$

$$W = \vec{F} \cdot \vec{S} = 25 \times 10^3 \times \frac{20}{100}$$

$$= 5 \times 10^3 = 5 \text{ kJ}$$

Question16

A force $\vec{F} = 2\hat{i} + b\hat{j} + \hat{k}$ is applied on a particle and it undergoes a displacement $\hat{i} - 2\hat{j} - \hat{k}$. What will be the value of b , if work done on the particle is zero.

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Options:

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. 0

D. 2

Answer: B

Solution:

To determine the value of b such that the work done on the particle is zero, follow these steps:

The work done by a force on a displacement is given by the dot product:

$$\text{Work} = \vec{F} \cdot \vec{d}.$$

Given the force

$$\vec{F} = 2\hat{i} + b\hat{j} + \hat{k},$$

and the displacement

$$\vec{d} = \hat{i} - 2\hat{j} - \hat{k},$$

compute the dot product:

$$\vec{F} \cdot \vec{d} = (2)(1) + (b)(-2) + (1)(-1).$$

Simplify the expression:

$$\vec{F} \cdot \vec{d} = 2 - 2b - 1 = 1 - 2b.$$

Since the work done is zero:

$$1 - 2b = 0.$$

Solve for b :

$$2b = 1,$$

$$b = \frac{1}{2}.$$

Thus, the value of b is $\frac{1}{2}$, which corresponds to Option B.

Question17

A ball having kinetic energy KE, is projected at an angle of 60° from the horizontal. What will be the kinetic energy of ball at the highest point of its flight?

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Options:

A. $\frac{(KE)}{2}$

B. $\frac{(KE)}{8}$

C. $\frac{(KE)}{4}$

D. $\frac{(KE)}{16}$

Answer: C

Solution:

Let's break down the problem step by step:

The ball is projected with kinetic energy

$$KE = \frac{1}{2}mv^2,$$

where v is the initial speed.

The ball is launched at an angle of 60° . So, its initial velocity components are:

$$\text{Horizontal: } v_x = v \cos(60^\circ) = \frac{v}{2}$$

$$\text{Vertical: } v_y = v \sin(60^\circ)$$

At the highest point of its flight, the vertical component of the velocity becomes zero (i.e., $v_y = 0$), but the horizontal component remains unchanged.

Therefore, the kinetic energy at the highest point is only due to the horizontal velocity:

$$KE_{\text{top}} = \frac{1}{2}mv_x^2$$

Substitute $v_x = \frac{v}{2}$:

$$KE_{\text{top}} = \frac{1}{2}m\left(\frac{v}{2}\right)^2 = \frac{1}{2}m\frac{v^2}{4} = \frac{1}{8}mv^2$$

Notice that the initial kinetic energy is

$$\text{KE} = \frac{1}{2}mv^2.$$

So we can write:

$$\text{KE}_{\text{top}} = \frac{1}{8}mv^2 = \frac{1}{4}\left(\frac{1}{2}mv^2\right) = \frac{1}{4}\text{KE}$$

Thus, the kinetic energy at the highest point is $\frac{\text{KE}}{4}$.

Answer: Option C.

Question18

A force $F = \alpha + \beta x^2$ acts on an object in the x -direction. The work done by the force is 5 J when the object is displaced by 1 m . If the constant $\alpha = 1$ N then β will be

JEE Main 2025 (Online) 24th January Morning Shift

Options:

A. 15 N/m^2

B. 10 N/m^2

C. 12 N/m^2

D. 8 N/m^2

Answer: C

Solution:

The work done by a force is given by the integral of the force over the displacement. The force is given by:

$$F(x) = \alpha + \beta x^2$$

To find the work done when the object is displaced from $x = 0$ to $x = 1$, we compute:

$$W = \int_0^1 (\alpha + \beta x^2) dx$$

Substituting $\alpha = 1$ N:

$$W = \int_0^1 (1 + \beta x^2) dx = \int_0^1 1 dx + \int_0^1 \beta x^2 dx$$

Calculating each integral separately:

$$\int_0^1 1 \, dx = [x]_0^1 = 1$$

$$\int_0^1 \beta x^2 \, dx = \beta \left[\frac{x^3}{3} \right]_0^1 = \beta \left[\frac{1^3}{3} - \frac{0^3}{3} \right] = \frac{\beta}{3}$$

Thus, the total work done is:

$$W = 1 + \frac{\beta}{3}$$

We are given that the work done is 5 J, so:

$$1 + \frac{\beta}{3} = 5$$

Subtract 1 from both sides:

$$\frac{\beta}{3} = 4$$

Multiply both sides by 3 to solve for β :

$$\beta = 12 \, \text{N/m}^2$$

Therefore, the value of β is $\boxed{12 \, \text{N/m}^2}$. Thus, Option C $12 \, \text{N/m}^2$ is the correct answer.

Question 19

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A: In a central force field, the work done is independent of the path chosen.

Reason R: Every force encountered in mechanics does not have an associated potential energy.

In the light of the above statements, choose the most appropriate answer from the options given below

JEE Main 2025 (Online) 28th January Morning Shift

Options:

A. **A** is false but **R** is true

- B. Both **A** and **R** are true but **R** is NOT the correct explanation of **A**
- C. **A** is true but **R** is false
- D. Both **A** and **R** are true and **R** is the correct explanation of **A**

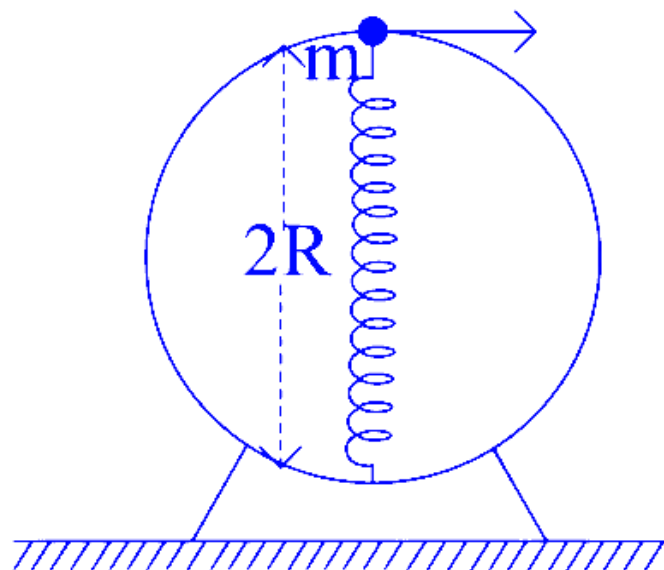
Answer: B

Solution:

In a central force field, the work done is independent of the path chosen, this is true because central forces are conservative forces. Every force encountered in mechanics does not have an associated potential energy this is also true because not all forces are conservative. Hence, option 2 is correct.

Question20

A bead of mass ' m ' slides without friction on the wall of a vertical circular hoop of radius ' R ' as shown in figure. The bead moves under the combined action of gravity and a massless spring (k) attached to the bottom of the hoop. The equilibrium length of the spring is ' R '. If the bead is released from top of the hoop with (negligible) zero initial speed, velocity of bead, when the length of spring becomes ' R ', would be (spring constant is ' k ', g is acceleration due to gravity)



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Options:

A. $\sqrt{2Rg + \frac{kR^2}{m}}$

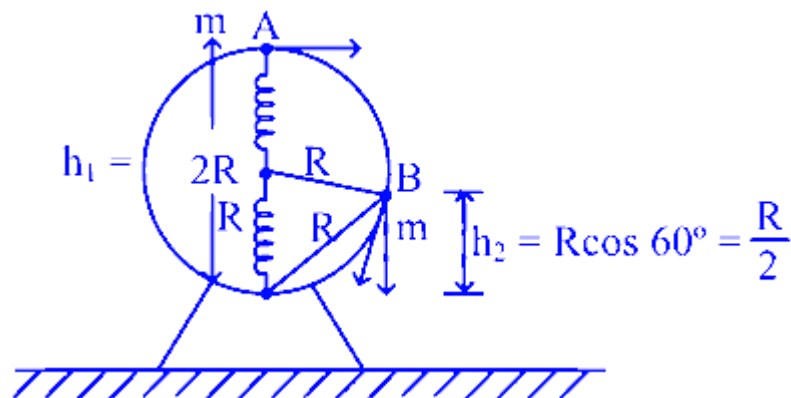
B. $\sqrt{3Rg + \frac{kR^2}{m}}$

C. $\sqrt{2Rg + \frac{4kR^2}{m}}$

D. $2\sqrt{gR + \frac{kR^2}{m}}$

Answer: B

Solution:



When mass is at position A,

Elongation of spring, $x_1 = 2R - R = R$

(As equilibrium length of spring is R)

$$h_1 = 2R$$

$v_1 = 0$ (As the bead is released from zero initial speed)

When mass is at position B,

$$x_2 = R - R = 0$$

$$h_2 = R/2, v_2 = v$$

So using energy conservation,

$$E_1 = E_f$$

$$\Rightarrow \frac{1}{2}kx_1^2 + \frac{1}{2}mv_1^2 + mgh_1 = \frac{1}{2}kx_2^2 + \frac{1}{2}mv_2^2 + mgh_2$$

$$\Rightarrow \frac{1}{2}kR^2 + mg(2R) = \frac{1}{2}k(o) + \frac{1}{2}mv^2 + mg\frac{R}{2}$$

$$\Rightarrow \frac{1}{2}kR^2 + 2mgR - \frac{mgR}{2} = \frac{1}{2}mv^2$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{1}{2}kR^2 + \frac{3mgR}{2}$$

$$\Rightarrow v^2 = \frac{kR^2}{m} + 3gR$$

$$\Rightarrow v = \sqrt{\frac{kR^2}{m} + 3gR}$$

Question21

A body of mass 4 kg is placed on a plane at a point P having coordinate $(3, 4)\text{m}$. Under the action of force $\vec{F} = (2\hat{i} + 3\hat{j})\text{N}$, it moves to a new point Q having coordinates $(6, 10)\text{m}$ in 4 sec . The average power and instanteous power at the end of 4 sec are in the ratio of :

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Options:

A.

4 : 3

B.

13 : 6

C.

1 : 2

D.

6 : 13

Answer: D

Solution:

$$\langle p \rangle = \frac{(2\hat{i} + 3\hat{j}) \cdot (3\hat{i} + 6\hat{j})}{4} = 6$$

$$\vec{a} = \left(\frac{\vec{F}}{m} = \frac{1}{2}\hat{i} + \frac{3}{4}\hat{j} \right)$$

$$\vec{v} \text{ at } t = 4\text{sec} = \left(\frac{1}{2}\hat{i} + \frac{3}{4}\hat{j} \right) \times 4 = (2\hat{i} + 3\hat{j})$$

$$P_{\text{ins}} = (2\hat{i} + 3)(2\hat{i} + 3\hat{j}) = 13$$

$$\frac{\langle P \rangle}{P_{\text{ins}}} = \frac{6}{13}$$

Note: Given data is not matching.

$$S = ut + \frac{1}{2}at^2$$

$$S = 0 + \frac{1}{2} \frac{(2\hat{i} + 3\hat{j})}{4} (4)^2 = 4\hat{i} + 6\hat{j}$$

$$\text{If } \vec{r}_i = 3\hat{i} + 4\hat{j} \text{ then } \vec{r}_f = 7\hat{i} + 10\hat{j}$$

But Final position given in the question is (6, 10).

Question22

A sand dropper drops sand of mass $m(t)$ on a conveyer belt at a rate proportional to the square root of speed (v) of the belt, i.e., $\frac{dm}{dt} \propto \sqrt{v}$. If P is the power delivered to run the belt at constant speed then which of the following relationship is true?

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

$$P \propto \sqrt{v}$$

B.

$$P \propto v$$

C.

$$P^2 \propto v^3$$

D.

$$P^2 \propto v^5$$

Answer: D

Solution:

Given, $\frac{dm}{dt} \propto \sqrt{v}$

So, $\frac{dm}{dt} = k\sqrt{v} \dots (i)$

where, $k = \text{constant}$

We know, Power = $P = FV$

$$\Rightarrow P = \frac{dp}{dt} v \text{ (As } F = \frac{dp}{dt} \text{ where, } p = \text{linear momentum)}$$

$$\Rightarrow P = \frac{d}{dt}(mv)v$$

$$\Rightarrow P = v \frac{dm}{dt} \cdot v \text{ (As speed is constant i.e. } v = \text{constant)}$$

$$\Rightarrow P = v^2(k\sqrt{v}) \text{ [From (i)]}$$

$$\Rightarrow P = kv^{5/2}$$

by squaring both sides,

$$\Rightarrow P^2 = k^2 v^5$$

$$\Rightarrow P^2 \propto v^5 \text{ (As } k^2 = \text{constant)}$$

Hence, option 4 is correct.

Question23

A particle is released from height S above the surface of the earth. At certain height its kinetic energy is three times its potential energy. The height from the surface of the earth and the speed of the particle at that instant are respectively.

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Options:

A. $\frac{S}{4}, \frac{3gS}{2}$

B. $\frac{S}{2}, \frac{3gS}{2}$

C. $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

D. $\frac{S}{2}, \sqrt{\frac{3gS}{2}}$

Answer: C

Solution:

Using the equation of motion for velocity at height x :

$$V^2 = 0 + 2g(S - x)$$

Simplifies to:

$$V^2 = 2g(S - x)$$

Potential Energy at height x :

$$\text{Potential Energy} = mgx$$

Relation between kinetic and potential energy:

Given that the kinetic energy is three times the potential energy,

$$mgx = 3 \times \frac{1}{2}mV^2$$

Simplifying gives:

$$gx = \frac{3}{2} \times 2g(S - x)$$

$$gx = 3g(S - x)$$

$$4x = S$$

Solving for x :

$$x = \frac{S}{4}$$

Finding the speed V at this instant:

$$V = \sqrt{2g\left(S - \frac{S}{4}\right)} = \sqrt{2g \times \frac{3S}{4}} = \sqrt{\frac{3gS}{2}}$$

Thus, the height from the surface of the Earth is $\frac{S}{4}$ and the speed of the particle at that height is $\sqrt{\frac{3gS}{2}}$.

Question24

A block of mass 25 kg is pulled along a horizontal surface by a force at an angle 45° with the horizontal. The friction coefficient

between the block and the surface is 0.25 . The block travels at a uniform velocity. The work done by the applied force during a displacement of 5 m of the block is :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. 490 J

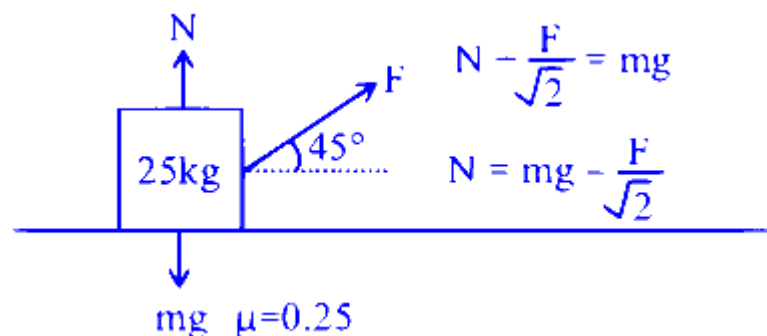
B. 970 J

C. 735 J

D. 245 J

Answer: D

Solution:



Block travels with uniform velocity

So $a = 0 \Rightarrow F \cos 45^\circ = \text{friction}$

$$\begin{aligned} \frac{F}{\sqrt{2}} &= \mu \left[mg - \frac{F}{\sqrt{2}} \right] \\ \frac{F}{\sqrt{2}} &= 0.25 \left[25 \times 9.8 - \frac{F}{\sqrt{2}} \right] \\ \Rightarrow 1.25 \frac{F}{\sqrt{2}} &= 61.25 \\ F &= \frac{61.25 \times \sqrt{2}}{1.25} = 49\sqrt{2} \\ W_{\text{ext}} &= FS \cos 45^\circ \\ &= 49\sqrt{2} \times 5 \times \frac{1}{\sqrt{2}} = 245 \text{ J} \end{aligned}$$

Question 25

An object of mass 1000 g experiences a time dependent force $\vec{F} = (2t\hat{i} + 3t^2\hat{j}) \text{ N}$. The power generated by the force at time t is:

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. $(2t^2 + 18t^3) \text{ W}$

B. $(3t^3 + 5t^5) \text{ W}$

C. $(2t^3 + 3t^5) \text{ W}$

D. $(2t^2 + 3t^3) \text{ W}$

Answer: C

Solution:

Given Information:

Force: $\vec{F} = (2t\hat{i} + 3t^2\hat{j}) \text{ N}$

Mass of the object, $m = 1000 \text{ g} = 1 \text{ kg}$

Determine Acceleration:

Using Newton's second law, $\vec{F} = m\vec{a}$.

Thus, the acceleration, $\vec{a} = 2t\hat{i} + 3t^2\hat{j}$.

Velocity Calculation:

To find velocity, integrate acceleration with respect to time:

$$\frac{d\vec{v}}{dt} = 2t\hat{i} + 3t^2\hat{j}$$

Integrate to find $\vec{v}$:

$$\vec{v} = \int (2t\hat{i} + 3t^2\hat{j}) dt = t^2\hat{i} + t^3\hat{j} + \vec{C}$$

(Assuming initial velocity is zero, $\vec{C} = 0$, hence $\vec{v} = t^2\hat{i} + t^3\hat{j}$).

Calculate Power:

Power is defined as the dot product of force and velocity vectors:

$$P = \vec{F} \cdot \vec{v} = (2t\hat{i} + 3t^2\hat{j}) \cdot (t^2\hat{i} + t^3\hat{j})$$

Computing the dot product gives:

$$P = (2t \cdot t^2) + (3t^2 \cdot t^3) = 2t^3 + 3t^5$$

The power generated at time t is $P = (2t^3 + 3t^5)$ W.

Question 26

Which one of the following forces cannot be expressed in terms of potential energy?

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

Frictional force

B.

Coulomb's force

C.

Restoring force

D.

Gravitational force

Answer: A

Solution:

The correct answer is:

Option A: Frictional force

Here's why:

Conservative vs. Non-Conservative Forces:

Conservative Forces: These are forces for which the work done is independent of the path taken; they only depend on the initial and final positions. For conservative forces, you can define a potential energy function. Examples include Coulomb's force, gravitational force, and the restoring force of a spring.

Non-Conservative Forces: These forces depend on the path taken and usually result in energy dissipation (often as heat). Friction is a typical example, and hence, it cannot be described solely by a potential energy function.

Example Explanations:

Coulomb's Force: It is a conservative force because the work done in moving a charge in an electric field only depends on the difference in electric potential between two points.

Gravitational Force: It is also conservative. The potential energy associated with gravitational force is given by gravitational potential energy, for example, $U = mgh$ near the Earth's surface.

Restoring Force: For a spring, the force is given by Hooke's law: $F = -kx$ and the corresponding potential energy is $U = \frac{1}{2}kx^2$. This is a conservative force.

Since friction is a non-conservative force that dissipates energy, it cannot be expressed in terms of potential energy.

Therefore, the correct answer is Option A: Frictional force.

Question27

**A spherical body of mass 2 kg starting from rest acquires a kinetic energy of 10000J at the end of 5th second. The force acted on the body is ____N.
[24-Jan-2023 Shift 1]**

Answer: 40

Solution:

$$\begin{aligned}\frac{1}{2} \times 2 \times v^2 &= 10000 \\ \Rightarrow v^2 &= 10000 \\ \Rightarrow v &= 100 \text{ m/s} \\ \Rightarrow v &= at = a \times 5 = 100 \\ \Rightarrow a &= 20 \text{ m/s}^2 \\ F &= ma = 2 \times 20 = 40 \text{ N}\end{aligned}$$

Question 28

A body of mass 1 kg begins to move under the action of a time dependent force $\vec{F} = (t\hat{i} + 3t^2\hat{j}) \text{ N}$. where $\hat{i}$ and $\hat{j}$ are the unit vectors along x and y axis. The power developed by above force, at the time $t = 2 \text{ s}$. will be _____ W.
[24-Jan-2023 Shift 2]

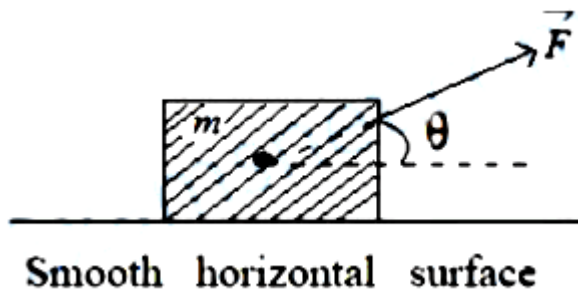
Answer: 100

Solution:

$$\begin{aligned}\vec{F} &= t\hat{i} + 3t^2\hat{j} \\ \frac{m d\vec{v}}{dt} &= t\hat{i} + 3t^2\hat{j} \\ m &= 1 \text{ kg}, \int_0^t d\vec{v} = \int_0^t t dt \hat{i} + \int_0^t 3t^2 dt \hat{j} \\ \vec{v} &= \frac{t^2}{2}\hat{i} + t^3\hat{j} \\ \text{Power} &= \vec{F} \cdot \vec{v} = \frac{t^3}{2} + 3t^5 \\ \text{At } t &= 2, \text{ power} = \frac{8}{2} + 3 \times 32 \\ &= 100\end{aligned}$$

Question 29

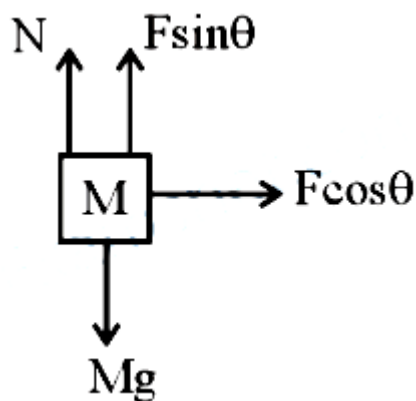
An object of mass ' m ' initially at rest on a smooth horizontal plane starts moving under the action of force $F = 2\text{N}$. In the process of its linear motion, the angle θ (as shown in figure) between the direction of force and horizontal varies as $\theta = kx$, where k is a constant and x is the distance covered by the object from its initial position. The expression of kinetic energy of the object will be $E = n/k \sin \theta$. The value of n is _____.



[25-Jan-2023 Shift 1]

Answer: 2

Solution:



$$F \cos \theta = ma$$

$$2 \cos(kx) = \frac{mvdv}{dx}$$

$$\int_0^v v dv = 2 \int_0^x \cos(kx) dx$$

$$\frac{mv^2}{2} = \frac{2}{k} \sin kx$$

$$\text{K.E.} = \frac{2}{k} \sin \theta$$

$$n = 2$$

Question30

A 0.4 kg mass takes 8 s to reach ground when dropped from a certain height ' P ' above surface of earth. The loss of potential energy in the last second of fall is J. [Take $g = 10\text{m/s}^2$]
[29-Jan-2023 Shift 1]

Answer: 300

Solution:

Displacement is 8th sec.

$$S_8 = 0 + \frac{1}{2} \times 10 \times (2 \times 8 - 1)$$

$$S_8 = 5 \times 15$$

$$\Delta U = 0.4 \times 10 \times 5 \times 15$$

$$\Delta U = 20 \times 15 = 300$$

Question31

Identify the correct statements from the following:

- (A) Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative.
- (B) Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative.
- (C) Work done by friction on a body sliding down an inclined plane is positive.
- (D) Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity is zero.
- (E) Work done by the air resistance on an oscillating pendulum is negative.

Choose the correct answer from the options given below:

[29-Jan-2023 Shift 2]

Options:

- A. B and E only
- B. A and C only
- C. B, D and E only
- D. B and D only

Answer: A

Question32

A body of mass 2 kg is initially at rest. It starts moving unidirectionally under the influence of a source of constant power P. Its displacement in 4 s is $\frac{1}{3}\alpha^2\sqrt{P}m$. The value of α will be _____
[30-Jan-2023 Shift 2]

Answer: 4

Solution:

$$\begin{aligned}\frac{1}{2}mV^2 &= Pt \\ V &= \sqrt{\frac{2Pt}{m}} \\ \frac{dx}{dt} &= \sqrt{\frac{2Pt}{m}} \\ x &= \sqrt{\frac{2P}{m}} \frac{2}{3} [t^{3/2}]_0^4 \\ x &= \frac{16\sqrt{P}}{3} = \frac{1}{3} \times 16\sqrt{P} \\ \alpha &= 4\end{aligned}$$

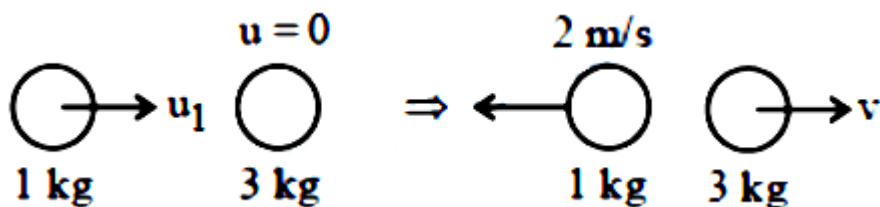
Question33

A body of mass 1 kg collides head on elastically with a stationary body of mass 3 kg. After collision, the smaller body reverses its direction of motion and moves with a speed of 2m/ s. The initial speed of the smaller body before collision is _____ ms^{-1} .

[25-Jan-2023 Shift 2]

Answer: 4

Solution:



$$1 \times u_1 = -2 + 3v \Rightarrow u_1 = -2 + 3v \dots (1)$$

$$1 = \frac{v+2}{u_1} \Rightarrow v+2 = u_1 \dots (2)$$

Solving (1) and (2)

$$u_1 = 4 \text{ m/s}$$

Question34

A nucleus disintegrates into two smaller parts, which have their velocities in the ratio 3 : 2. The ratio of their nuclear sizes will be

$\left(\frac{x}{3}\right)^{\frac{1}{3}}$. The value of 'x' is:

[25-Jan-2023 Shift 2]

Answer: 2

Solution:



$$\frac{v_1}{v_2} = \frac{3}{2}$$

$$m_1 v_1 = m_2 v_2 \Rightarrow \frac{m_1}{m_2} = \frac{2}{3}$$

Since, Nuclear mass density is constant

$$\frac{m_1}{\frac{4}{3}\pi r_1^3} = \frac{m_2}{\frac{4}{3}\pi r_2^3}$$

$$\left(\frac{r_1}{r_2} \right)^3 = \frac{m_1}{m_2}$$

$$\frac{r_1}{r_2} = \left(\frac{2}{3} \right)^{\frac{1}{3}}$$

$$\text{So, } x = 2$$

Question35

A ball of mass 200g rests on a vertical post of height 20m. A bullet of mass 10g, travelling in horizontal direction, hits the centre of the ball. After collision both travels independently. The ball hits the ground at a distance 30m and the bullet at a distance of 120m from the foot of the post. The value of initial velocity of the bullet will be (if $g = 10\text{m/s}^2$) :
[30-Jan-2023 Shift 1]

Options:

A. 120m/s

B. 60m/s

C. 400m/s

D. 360m/s

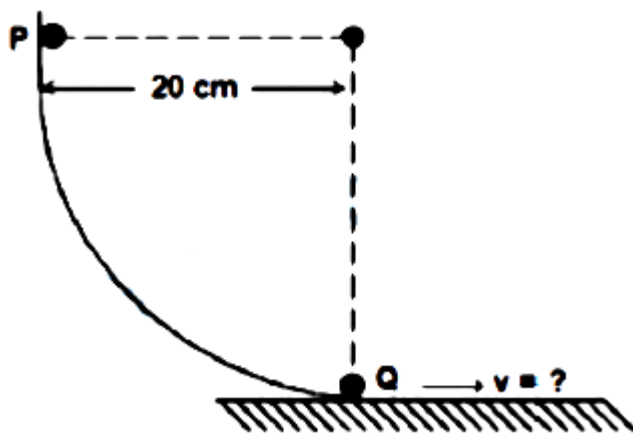
Answer: D

Solution:

$$v_1 = \frac{30}{\sqrt{\frac{2h}{g}}}, \quad v_2 = \frac{120}{\sqrt{\frac{2h}{g}}}$$
$$(0.01)u = (0.2) \frac{30\sqrt{g}}{\sqrt{2h}} + (0.01) \frac{120\sqrt{g}}{\sqrt{2h}}$$
$$u = 300 + 60 = 360\text{ms}^{-1}$$

Question36

As per the given figure, a small ball P slides down the quadrant of a circle and hits the other ball Q of equal mass which is initially at rest. Neglecting the effect of friction and assume the collision to be elastic, the velocity of ball Q after collision will be :($g = 10\text{m/s}^2$)



[30-Jan-2023 Shift 1]

Options:

- A. 0
- B. 0.25m/s
- C. 2m/s
- D. 4m/s

Answer: C

Solution:

The velocities will be interchanged after collision.

$$\text{Speed of P just before collision} = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.2} = 2 \text{ m/s}$$

Question 37

A machine gun of mass 10 kg fires 20g bullets at the rate of 180 bullets per minute with a speed of 100 m s^{-1} each. The recoil velocity of the gun is :
[30-Jan-2023 Shift 2]

Options:

A. 0.02 m/s

B. 2.5 m/s

C. 1.5 m/s

D. 0.6 m/s

Answer: D

Solution:

$$20 \times 10^{-3} \times \frac{180}{60} \times 100 = 10 \text{ V}$$
$$\Rightarrow v = 0.6 \text{ m/s}$$

Question 38

100 balls each of mass m moving with speed v simultaneously strike a wall normally and reflected back with same speed, in time t s. The total force exerted by the balls on the wall is
[31-Jan-2023 Shift 1]

Options:

A. $\frac{100 \text{ mv}}{t}$

B. $\frac{200 \text{ mv}}{t}$

C. 200 mvt

D. $\frac{\text{mv}}{100t}$

Answer: B

Solution:

$$P_i = Nm v \hat{i} \quad \vec{P}_f = -Nm v \hat{i}$$

N is Number of balls strikes with wall

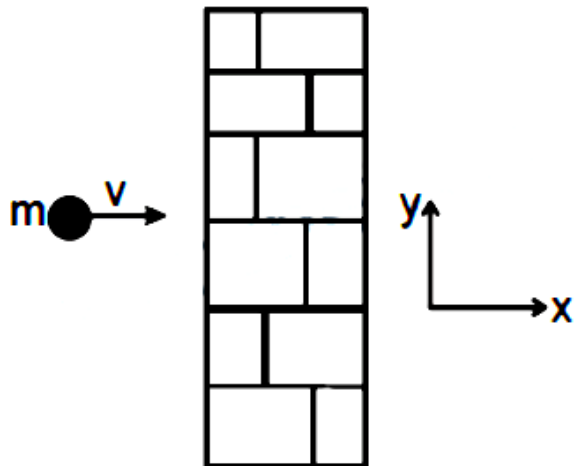
$$N = 100$$

$$\Delta \vec{p} = \vec{P}_f - \vec{P}_i = -2 Nm v \hat{i}$$

$$= -200 Nm \hat{i}$$

$$\vec{F}_{\text{Total}} = \Delta \frac{\vec{p}}{\Delta t} = - \frac{200 \text{ mvt}}{t}$$

$$|\vec{F}| = \frac{200 \text{ mv}}{t}$$



Question39

A solid sphere of mass 1 kg rolls without slipping on a plane surface. Its kinetic energy is $7 \times 10^{-3} \text{ J}$. The speed of the centre of

mass of the sphere is _____ cm s^{-1} .
[31-Jan-2023 Shift 1]

Answer: 10

Solution:

$$\frac{1}{2}mv^2 + \frac{1}{2}I^2 = 7 \times 10^{-3}$$

$$\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}MR^2\right)\left(\frac{V}{R}\right)^2 = 7 \times 10^{-3}$$

$$\frac{1}{2}MV^2\left[1 + \frac{2}{5}\right] = 7 \times 10^{-3}$$

$$\frac{1}{2}(1)(V^2)\left(\frac{7}{5}\right) = 7 \times 10^{-3}$$

$$V^2 = 10^{-2}$$

$$V = 10^{-1} = 0.1 \text{ m/s} = 10 \text{ cm/s}$$

Ans: 10

Question40

A ball is dropped from a height of 20m. If the coefficient of restitution for the collision between ball and floor is 0.5, after hitting the floor, the ball rebounds to a height of _____ m.
[31-Jan-2023 Shift 2]

Answer: 5

Solution:

We know, $h' = e^2h$

$$h' = (0.5)^2 \times 20\text{m} = 5\text{m}$$

Question41

A small particle moves to position $5\hat{i} - 2\hat{j} + \hat{k}$ from its initial position $2\hat{i} + 3\hat{j} - 4\hat{k}$ under the action of force $5\hat{i} + 2\hat{j} + 7\hat{k}$ N. The value of work done will be _____ J.
[1-Feb-2023 Shift 1]

Answer: 40

Solution:

$$\begin{aligned} W &= \vec{F} \cdot (\vec{r}_f - \vec{r}_i) \\ &= (5\hat{i} + 2\hat{j} + 7\hat{k}) \cdot ((5\hat{i} - 2\hat{j} + \hat{k}) - (2\hat{i} + 3\hat{j} - 4\hat{k})) \\ W &= 40\text{J} \end{aligned}$$

Question42

A force $F = (5 + 3y^2)$ acts on a particle in the y direction, where F is newton and y is in meter. The work done by the force during a displacement from $y = 2\text{m}$ to $y = 5\text{m}$ is _____ J.
[1-Feb-2023 Shift 2]

Answer: 132

Solution:

$$\begin{aligned} F &= 5 + 3y^2 \\ W &= \int_2^5 (5 + 3y^2) dy \end{aligned}$$

$$= \left[5y + \frac{3y^3}{3} \right]_2^5$$

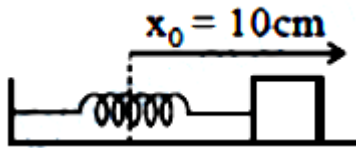
$$= 132\text{J}$$

Question43

A block is fastened to a horizontal spring. The block is pulled to a distance $x = 10\text{ cm}$ from its equilibrium position (at $x = 0$) on a frictionless surface from rest. The energy of the block at $x = 5\text{ cm}$ is 0.25J . The spring constant of the spring is _____ Nm^{-1}
[1-Feb-2023 Shift 2]

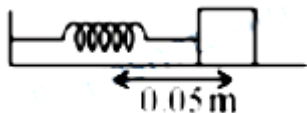
Answer: 67

Solution:



$$U_i = \frac{1}{2}kx_0^2$$

$$K_i = 0$$



$$U_f = \frac{1}{2}k\left(\frac{x_0}{2}\right)^2$$

$$K_f = 0.25\text{J}$$

$$\frac{1}{2}kx_0^2 + 0 = \frac{1}{2}k\frac{x_0^2}{4} + 0.25$$

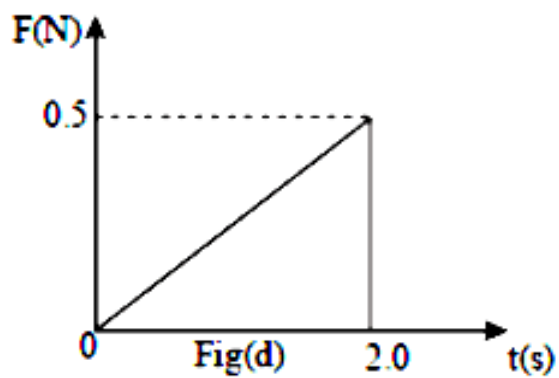
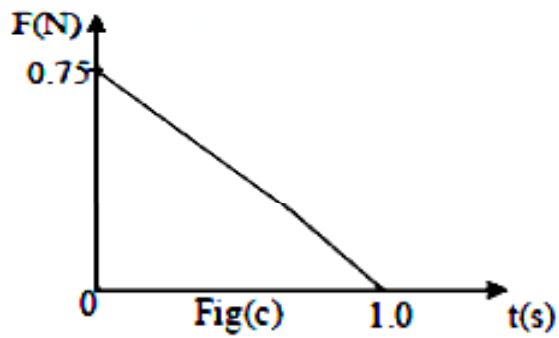
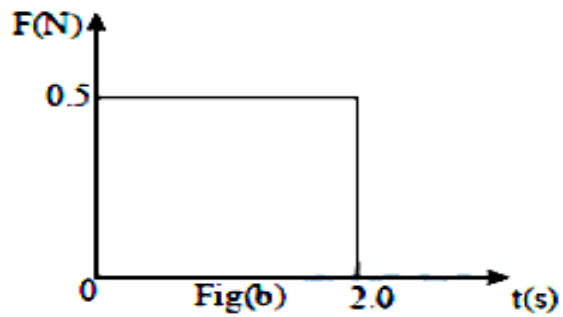
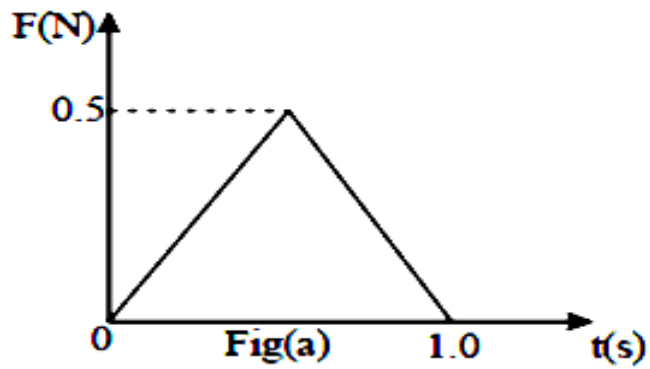
$$\frac{1}{2}kx_0^2 \frac{3}{4} = \frac{1}{4}$$

$$\frac{1}{2}k \frac{3}{100} = 1 \Rightarrow k = \frac{200}{3}\text{N/m}$$

$$= 67\text{N/m}$$

Question44

Figures (a), (b), (c) and (d) show variation of force with time.



The impulse is highest in figure.
[1-Feb-2023 Shift 2]

Options:

A. Fig (c)

B. Fig (b)

C. Fig (a)

D. Fig (d)

Answer: B

Solution:

Impulse = Area under $F = t$ curve

(a) $\frac{1}{2} \times 1 \times 0.5 = \frac{1}{4} \text{ N} \cdot \text{s}$

(b) $0.5 \times 2 = 1 \text{ N} \cdot \text{s}$ (maximum)

(c) $\frac{1}{2} \times 1 \times 0.75 = \frac{3}{8} \text{ N} \cdot \text{s}$

(d) $\frac{1}{2} \times 2 \times 0.5 = \frac{1}{2} \text{ N} \cdot \text{s}$

Question45

A small particle of mass m moves in such a way that its potential energy $U = \frac{1}{2}m\omega^2 r^2$ where ω is constant and r is the distance of the particle from origin. Assuming Bohr's quantization of momentum and circular orbit, the radius of n^{th} orbit will be proportional to,
[6-Apr-2023 shift 2]

Options:

A. n

B. n^2

C. $\frac{1}{n}$

D. $\sqrt{n}$

Answer: D

Solution:

Given $U = \frac{1}{2}m\omega^2 r^2$, to find radius r as $f(n)$, where n is orbit

Using Bohr's postulate : angular momentum $L = mvr = \frac{nh}{2\pi}$

$$\text{or } m r \omega^2 = \frac{nh}{2\pi}$$

$$\Rightarrow r \propto \sqrt{n}$$

Question46

A body is dropped on ground from a height ' h_1 ' and after hitting the ground, it rebounds to a height ' h_2 '. If the ratio of velocities of the body just before and after hitting ground is 4 , then percentage loss in kinetic energy of the body is $\frac{x}{4}$. The value of x is _____.

[6-Apr-2023 shift 2]

Answer: 375

Solution:

Let u and v be speeds, just before and after body strikes the ground.

Given $\frac{u}{v} = \frac{4}{1}$

$$\text{loss in KE} : \Delta \text{KE} = \frac{\frac{1}{2}mu^2 = \frac{1}{2}mv^2}{\frac{1}{2}mu^2}$$

$$\Delta \text{KE} = 1 - \left(\frac{v}{u}\right)^2 = 1 - \frac{1}{16} = \frac{15}{16}$$

$$\text{Percentage loss} = \frac{15}{16} \times 100 = 375$$

Question47

The momentum of a body is increased by 50%. The percentage increase in the kinetic energy of the body is _____ %.
[8-Apr-2023 shift 1]

Answer: 125

Solution:

$$K_i = \frac{P_i^2}{2m}$$

$$K_f = \frac{\left(P_i + \frac{P_i}{2}\right)^2}{2m} \Rightarrow K_f = \frac{9}{4} \frac{P_i^2}{2m}$$

$$\text{Percentage increase in K.E.} = \frac{K_f - K_i}{K_i} \times 100$$

$$\begin{aligned} & \frac{\frac{9}{4} - 1}{1} \times 100 \\ &= \frac{5}{4} \times 100 = 125\% \end{aligned}$$

Question48

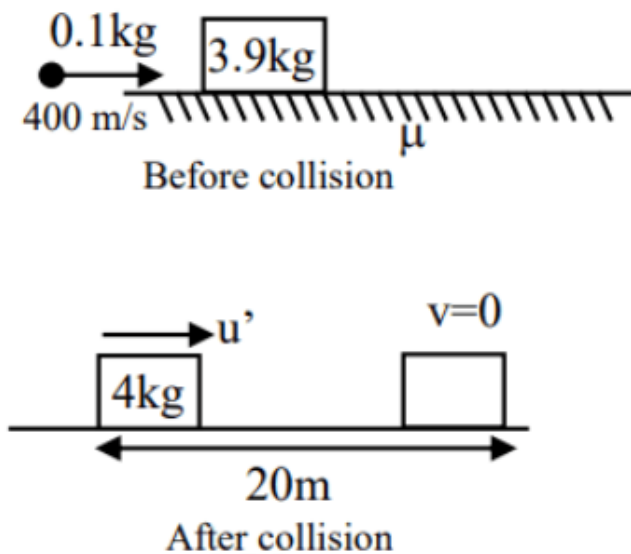
A bullet of mass 0.1 kg moving horizontally with speed 400ms^{-1} hits a wooden block of mass 3.9 kg kept on a horizontal rough surface. The bullet gets embedded into the block and moves 20m before coming to rest. The coefficient of friction between the block and the surface is (. Given $g = 10\text{m/s}^2$)
[8-Apr-2023 shift 2]

Options:

- A. 0.90
- B. 0.65
- C. 0.25
- D. 0.50

Answer: C

Solution:



Apply momentum conservation just before and just after the collision

$$0.1 \times 400 = (3.9 + .1)u$$

$$\Rightarrow u = 10\text{ m/s}$$

$$\Delta \text{KE} = W_{\text{all FORCE}}$$

$$\because f = \mu mg \text{ (kinetic friction)}$$

$$\Rightarrow 0 - \frac{1}{2}(4)(10)^2 = -\mu(4)g \times 20$$

$$\Rightarrow \mu = 0.25$$

Question49

A body of mass 5 kg is moving with a momentum of 10 kg ms^{-1} . Now a force of 2 N acts on the body in the direction of its motion for 5 s . The increase in the Kinetic energy of the body is _____ J.

[8-Apr-2023 shift 2]

Answer: 30

Solution:

$$(\text{KE}) = \frac{P^2}{2M}$$

$$\Rightarrow \frac{1}{2}mu^2 = \frac{(10)^2}{2 \times 5}$$

$$= \frac{1}{2} \times 5 \times u^2 = \frac{100}{10}$$

Initial speed $u = 2 \text{ m/s}$

$$\Delta KE = W_{\text{all forces}}$$

$$= \vec{F} \cdot \vec{S} \quad (\theta = 0^\circ)$$

$$= F \left(ut + \frac{1}{2} at^2 \right)$$

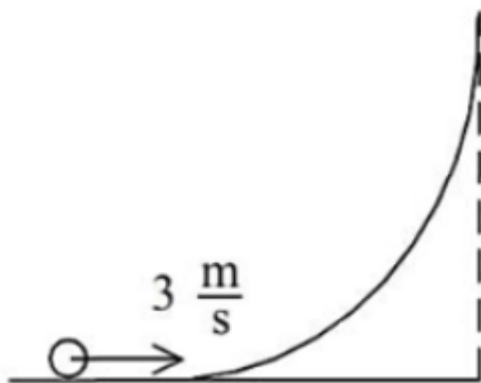
$$= 2 \cdot \left[2 \times 5 + \frac{1}{2} \times \frac{2}{5} \times 5^2 \right]$$

$$= 30 \text{ J}$$

Question 50

A hollow spherical ball of uniform density rolls up a curved surface with an initial velocity 3 m/s (as shown in figure).

Maximum height with respect to the initial position covered by it will be _____ cm (take, $g = 10 \text{ m/s}^2$)

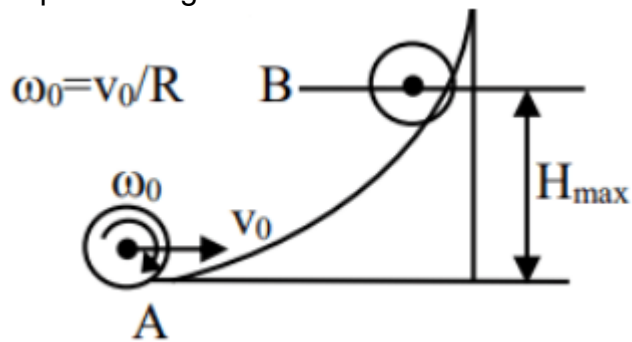


[8-Apr-2023 shift 2]

Answer: 75

Solution:

For pure rolling



$$(M \cdot E)_A = (M \cdot E)_B$$

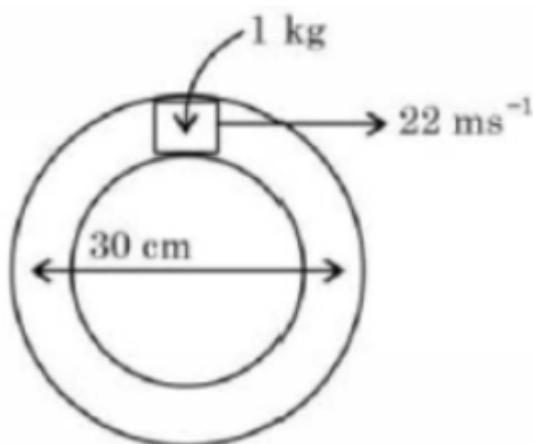
$$\Rightarrow \frac{1}{2} m v_0^2 + \frac{1}{2} \times \left(\frac{2}{3} m R^2 \right) \left(\frac{v_0}{R} \right)^2 = m g H_{\max}$$

$$\Rightarrow H_{\max} = \frac{5}{6} \frac{v_0^2}{g} = \frac{5}{6} \times \frac{3^2}{10} = 0.75 \text{ m}$$

$$\Rightarrow H_{\max} = 75 \text{ cm}$$

Question51

A closed circular tube of average radius 15 cm, whose inner walls are rough, is kept in vertical plane. A block of mass 1 kg just fit inside the tube. The speed of block is 22 m/s, when it is introduced at the top of tube. After completing five oscillations, the block stops at the bottom region of tube. The work done by the tube on the block is _____ J. (Given : $g = 10 \text{ m/s}^2$)



[10-Apr-2023 shift 1]

Answer: 245

Solution:

$$R_{\text{arg}} = 15 \text{ cm} = .15\text{m}$$

By WET

$$W_f + W_{\text{gravity}} = \Delta K = K_f - K_i$$

$$W_f + 10 \times .3 = 0 - \frac{1}{2} \times 1 \times (22)^2$$

$$W_f = -3 - \frac{484}{2} = 3 - 242 = -245$$

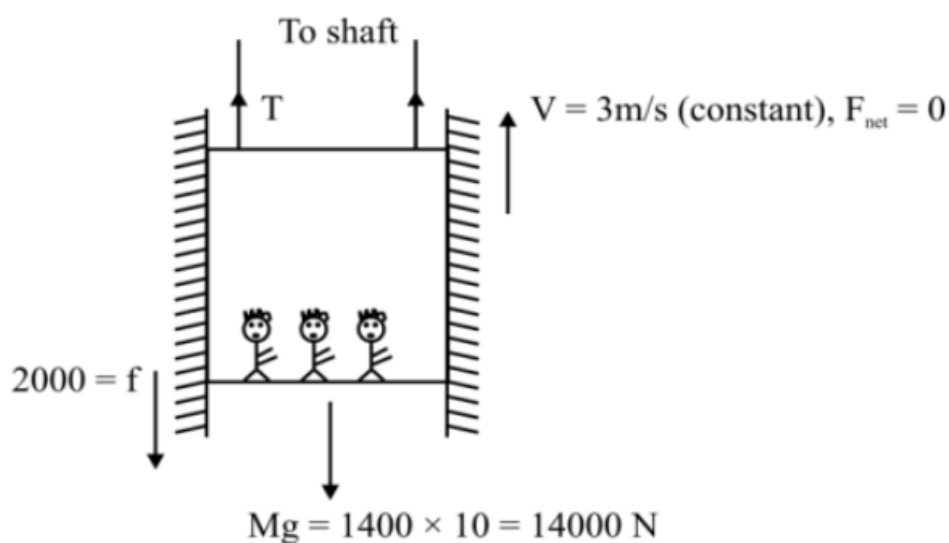
Work by friction = -245 By NTA (+245)

Question52

If the maximum load carried by an elevator is 1400 kg (600 kg– Passengers +800 kg - elevator), which is moving up with a uniform speed of 3m s^{-1} and the frictional force acting on it is 2000N, then the maximum power used by the motor is _____ kW($g = 10\text{m/ s}^2$)
[10-Apr-2023 shift 2]

Answer: 48

Solution:



Tension in the string $\Rightarrow 16000\text{N}$

Maximum power = $(F)(V)$

$$= 16000 \times 3$$

$$= 48000$$

$$= 48 \text{ kW}$$

Ans. 48

Question 53

A force $\vec{F} = (2 + 3x)\hat{i}$ acts on a particle in the x direction where F is in newton and x is in meter. The work done by this force during a displacement from $x = 0$ to $x = 4\text{m}$, is _____ mJ.
[11-Apr-2023 shift 1]

Answer: 32

Solution:

$$\vec{F} = (2 + 3x)\hat{i}$$
$$W = \int_0^4 \vec{F} \cdot d\vec{x} = \int_0^4 (2 + 3x) \cdot dx$$

$$W = \left(2x + \frac{3x^2}{2} \right) \Big|_0^4 = (8 + 24)$$

$$W = 32\text{J}$$

Question 54

A block of mass 5 kg starting from rest pulled up on a smooth incline plane making an angle of 30° with horizontal with an

effective acceleration of 1ms^{-2} . The power delivered by the pulling force at $t = 10\text{ s}$ from the starts is _____ W.

[use $g = 10\text{ms}^{-2}$] (Calculate the nearest integer value)

[11-Apr-2023 shift 2]

Answer: 300

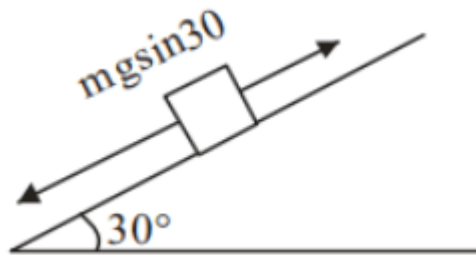
Solution:

$$F - mg \sin 30 = ma$$

$$F = 5 \times 1 + 25 = 30\text{N}$$

$$V = u + at = 0 + 1 \times 10 = 10$$

$$P = FV = 30 \times 10 = 300 \text{ watt}$$



Question55

To maintain a speed of 80 km/h by a bus of mass 500 kg on a plane rough road for 4 km distance, the work done by the engine of the bus will be _____ KJ. [The coefficient of friction between tyre of bus and road is 0.04 .]

[12-Apr-2023 shift 1]

Answer: 784

Solution:

$$\begin{aligned} F &= 0.04 \times 500 \times 9.8 \\ &= 20 \times 9.8 = 196 \\ WD &= F \times \text{disc} \\ &= 196 \times 4000 \\ &= 784 \text{ kJ} \end{aligned}$$

Question56

The ratio of powers of two motors is $\frac{3\sqrt{x}}{\sqrt{x}+1}$, that are capable of raising 300 kg water in 5 minutes and 50 kg water in 2 minutes respectively from a well of 100m deep. The value of x will be [13-Apr-2023 shift 1]

Options:

- A. 16
- B. 2
- C. 4
- D. 2.4

Answer: A

Solution:

$$\begin{aligned} P &= \frac{\text{Work}}{\text{Time}} \\ P_1 &= \frac{mgh}{t_1} = \frac{(300)g(100)}{5} \\ P_2 &= \frac{(50)g(100)}{2} \\ \frac{P_1}{P_2} &= \frac{600}{250} = \frac{12}{5} = \frac{3 \times 4}{4+1} \end{aligned}$$

$$\frac{P_1}{P_2} = \frac{3\sqrt{16}}{\sqrt{16}+1}$$

$$x = 16$$

Question57

A body of mass (5 ± 0.5) kg is moving with a velocity of (20 ± 0.4) m/ s. Its kinetic energy will be [13-Apr-2023 shift 1]

Options:

A. (1000 ± 140) J

B. (500 ± 140) J

C. (500 ± 0.14) J

D. (1000 ± 0.14) J

Answer: A

Solution:

Kinetic energy, $KE = \frac{1}{2} mv^2$

$$KE = \frac{1}{2} \times 5 \times 20^2$$

$$KE = 1000J$$

$$\frac{\Delta K}{K} = \frac{\Delta m}{m} + \frac{2 \Delta v}{v}$$

$$\frac{\Delta K}{1000} = \frac{0.5}{5} + 2 \times \frac{0.4}{20}$$

$$\Delta K = 1000(0.1 + 0.04)$$

$$\Delta K = 1000 \times 0.14$$

$$\Delta K = 140J$$

$$KE = (1000 \pm 140)J$$

Question58

A car accelerates from rest to u m/ s. The energy spent in this process is E J. The energy required to accelerate the car from

um/ s to 2um/ s is nE J. The value of n is _____.
[13-Apr-2023 shift 2]

Answer: 3

Solution:

$$\begin{aligned}E_1 &= \frac{1}{2}mu^2 - 0 = \frac{1}{2}mu^2 = E \\E_2 &= \frac{1}{2}m(2u)^2 - \frac{1}{2}mu^2 \\&= \frac{3}{2}mu^2 = 3E\end{aligned}$$

Question59

A body is released from a height equal to the radius (r) of the earth. The velocity of the body when it strikes the surface of the earth will be :

(Given g = acceleration due to gravity on the earth.)

[15-Apr-2023 shift 1]

Options:

A. $\sqrt{gR}$

B. $\sqrt{\frac{gR}{2}}$

C. $\sqrt{4gR}$

D. $\sqrt{2gR}$

Answer: A

Solution:

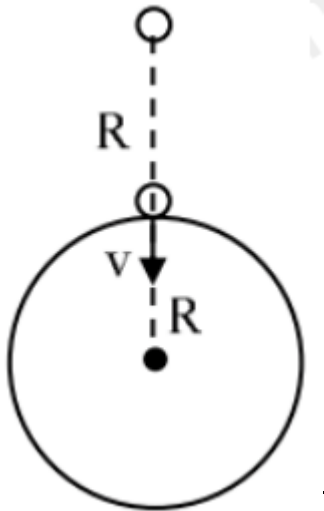
By energy conservation,

$$K_1 + U_1 = K_2 + U_2$$

$$0 - \frac{GMm}{2R} = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

$$v = \sqrt{\frac{GM}{R} \times \frac{R}{R}}$$

$$v = \sqrt{gR}$$



Question60

A block of mass 10 kg is moving along x-axis under the action of force $F = 5x$ N. The work done by the force in moving the block from $x = 2$ m to 4 m will be _____ J.

[15-Apr-2023 shift 1]

Answer: 30

Solution:

Work done, $w = \int F dx$

$$w = \int_2^4 5x dx$$

$$w = \frac{5}{2} [x^2]_2^4$$

$$w = \frac{5}{2} (16 - 4)$$

$$w = 30 \text{ J}$$

Question61

A particle of mass m moving with velocity v collides with a stationary particle of mass $2m$. After collision, they stick together and continue to move together with velocity
[10-Apr-2023 shift 1]

Options:

A. $\frac{v}{2}$

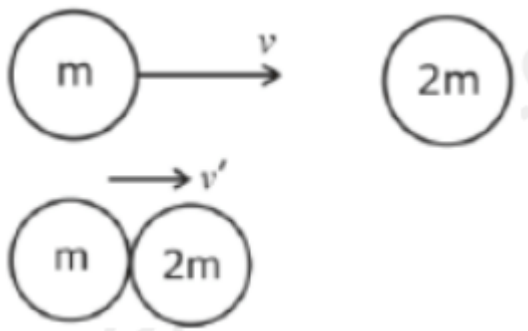
B. $\frac{v}{3}$

C. $\frac{v}{4}$

D. v

Answer: B

Solution:



$$p_i = p_f \Rightarrow mv + 2m(0) = 3m(v')$$

$$v' = \frac{v}{3}$$

Question62

An average force of 125N is applied on a machine gun firing bullets each of mass 10g at the speed of 250m/s to keep it in position. The number of bullets fired per second by the machine gun is :

[11-Apr-2023 shift 1]

Options:

A. 25

B. 5

C. 100

D. 50

Answer: D

Solution:

$$F = 125\text{N}$$

$$F = \frac{dp}{dt} \quad n \rightarrow \text{No. of bullets}$$

$$F = \frac{d(nmv)}{dt} = mv \frac{dn}{dt}$$

$$125 = \frac{10n}{1000} \times 250 \times \frac{dn}{dt}$$

$$\frac{125 \times 1000}{2500} = \frac{dn}{dt}$$

$$\frac{dn}{dt} = 50$$

option $\rightarrow$ (4)

Question63

A nucleus disintegrates into two nuclear parts, in such a way that ratio of their nuclear sizes is $1 : 2^{1/3}$. Their respective speed have a ratio of $n : 1$. The value of n is _____
[11-Apr-2023 shift 2]

Answer: 2

Solution:

From LCM :

$$m_1 V_1 = m_2 V_2$$

$$\frac{V_1}{V_2} = \frac{m_2}{m_1} = \frac{r_2^3}{r_1^3} = \left(\frac{r_2}{r_1} \right)^3$$

$$\frac{V_1}{V_2} = \left(\frac{\frac{1}{2^3}}{1} \right)^3 = \frac{2}{1}$$

$$n = 2$$

Question64

Two bodies are having kinetic energies in the ratio 16 : 9. If they have same linear momentum, the ratio of their masses respectively is:
[13-Apr-2023 shift 1]

Options:

A. 16 : 9

B. 4 : 3

C. 9 : 16

D. 3 : 4

Answer: C

Solution:

$$\text{Kinetic energy, K E} = \frac{P^2}{2m}$$

$$\frac{k_1}{k_2} = \frac{m_2}{m_1}$$

$$\frac{16}{9} = \frac{m_2}{m_1}$$

$$\frac{m_1}{m_2} = \frac{9}{16}$$

Question65

**A bullet of 10g leaves the barrel of gun with a velocity of 600m/ s. If the barrel of gun is 50 cm long and mass of gun is 3 kg, then value of impulse supplied to the gun will be:
[13-Apr-2023 shift 1]**

Options:

A. 12 Ns

B. 6 Ns

C. 3 Ns

D. 36 Ns

Answer: B

Solution:

$$\begin{aligned}\text{Impulse, } |\vec{I}| &= |\Delta \vec{p}| \\ &= mV - 0 \\ &= (10 \times 10^{-3} \text{ kg})(600 \text{ m/s}) \\ I &= 6 \text{ N-s}\end{aligned}$$

Question66

A block of mass 10 kg starts sliding on a surface with an initial velocity of 9.8 ms^{-1} . The coefficient of friction between the surface and block is 0.5. The distance covered by the block before coming to rest is:

[use $g = 9.8 \text{ ms}^{-2}$]

[24-Jun-2022-Shift-1]

Options:

A. 4.9m

B. 9.8m

C. 12.5m

D. 19.6m

Answer: B

Solution:

$$\begin{aligned} S &= \frac{u^2}{2a} = \frac{u^2}{2(\mu g)} \\ &= \frac{(9.8)^2}{2 \times 0.5 \times (9.8)} \\ &= \frac{9.8}{1} \\ &= 9.8\text{m} \end{aligned}$$

Question67

A boy ties a stone of mass 100g to the end of a 2m long string and whirls it around in a horizontal plane. The string can withstand the maximum tension of 80N. If the maximum speed with which the stone can revolve is $\frac{K}{\pi}$ rev./min. The value of K is :

(Assume the string is massless and unstretchable)

[24-Jun-2022-Shift-1]

Options:

A. 400

B. 300

C. 600

D. 800

Answer: C

Solution:

$$\begin{aligned} T &= m\omega^2 r \\ \Rightarrow 80 &= 0.1 \times \left(2\pi \times \frac{K}{\pi} \times \frac{1}{60} \right)^2 \times 2 \end{aligned}$$

$$\Rightarrow \frac{800}{2} = \frac{K^2}{900}$$

$$\Rightarrow K = 30 \times 20 = 600$$

Question68

A particle experiences a variable force $\vec{F} = (4x\hat{i} + 3y^2\hat{j})$ in a horizontal $x - y$ plane. Assume distance in meters and force is newton. If the particle moves from point (1, 2) to point (2, 3) in the $x - y$ plane, then Kinetic Energy changes by :
[24-Jun-2022-Shift-1]

Options:

A. 50.0J

B. 12.5J

C. 25.0J

D. 0J

Answer: C

Solution:

$$W = \int \vec{F} \cdot d\vec{r}$$

$$= \int_1^2 4x dx + \int_2^3 3y^2 dy$$

$$= [2x^2]_1^2 + [y^3]_2^3$$

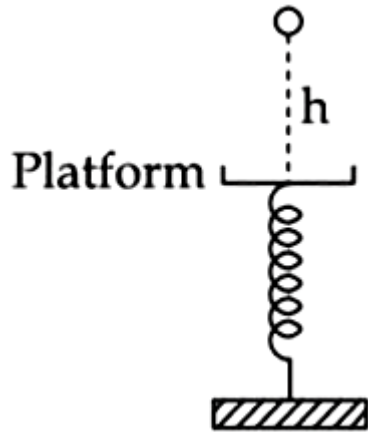
$$= 2 \times 3 + (27 - 8)$$

$$= 25J$$

Question69

A ball of mass 100g is dropped from a height $h = 10$ cm on a platform fixed at the top of a vertical spring (as shown in figure). The ball stays on the platform and the platform is depressed by a distance

$\frac{h}{2}$. The spring constant is Nm^{-1} .



(Use $g = 10\text{ms}^{-2}$)
[24-Jun-2022-Shift-1]

Answer: 120

Solution:

$$\begin{aligned}mg\left(h + \frac{h}{2}\right) &= \frac{1}{2}k\left(\frac{h}{2}\right)^2 \\ \Rightarrow 0.1 \times 10 \times (0.15) &= \frac{1}{2}k(0.05)^2 \\ \Rightarrow k &= 120\text{N/m}\end{aligned}$$

Question70

A stone of mass m , tied to a string is being whirled in a vertical circle with a uniform speed. The tension in the string is
[24-Jun-2022-Shift-2]

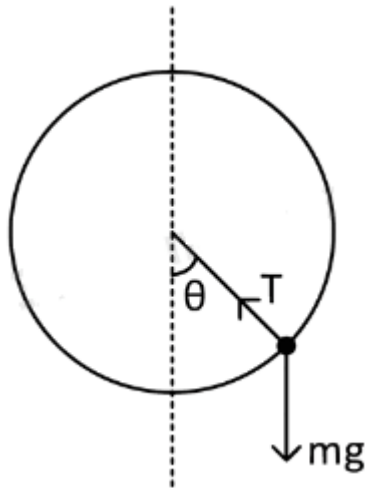
Options:

- A. the same throughout the motion.
- B. minimum at the highest position of the circular path.
- C. minimum at the lowest position of the circular path.

D. minimum when the rope is in the horizontal position.

Answer: B

Solution:



$$\text{At any } \theta : T - mg \cos \theta = \frac{mv^2}{R}$$

$$\Rightarrow T = mg \cos \theta + \frac{mv^2}{R}$$

Since v is constant,

$\Rightarrow T$ will be minimum when $\cos \theta$ is minimum.

$\Rightarrow \theta = 180^\circ$ corresponds to T_{minimum}

Question71

Potential energy as a function of r is given by $U = \frac{A}{r^{10}} - \frac{B}{r^5}$, where r is the interatomic distance, A and B are positive constants. The equilibrium distance between the two atoms will be :
[24-Jun-2022-Shift-2]

Options:

A. $\left(\frac{A}{B} \right)^{\frac{1}{5}}$

B. $\left(\frac{B}{A} \right)^{\frac{1}{5}}$

$$C. \left(\frac{2A}{B} \right)^{\frac{1}{5}}$$

$$D. \left(\frac{B}{2A} \right)^{\frac{1}{5}}$$

Answer: C

Solution:

For equilibrium

$$-\frac{dU}{dr} = 0 = \frac{10A}{r^{11}} - \frac{5B}{r^6}$$

$$\Rightarrow r^5 = \frac{2A}{B}$$

$$\text{And } r = \left(\frac{2A}{B} \right)^{1/5}$$

Question 72

**A 0.5 kg block moving at a speed of 12ms^{-1} compresses a spring through a distance 30 cm when its speed is halved. The spring constant of the spring will be _____ Nm^{-1}
[25-Jun-2022-Shift-1]**

Answer: 600

Solution:

$$\frac{1}{2}mV^2 = \frac{1}{2}kx^2 + \frac{1}{2}m\left(\frac{V}{2}\right)^2$$

$$\Rightarrow \frac{3}{8}mv^2 = \frac{1}{2}kx^2$$

$$\Rightarrow k = \frac{3}{4} \times \frac{1}{2} \times \frac{144}{9} \times 100$$

$$= 600$$

$$\Rightarrow 600$$

Question73

The ratio of specific heats $\left(\frac{C_P}{C_V} \right)$ in terms of degree of freedom

(f) is given by :

[25-Jun-2022-Shift-2]

Options:

A. $\left(1 + \frac{f}{3} \right)$

B. $\left(1 + \frac{2}{f} \right)$

C. $\left(1 + \frac{f}{2} \right)$

D. $\left(1 + \frac{1}{f} \right)$

Answer: B

Solution:

$$\frac{C_P}{C_V} = \gamma$$

$$C_V = \left(\frac{f}{2} \right) R \text{ and } C_P - C_V = R$$

$$\Rightarrow \frac{C_P}{C} = \frac{1 + f/2}{f/2} = 1 + \frac{2}{f}$$

Question74

Arrange the four graphs in descending order of total work done; where W_1 , W_2 , W_3 and W_4 are the work done corresponding to figure a, b, c and d respectively.

Figure-a

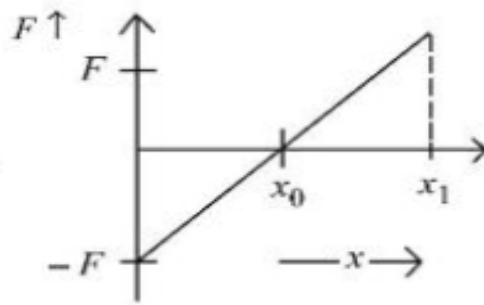


Figure-b

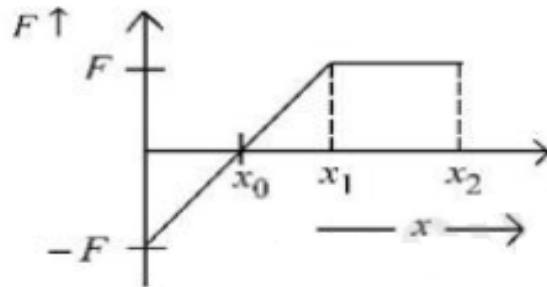


Figure-c

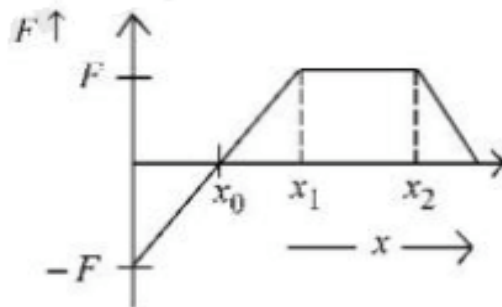
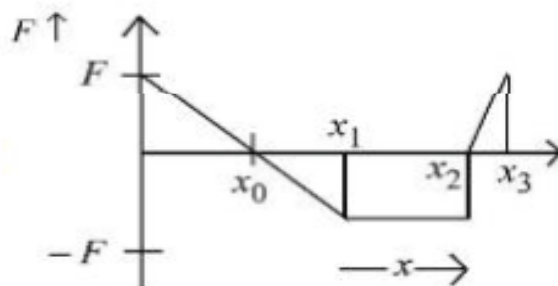


Figure-d



[26-Jun-2022-Shift-2]

Options:

A. $W_3 > W_2 > W_1 > W_4$

B. $W_3 > W_2 > W_4 > W_1$

C. $W_2 > W_3 > W_4 > W_1$

D. $W_2 > W_3 > W_1 > W_4$

Answer: A

Solution:

$$W_a = 0, W_b = +ve, W_c = +ve > W_b, W_d = -ve$$

$$\Rightarrow W_c > W_b > W_a > W_d$$

$$\Rightarrow W_3 > W_2 > W_1 > W_4$$

Question 75

A stone tied to a spring of length L is whirled in a vertical circle with the other end of the spring at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u . The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x(u^2 - gL)}$. The value of x is -
[27-Jun-2022-Shift-2]

Options:

A. 3

B. 2

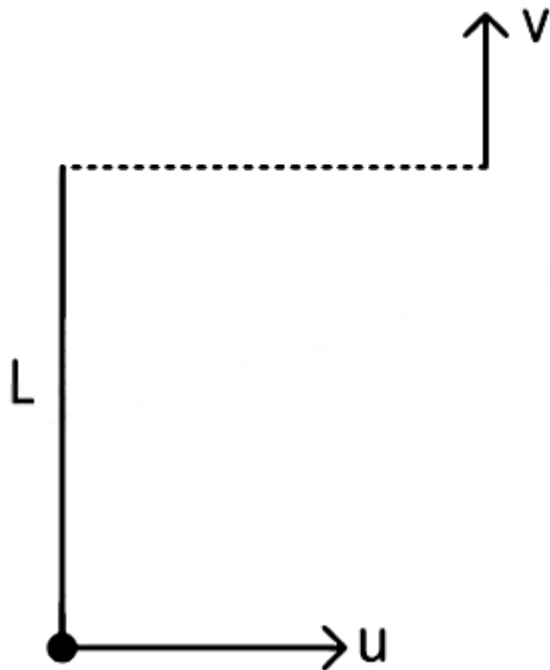
C. 1

D. 5

Answer: B

Solution:

$$\vec{v} = \sqrt{u^2 - 2gL} \hat{j}$$

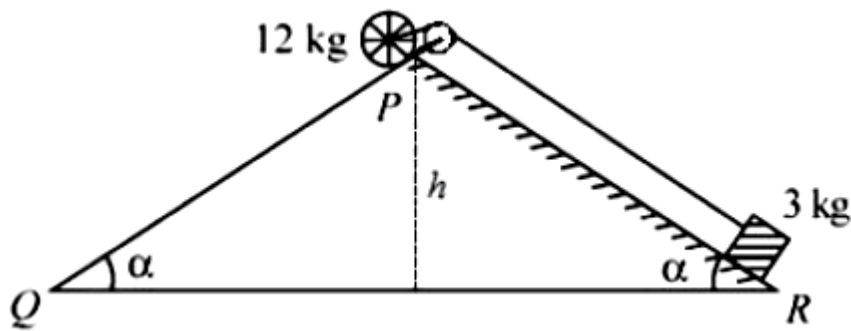


$$\begin{aligned} \vec{u} &= u \hat{i} \\ \therefore |\vec{v} - \vec{u}| &= \sqrt{(u^2 - 2gL) + u^2} \\ &= \sqrt{2u^2 - 2gL} \\ \therefore x &= 2 \end{aligned}$$

Question 76

A rolling wheel of 12 kg is on an inclined plane at position P and connected to a mass of 3 kg through a string of fixed length and pulley as shown in figure. Consider PR as friction free surface. The velocity of centre of mass of the wheel when it reaches at the bottom Q of the inclined plane PQ will be $\frac{1}{2}\sqrt{xgh}$ m/s. The value

of x is _____



[27-Jun-2022-Shift-2]

Answer: None

Solution:

For rolling wheel

$$[12g\sin \alpha - 3g\sin \alpha] \times R = (2 \times 12R^2 + 3R^2) \times \frac{a}{R}$$

$$\Rightarrow \frac{9g\sin \alpha}{27} = a$$

$$\Rightarrow a = \frac{g\sin \alpha}{3}$$

$$\therefore v = \sqrt{2 \times \frac{g\sin \alpha}{3} \times \frac{h}{\sin \alpha}} = \sqrt{\frac{2}{3}gh}$$

$$= \frac{1}{2} \times \sqrt{\frac{8}{3}gh}$$

$$\therefore x = \frac{8}{3} = 2.67$$

Question77

A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration is varying with time t as $a = k^2rt^2$, where k is a constant. The power delivered to the particle by the force acting on it is given as

[28-Jun-2022-Shift-1]

Options:

A. zero

B. $mk^2r^2t^2$

C. mk^2r^2t

D. mk^2rt

Answer: C

Solution:

$$a_r = k^2rt^2 = \frac{v^2}{r}$$

$$\Rightarrow v^2 = k^2r^2t^2 \text{ or } v = krt$$

$$\text{and } \frac{d|v|}{dt} = kr$$

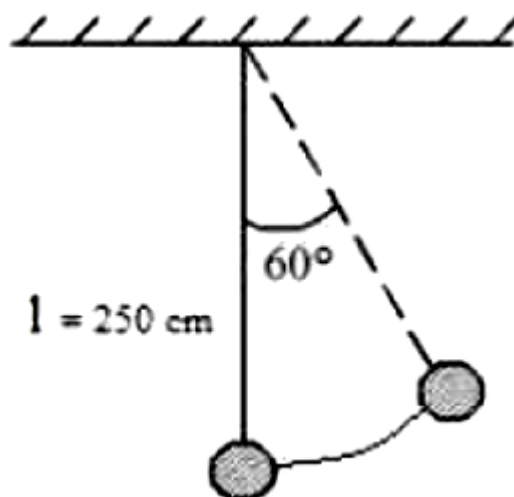
$$\Rightarrow a_t = kr$$

$$\Rightarrow |\vec{F} \cdot \vec{v}| = (mkr)(krt)$$

$$= mk^2r^2t = \text{power delivered}$$

Question 78

A pendulum is suspended by a string of length 250 cm. The mass of the bob of the pendulum is 200g. The bob is pulled aside until the string is at 60° with vertical as shown in the figure. After releasing the bob, the maximum velocity attained by the bob will be ____ ms^{-1} . (if $g = 10\text{m/s}^2$)



[28-Jun-2022-Shift-1]

Answer: 5

Solution:

$$\begin{aligned}\frac{1}{2}mv^2 &= mgl(1 - \cos \theta) \\ \Rightarrow v &= \sqrt{2gl(1 - \cos \theta)} \\ &= \sqrt{2 \times 10 \times 2.5 \times \frac{1}{2}} \\ &= 5 \text{ m/s}\end{aligned}$$

Question 79

A particle of mass 500 gm is moving in a straight line with velocity $v = bx^{5/2}$. The work done by the net force during its displacement from $x = 0$ to $x = 4\text{m}$ is : (Take $b = 0.25\text{m}^{-3/2}\text{s}^{-1}$).
[29-Jun-2022-Shift-1]

Options:

- A. 2J
- B. 4J
- C. 8J
- D. 16J

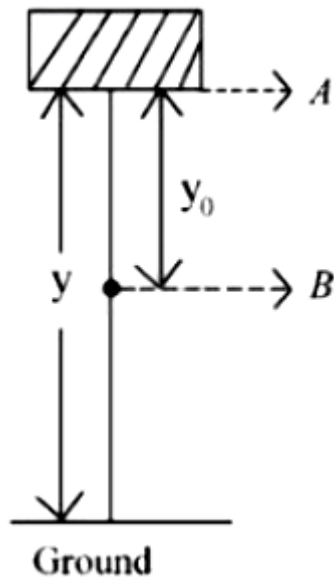
Answer: D

Solution:

$$\begin{aligned}W_{\text{total}} &= \Delta K \\ &= \frac{1}{2} \left(\frac{1}{2} \right) [\{ b(4)^{5/2} \}^2 - 0] \\ &= \frac{b^2}{4} \times 4^5 \\ \Rightarrow W_{\text{total}} &= 16\text{J}\end{aligned}$$

Question80

In the given figure, the block of mass m is dropped from the point 'A'. The expression for kinetic energy of block when it reaches point 'B' is



[29-Jun-2022-Shift-2]

Options:

- A. $\frac{1}{2}mgy_0^2$
- B. $\frac{1}{2}mgy^2$
- C. $mg(y - y_0)$
- D. mgy_0

Answer: D

Solution:

Loss in potential energy = gain in kinetic energy

$$-(mg(y - y_0) - mgy) = K E - 0$$

$$\Rightarrow KE = mgy_0$$

Question81

A batsman hits back a ball of mass 0.4 kg straight in the direction of the bowler without changing its initial speed of 15ms^{-1} . The impulse imparted to the ball is ____Ns.
[26-Jun-2022-Shift-2]

Answer: 0

Solution:

$$\begin{aligned}\text{Impulse} &= \text{change in momentum} \\ &= m[v - (-v)] = 2mv \\ &= 2 \times 0.4 \times 15 = 12 \text{ N s}\end{aligned}$$

Question82

Two blocks of masses 10 kg and 30 kg are placed on the same straight line with coordinates (0, 0) cm and (x, 0) cm respectively. The block of 10 kg is moved on the same line through a distance of 6 cm towards the other block. The distance through which the block of 30 kg must be moved to keep the position of centre of mass of the system unchanged is :
[27-Jun-2022-Shift-1]

Options:

- A. 4 cm towards the 10 kg block
- B. 2 cm away from the 10 kg block
- C. 2 cm towards the 10 kg block

D. 4 cm away from the 10 kg block

Answer: C

Solution:

For COM to remain unchanged,

$$m_1 x_1 = m_2 x_2$$

$$\Rightarrow 10 \times 6 = 30 \times x_2$$

$$\Rightarrow x_2 = 2 \text{ cm towards } 10 \text{ kg block.}$$

Question83

What percentage of kinetic energy of a moving particle is transferred to a stationary particle when it strikes the stationary particle of 5 times its mass?

(Assume the collision to be head-on elastic collision)

[27-Jun-2022-Shift-1]

Options:

A. 50.0%

B. 66.6%

C. 55.6%

D. 33.3%

Answer: C

Solution:

For a head on elastic collision

$$v_2 = \frac{mu_1}{m+5m} + \frac{mu_1}{m+5m}$$

$$= \frac{2u_1}{6} \text{ or } \frac{u_1}{3}$$

$$\text{Initial kinetic energy of first mass} = \frac{1}{2}mu_1^2$$

Final kinetic energy of second mass

$$= \frac{1}{2} \times 5m \left(\frac{u_1}{3} \right)^2$$

$$= \frac{5}{9} \left(\frac{1}{2} m u_1^2 \right)$$

⇒ kinetic energy transferred = 55% of initial kinetic energy of first colliding mass

Question84

A pendulum of length 2m consists of a wooden bob of mass 50g. A bullet of mass 75g is fired towards the stationary bob with a speed v . The bullet emerges out of the bob with a speed $\frac{v}{3}$ and the bob just completes the vertical circle. The value of v is _____ ms^{-1} . (if $g = 10\text{m/s}^2$).

[27-Jun-2022-Shift-1]

Answer: 10

Solution:

$$v_{\text{bob}} = \sqrt{5gl} = \sqrt{5 \times 10 \times 2} = 10\text{m/s}$$

Conserving momentum :

$$75 \times v = 75 \times \frac{v}{3} + 50 \times 10$$

$$\Rightarrow 50v = 50 \times 10$$

$$\Rightarrow v = 10\text{m/s}$$

Question85

The position vector of 1 kg object is $\vec{r} = (3\hat{i} - \hat{j})\text{m}$ and its velocity $\vec{v} = (3\hat{j} + \hat{k})\text{ms}^{-1}$. The magnitude of its angular momentum is $\sqrt{x}\text{Nm}$ where x is _____

[28-Jun-2022-Shift-1]

Answer: 91

Solution:

$$\begin{aligned} |\vec{i}| &= |\vec{r} \times (m\vec{v})| \\ &= |(3\hat{i} - \hat{j}) \times (3\hat{j} + \hat{k})| \\ &= |-\hat{i} - 3\hat{j} + 9\hat{k}| \\ &= \sqrt{91} \end{aligned}$$

Question86

**A body of mass M at rest explodes into three pieces, in the ratio of masses 1 : 1 : 2. Two smaller pieces fly off perpendicular to each other with velocities of 30ms^{-1} and 40ms^{-1} respectively. The velocity of the third piece will be :
[29-Jun-2022-Shift-1]**

Options:

A. 15ms^{-1}

B. 25ms^{-1}

C. 35ms^{-1}

D. 50ms^{-1}

Answer: B

Solution:

Conserving momentum :

$$\begin{aligned} m(30\hat{i}) + m(40\hat{j}) + 2m(\vec{v}) &= \vec{0} \\ \Rightarrow \vec{v} &= -15\hat{i} - 20\hat{j} \\ \Rightarrow |\vec{v}| &= 25\text{m/s} \end{aligned}$$

Question87

A body of mass 0.5kg travels on straight line path with velocity $v = (3x^2 + 4)\text{m/s}$. The net workdone by the force during its displacement from $x = 0$ to $x = 2\text{m}$ is :
[25-Jul-2022-Shift-1]

Options:

- A. 64J
- B. 60J
- C. 120J
- D. 128J

Answer: B

Solution:

$$\begin{aligned}v &= 3x^2 + 4 \\ \text{at } x = 0, v_1 &= 4\text{m/s} \\ x = 2, v_2 &= 16\text{m/s} \\ \Rightarrow \text{Work done} &= \Delta \text{ kinetic energy} \\ &= \frac{1}{2} \times m(v_2^2 - v_1^2) \\ &= \frac{1}{2}(256 - 16) \\ &= 60\text{J}\end{aligned}$$

Question88

A bag of sand of mass 9.8kg is suspended by a rope. A bullet of 200g travelling with speed 10ms^{-1} gets embedded in it, then loss of kinetic energy will be :
[25-Jul-2022-Shift-2]

Options:

- A. 4.9J
- B. 9.8J
- C. 14.7J
- D. 19.6J

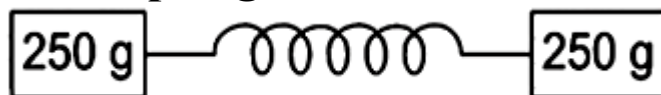
Answer: B

Solution:

$$\begin{aligned}
 \text{Loss in K E} &= \frac{1}{2} \times \frac{m_1 m_2}{m_1 + m_2} \times v^2 \\
 &= \frac{1}{2} \times \frac{9.8 \times 0.2}{10} \times (10)^2 \\
 &= 9.8\text{J}
 \end{aligned}$$

Question89

As per the given figure, two blocks each of mass 250g are connected to a spring of spring constant 2Nm^{-1} . If both are given velocity v in opposite directions, then maximum elongation of the spring is:



[26-Jul-2022-Shift-1]

Options:

- A. $\frac{v}{2\sqrt{2}}$
- B. $\frac{v}{2}$
- C. $\frac{v}{4}$
- D. $\frac{v}{\sqrt{2}}$

Answer: B

Solution:

$\therefore$ Loss in K E = Gain in spring energy

$$\Rightarrow \frac{1}{2}mv^2 \times 2 = \frac{1}{2}kx_m^2$$

$$\Rightarrow 2 \times \frac{1}{4} \times v^2 = 2 \times x_m^2$$

$$\Rightarrow x_m = \sqrt{\frac{v^2}{4}} = \frac{v}{2}$$

Question90

A body of mass 8 kg and another of mass 2 kg are moving with equal kinetic energy. The ratio of their respective momentum will be :

[26-Jul-2022-Shift-2]

Options:

A. 1 : 1

B. 2 : 1

C. 1 : 4

D. 4 : 1

Answer: B

Solution:

$$K.E = \frac{P^2}{2m}$$

$$K_1 = \frac{P_1^2}{2(8)}; K_2 = \frac{P_2^2}{2(2)}$$

$$K_1 = K_2$$

$$So_0$$

$$4P_2^2 = P_1^2$$

$$\frac{P_1}{P_2} = 2$$

Question91

Sand is being dropped from a stationary dropper at a rate of 0.5kg s^{-1} on a conveyor belt moving with a velocity of 5m s^{-1} . The power needed to keep the belt moving with the same velocity will be :

[27-Jul-2022-Shift-1]

Options:

A. 1.25W

B. 2.5W

C. 6.25W

D. 12.5W

Answer: D

Solution:

$$\frac{dm}{dt} = 0.5 \text{ kg/s}$$

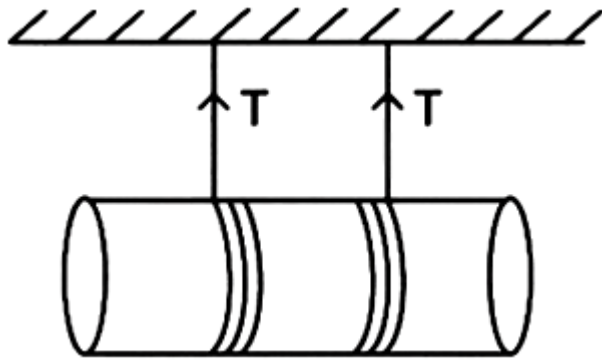
$$v = 5 \text{ m/s}$$

$$F = \frac{v dm}{dt} = 2.5 \text{ kg m/s}^2$$

$$P = \vec{F} \cdot \vec{v} = (2.5)(5)W \\ = 12.5W$$

Question92

A solid cylinder length is suspended symmetrically through two massless strings, as shown in the figure. The distance from the initial rest position, the cylinder should be unbinding the strings to achieve a speed of 4m s^{-1} , is _____ cm. (. take $g = 10\text{m s}^{-2}$)



[27-Jul-2022-Shift-2]

Answer: 120

Solution:

From energy conservation

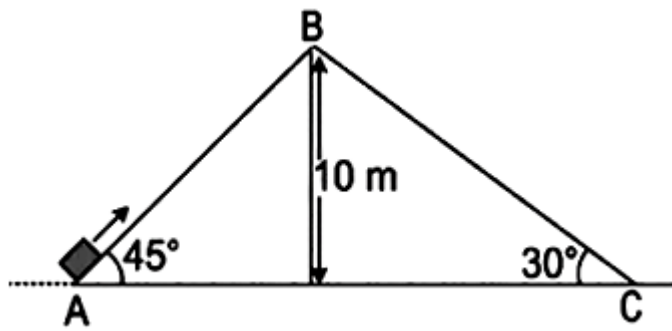
$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2} \frac{mR^2}{2} \omega^2$$

$$10h = \frac{16}{2} + \frac{16}{4} \Rightarrow h = 1.2\text{m} = 120\text{ cm}$$

Question93

Two inclined planes are placed as shown in figure. A block is projected from the Point A of inclined plane AB along its surface with a velocity just sufficient to carry it to the top Point B at a height 10m. After reaching the Point B the block slides down on inclined plane BC. Time it takes to reach to the point C from point A is $t(\sqrt{2} + 1)$ s. The value of t is _____.
(use $g = 10\text{m/s}^2$)



[27-Jul-2022-Shift-2]

Answer: 2

Solution:

$$\text{From E.C.} = \frac{1}{2} m v_0^2 = mgh$$

$$v_0 = 10\sqrt{2}$$

For A → B

at B, $v = 0$

$$a = -g \sin 45^\circ = \frac{-10}{\sqrt{2}}$$

$$v = u + at_1 \Rightarrow 0 = 10\sqrt{2} - \frac{10}{\sqrt{2}} t_1 \Rightarrow t_1 = 2 \text{ sec}$$

For B → C

$$s = ut_2 + \frac{1}{2} at_2^2$$

$$\frac{10}{\sin 30^\circ} = \frac{1}{2} (10 \sin 30^\circ) t_2^2$$

$$t_2 = 2\sqrt{2}$$

So total time

$$T = t_1 + t_2$$

$$T = t_1 + t_2$$

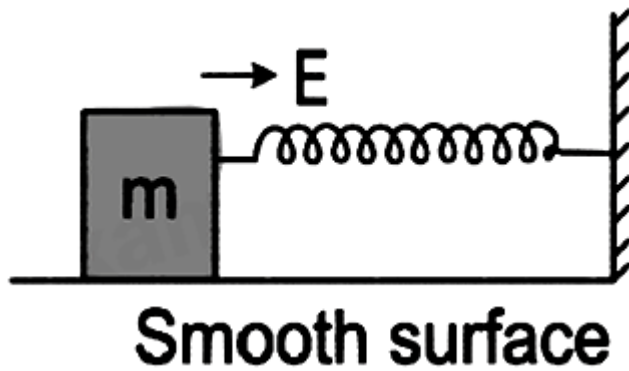
$$= 2\sqrt{2} + 2$$

$$= 2(\sqrt{2} + 1) \text{ sec}$$

Question94

A block of mass '1m' (as shown in figure) moving with kinetic energy E compresses a spring through a distance 25 cm when, its speed is halved. The value of spring constant of used spring will

be $nE \text{ Nm}^{-1}$ for $n = \underline{\hspace{2cm}}$.



[28-Jul-2022-Shift-1]

Answer: 24

Solution:

Using work - energy theorem

$$W_{\text{net}} = (K_f - K_i)$$

$$\Rightarrow -\frac{1}{2}Kx^2 = \frac{1}{2}m\left(\frac{v}{2}\right)^2 - \frac{1}{2}mv^2 = \frac{E}{4} - E$$

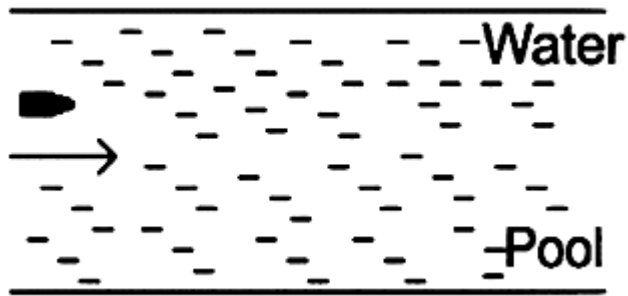
$$\Rightarrow \frac{1}{2}Kx^2 = \frac{3E}{4} \Rightarrow K = \frac{3E}{2x^2}$$

$$\Rightarrow K = \frac{3E}{2 \times \left(\frac{1}{4}\right)^2} = 24E$$

$$n = 24$$

Question95

A bullet of mass 200g having initial kinetic energy 90J is shot inside a long swimming pool as shown in the figure. If it's kinetic energy reduces to 40J within 1 s, the minimum length of the pool, the bullet has to travel so that it completely comes to rest is



[28-Jul-2022-Shift-2]

Options:

- A. 45 m
- B. 90 m
- C. 125 m
- D. 25 m

Answer: A

Solution:

$$\begin{aligned}
 \frac{1}{2}mx^2 &= 90 \\
 \Rightarrow \frac{1}{2} \times 0.2 \times x^2 &= 90 \\
 x^2 &= 900 \\
 x &= 30\text{m/ s} \\
 \frac{1}{2}mv^2 &= 40 \Rightarrow v = \frac{2}{3} \times 30 = 20\text{m/ s} \\
 20 &= 30 - a \times 1 \Rightarrow a = -10\text{m/ s}^2 \\
 0 - x^2 &= 2as \\
 s &= \frac{x^2}{-2a} = \frac{30 \times 30}{2 \times 10} \\
 &= 45\text{m}
 \end{aligned}$$

Question96

Sun light falls normally on a surface of area 36cm^2 and exerts an average force of $7.2 \times 10^{-9}\text{N}$ within a time period of 20 minutes. Considering a case of complete absorption, the energy flux of

incident light is
[28-Jul-2022-Shift-2]

Options:

A. $25.92 \times 10^2 \text{W/cm}^2$

B. $8.64 \times 10^{-6} \text{W/cm}^2$

C. 6.0W/cm^2

D. 0.06W/cm^2

Answer: D

Solution:

$$\frac{I}{C} \times \text{area} = \text{force}$$

$$\frac{I}{C} \times 36 \times 10^{-4} = 7.2 \times 10^{-9}$$

$$I = \frac{7.2 \times 10^{-9} \times 3 \times 10^8}{36 \times 10^{-9} \times 10}$$

$$= \frac{6 \times 10^{-1}}{10^{-3}}$$

$$I = 6 \times 10^2 \frac{\text{W}}{\text{m}^2}$$

$$= 0.06 \frac{\text{W}}{\text{cm}^2}$$

Question97

If momentum of a body is increased by 20%, then its kinetic energy increases by
[29-Jul-2022-Shift-2]

Options:

A. 36%

B. 40%

C. 44%

D. 48%

Answer: C

Solution:

Let, initial momentum of body (p_i) = p

∴ Final momentum (p_f) = $p_i + 20\%$ of p_i

$$= p + 0.2p$$

$$= 1.2p$$

We know,

$$\text{Kinetic energy (E)} = \frac{p^2}{2m}$$

$$\therefore E_i = \frac{p^2}{2m}$$

$$\text{and } E_f = \frac{(1.2p)^2}{2m} = \frac{1.44p^2}{2m}$$

∴ % Change in kinetic energy

$$= \frac{E_f - E_i}{E_i} \times 100$$

$$= \frac{\frac{1.44p^2}{2m} - \frac{p^2}{2m}}{\frac{p^2}{2m}} \times 100$$

$$= \frac{\frac{p^2}{2m}(1.44 - 1)}{\frac{p^2}{2m}} \times 100 = 0.44 \times 100 = 44$$

Question98

Two billiard balls of mass 0.05kg each moving in opposite directions with 10ms^{-1} collide and rebound with the same speed. If the time duration of contact is $t = 0.005\text{s}$, then what is the force exerted on the ball due to each other?

[25-Jul-2022-Shift-2]

Options:

A. 100N

B. 200N

C. 300N

D. 400N

Answer: B

Solution:

Change in momentum of one ball

$$= 2 \times (0.05)(10)\text{kgm/s}$$

$$= 1\text{kgm/s}$$

$$\Rightarrow F_{\text{avg}} = \frac{1}{\Delta t} = \frac{1}{0.005}\text{N}$$

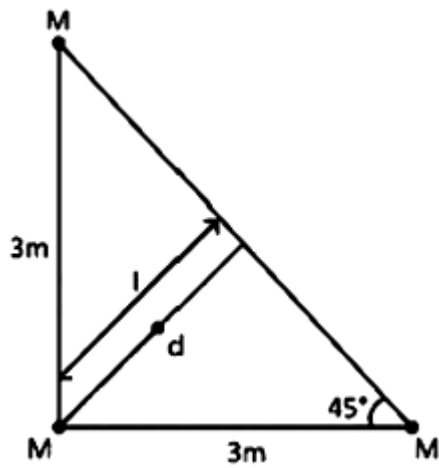
$$= 200\text{N}$$

Question99

Three identical spheres each of mass M are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 3m each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be $\sqrt{x}\text{m}$. The value of x is ____
[25-Jul-2022-Shift-2]

Answer: 2

Solution:



$$d_{\text{cm}} = 3 \sin 45^\circ = \frac{3}{\sqrt{2}}$$

$$d_{\text{cm}} = \frac{2}{3} \times \frac{3}{\sqrt{2}} = \sqrt{2} = \sqrt{x}$$

$$x = 2$$

Question100

A ball of mass 0.15 kg hits the wall with its initial speed of 12ms^{-1} and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is 100N , calculate the time duration of the contact of ball with the wall.

[26-Jul-2022-Shift-2]

Options:

A. 0.018s

B. 0.036s

C. 0.009s

D. 0.072s

Answer: B

Solution:

$$\vec{P}_i = 0.15 \times 12 (\hat{i})$$

$$\vec{P}_f = 0.15 \times 12 (-\hat{i})$$

$$|\Delta \vec{P}| = 3.6 \text{ kg} \cdot \text{m/s}$$

$$3.6 = F \Delta t$$

$$3.6 = 100 \Delta t$$

$$\Delta t = 0.036 \text{ sec}$$

Question101

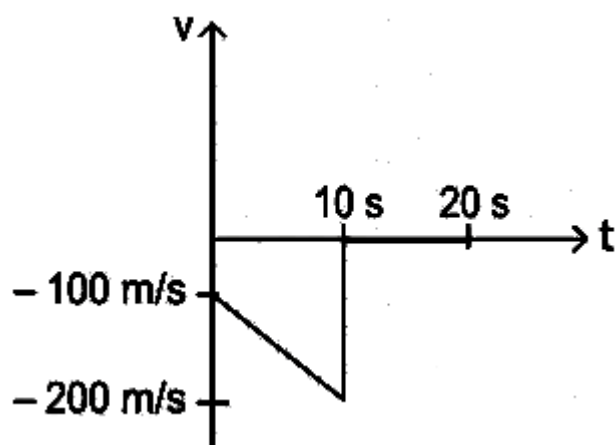
A bullet is shot vertically downwards with an initial velocity of 100 m/s from a certain height. Within 10 s , the bullet reaches the ground and instantaneously comes to rest due to the perfectly inelastic collision. The velocity-time curve for total time $t = 20 \text{ s}$ will be:

(Take $g = 10 \text{ m/s}^2$).

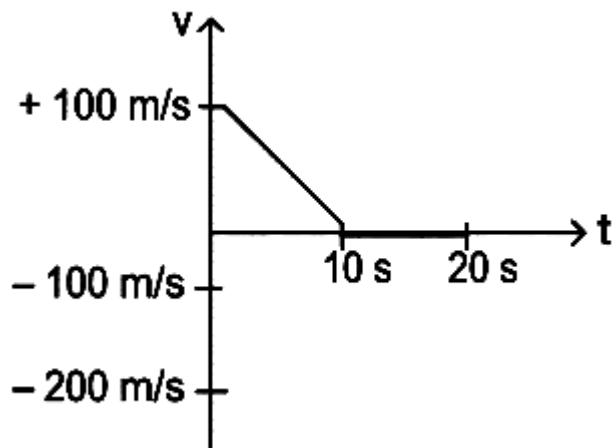
[27-Jul-2022-Shift-1]

Options:

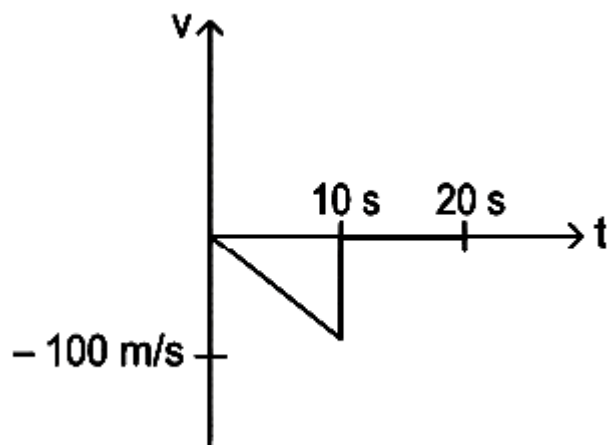
A.



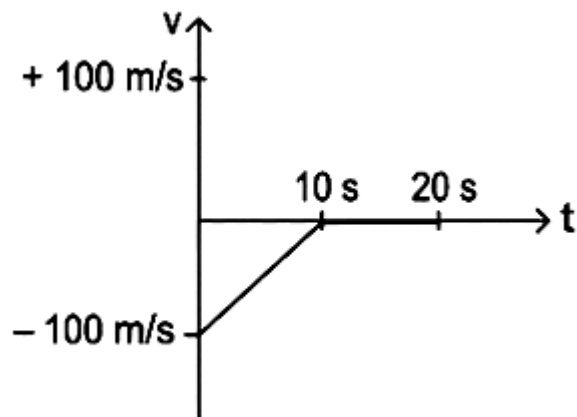
B.



C.



D.



Answer: A

Solution:

$$|v_{10}| = (100 + 10 \times 10) \text{ m/s}$$

$$v_{10} = -200 \text{ m/s and } v_0 = -100 \text{ m/s}$$

from 10 s to 20 s velocity remains zero

⇒ from $t = 0 \text{ s}$ to 10 s velocity increases in magnitude linearly.

⇒ graph given in option A fits correctly.

Question102

The velocity of the bullet becomes one third after it penetrates 4 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling at $(4 + x)$ cm inside the block. The value of x is :

[27-Jul-2022-Shift-2]

Options:

A. 2.0

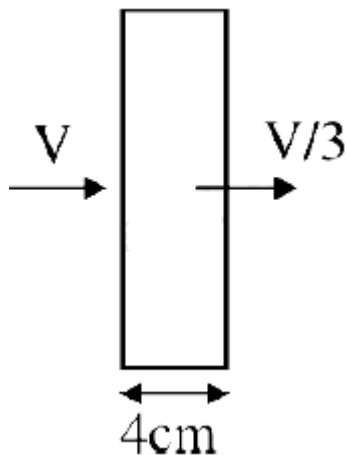
B. 1.0

C. 0.5

D. 1.5

Answer: C

Solution:



$$\left(\frac{V}{3}\right)^2 = V^2 - 2a(4) \Rightarrow a = \frac{8V^2}{9(8)} = \frac{V^2}{9}$$

$$0 = V^2 - 2a(4 + x)$$

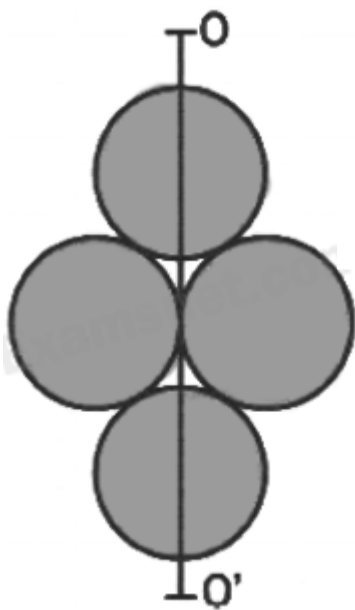
$$\Rightarrow V^2 = 2\left(\frac{V^2}{9}\right)(4 + x)$$

$$4.5 = 4 + x$$

$$x = 0.5$$

Question103

Four identical discs each of mass ' M ' and diameter 'a' are arranged in a small plane as shown in figure. If the moment of inertia of the system about OO' is $\frac{x}{4}Ma^2$. Then, the value of x will be _____.



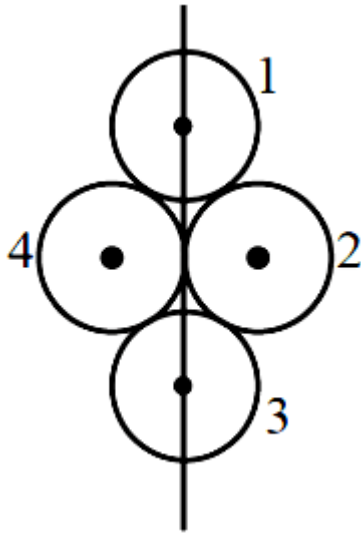
[28-Jul-2022-Shift-1]

Answer: 3

Solution:

$$I_1 = I_3 = \frac{MR^2}{4}$$

$$I_2 = \frac{MR^2}{4} + MR^2 = \frac{5}{4}MR^2 = I_4$$



$$\text{So } I = I_1 + I_2 + I_3 + I_4$$

$$= \frac{MR^2}{2} + \frac{5}{2}MR^2$$

$$= 3MR^2, \text{ Putting } R = \frac{a}{2}$$

$$I = \frac{3Ma^2}{4}, \text{ So } x = 3$$

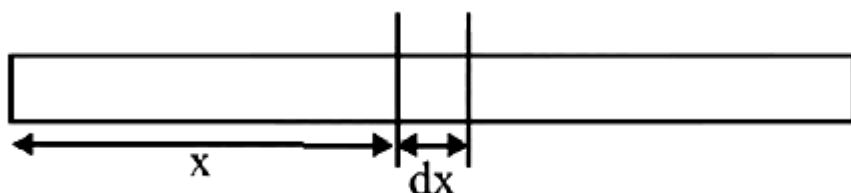
Question104

The distance of centre of mass from end A of a one dimensional rod (AB) having mass density $\rho = \rho_0 \left(1 - \frac{x^2}{L^2} \right)$ kg/m and length L (in meter) is $\frac{3L}{\alpha}$ m. The value of α is _____ (where x is the distance from end A)

[28-Jul-2022-Shift-2]

Answer: 8

Solution:



$$dm = \lambda \cdot dx = \lambda_0 \left(1 - \frac{x^2}{\ell^2} \right)$$

$$X_{cm} = \frac{\int x dm}{\int dm_\ell}$$

$$= \frac{\lambda_0 \int_0^\ell \left(x - \frac{x^3}{\ell^2} \right) dx}{\int_0^\ell \lambda_0 \left(1 - \frac{x^2}{\ell^2} \right) dx} = \frac{\frac{\ell^2}{2} - \frac{\ell^4}{4\ell^2}}{\ell - \frac{\ell^3}{3\ell^2}} = \frac{3\ell}{8}$$

Question 105

Two bodies of mass 1 kg and 3 kg have position vectors $\hat{i} + 2\hat{j} + \hat{k}$ and $-3\hat{i} - 2\hat{j} + \hat{k}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector :

[29-Jul-2022-Shift-1]

Options:

A. $\hat{i} + 2\hat{j} + \hat{k}$

B. $-3\hat{i} - 2\hat{j} + \hat{k}$

C. $-2\hat{i} + 2\hat{k}$

D. $2\hat{i} - \hat{j} + 2\hat{k}$

Answer: A

Solution:

$$\vec{r}_{com} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

$$= \frac{(1-9)\hat{i} + (2-6)\hat{j} + (1+3)\hat{k}}{4}$$

$$= \frac{-8\hat{i} - 4\hat{j} + 4\hat{k}}{4}$$

$$\vec{r}_{\text{com}} = -2\hat{i} - \hat{j} + \hat{k}$$

$$|\vec{r}| = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$|\hat{i} + 2\hat{j} + \hat{k}| = \sqrt{6}$$

Question106

Two particles having masses 4g and 16g respectively are moving with equal kinetic energies. The ratio of the magnitudes of their linear momentum is $n : 2$. The value of n will be
[25 Feb 2021 Shift 2]

Answer: 1

Solution:

Given, mass of particle A, $m_A = 4\text{g}$

Mass of particle B, $m_B = 16\text{g}$

Kinetic energy of A and B is same

i.e. $K E_A = K E_B$

As, kinetic energy ($K E$) = $p^2/2m$

where, p is momentum and m is mass.

$$\therefore \frac{(p_A)^2}{m_A} = \frac{(p_B)^2}{m_B} \Rightarrow \frac{p_A^2}{4} = \frac{p_B^2}{16} \Rightarrow \frac{p_A}{p_B} = \frac{1}{2}$$

$\therefore$ linear momentum is $n : 2$.

$$\therefore n = 1$$

Question107

Two solids A and B of mass 1kg and 2kg respectively are moving with equal linear momentum. The ratio of their kinetic energies $(K E)_A : (K E)_B$ will be $\frac{A}{1}$, so the value of A will be
[24 Feb 2021 Shift 2]

Answer: 2

Solution:

Given, $m_A = 1\text{kg}$, $m_B = 2\text{kg}$, $(K E)_A : (K E)_B = A : 1$

Linear momentum of A and B are equal.

$$\Rightarrow p_A = p_B$$

$$\therefore \text{Kinetic energy } (K E) = p^2 / 2m$$

$$\therefore \frac{K E_A}{K E_B} = \frac{m_B}{m_A} = \frac{2}{1} = \frac{A}{1} \Rightarrow A = 2$$

Question108

A small bob tied at one end of a thin string of length 1m is describing a vertical circle, so that the maximum and minimum tension in the string are in the ratio 5 : 1. The velocity of the bob at the highest position is m/s. (Take, $g = 10\text{m/s}^2$)
[25 Feb 2021 Shift 1]

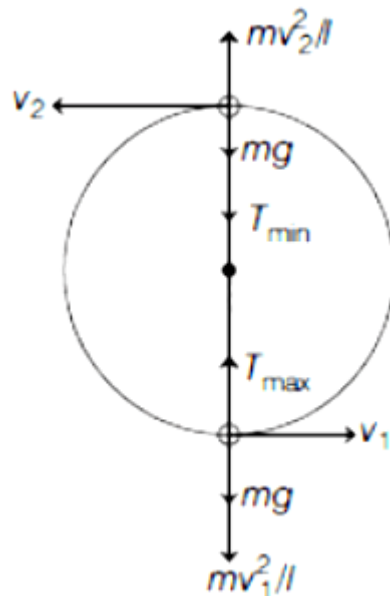
Answer: 5

Solution:

Given, length of string, $l = 1\text{m}$

$T_{\max}$ and $T_{\min}$ be the tension in string and v_1 and v_2 be the velocities of bob at bottom and top

in vertical circle.



$$T_{\max} = mg + mv_1^2/l$$

$$\text{and } T_{\min} = mv_2^2/l - mg$$

$$\therefore \frac{T_{\max}}{T_{\min}} = \frac{mg + mv_1^2/l}{mv_2^2/l - mg} = \frac{5}{1}$$

$$\Rightarrow mg + mv_1^2/l = 5mv_2^2/l - 5mg$$

$$\text{Here, } v_1 = \sqrt{v_2^2 + 4gl}$$

$$\Rightarrow mg + \frac{m}{l}(v_2^2 + 4gl) = \frac{5mv_2^2}{l} - 5mg$$

$$\Rightarrow g + \frac{v_2^2 + 4gl}{l} = \frac{5v_2^2}{l} - 5g$$

$$\Rightarrow 6gl = 5v_2^2 - v_2^2 - 4gl \Rightarrow 10gl = 4v_2^2$$

$$\Rightarrow v_2 = \sqrt{\frac{10gl}{4}} = \sqrt{\frac{10 \times 10 \times 1}{4}}$$

$$= \sqrt{25} = 5 \text{ m/s}$$

Question 109

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A Body P having mass M moving with speed u has head-on collision elastically with another body Q having mass m initially at rest. If $m \ll M$, body Q will have a maximum speed equal to $2u$ after collision.

Reason R During elastic collision, the momentum and kinetic

energy are both conserved.

In the light of the above statements, choose the most appropriate answer from the options given below.
[26 Feb 2021 Shift 1]

Options:

- A. A is not correct but R is correct.
- B. Both A and R are correct but R is not the correct explanation of A.
- C. Both A and R are correct and R is the correct explanation of A.
- D. A is correct but R is not correct.

Answer: C

Solution:

Let v_1 and v_2 are the speed of P and Q after collision.

By using law of conservation of momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$M u + m \cdot 0 = M v_1 + m v_2$$

$$\Rightarrow \frac{M(u - v_1)}{m} = v_2$$

and by using law of conservation of energy,

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\Rightarrow M u^2 + 0 = M v_1^2 + m v_2^2$$

$$\Rightarrow M(u^2 - v_1^2) = m v_2^2$$

$$\Rightarrow M(u - v_1)(u + v_1)/m = v_2^2$$

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2$$

$$M u^2 + 0 = M v_1^2 + m v_2^2$$

$$\Rightarrow M(u^2 - v_1^2) = m v_2^2$$

$$\Rightarrow M(u - v_1)(u + v_1)/m = v_2^2$$

Substituting the value of $M \frac{(u - v_1)}{m}$ from Eq. (i) in

Eq. (ii), we get

$$v_2(u + v_1) = v_2^2$$

$$u + v_1 = v_2$$

$$\Rightarrow M \gg m$$

$$v_1 = u$$

$$v_2 = 2u$$

Eq. (ii), we get

$$v_2(u + v_1) = v_2^2$$

$$u + v_1 = v_2$$

$$M > m$$

$$v_1 = u$$

$$v_2 = 2u$$

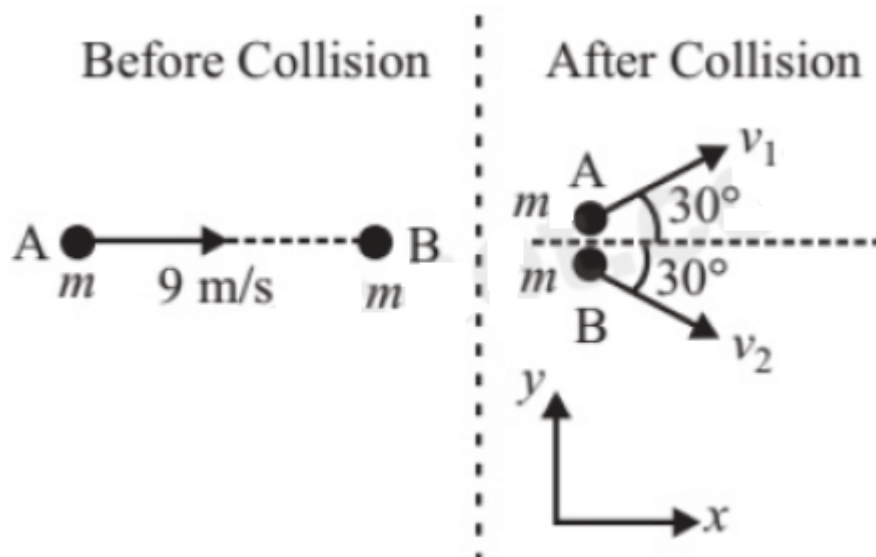
Hence, option (c) is the correct.

Question110

A ball with a speed of 9m/s collides with another identical ball at rest. After the collision, the direction of each ball makes an angle of 30° with the original direction. The ratio of velocities of the balls after collision is $x : y$, where x is [24feb2021shift1]

Answer: 1

Solution:



From conservation of momentum along y -axis.

$$\vec{P}_{iy} = \vec{P}_{fy}$$

$$0 + 0 = mv_1 \sin 30^\circ \hat{j} + mv_2 \sin 30^\circ (-\hat{j})$$

$$mv_2 \sin 30^\circ = mv_1 \sin 30^\circ$$

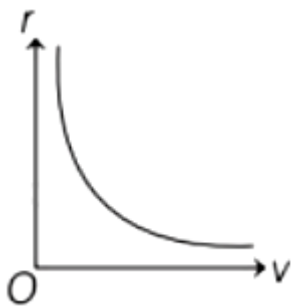
$$v_2 = v_1 \quad \text{or} \quad \frac{v_1}{v_2} = 1$$

Question111

A particle of mass m moves in a circular orbit under the central potential field, $U(r) = \frac{-C}{r}$, where C is a positive constant. The correct radius-velocity graph of the particle's motion is [18 Mar 2021 Shift 2]

Options:

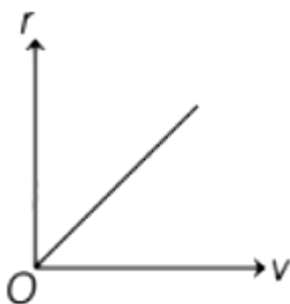
A.



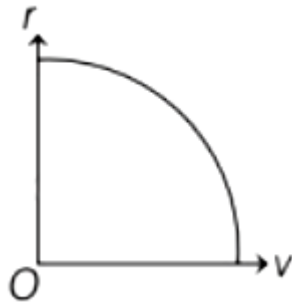
B.



C.



D.



Answer: A

Solution:

The central potential field when particle moves in circular orbit,

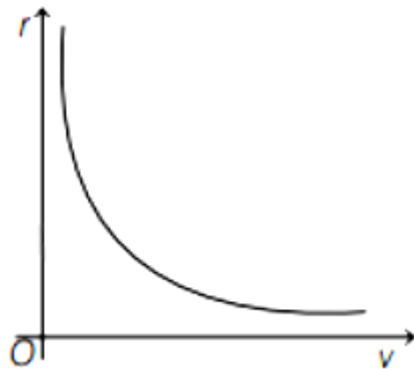
$$U(r) = -\frac{C}{r}$$

We know that,

$$F = -\frac{dU}{dr}$$

$$\Rightarrow F = -\frac{d}{dr}\left(-\frac{C}{r}\right) \Rightarrow |F| = -\frac{C}{r^2}$$

$$\Rightarrow \frac{mv^2}{r} = -\frac{C}{r^2} \Rightarrow v^2 \propto \frac{1}{r}$$

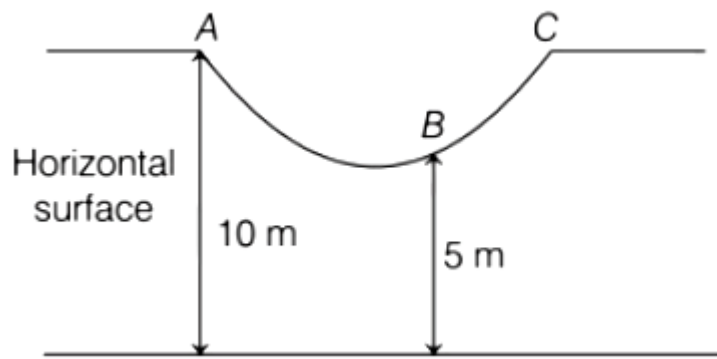


The graph between velocity and radius is hyperbolic.

Question112

As shown in the figure, a particle of mass 10kg is placed at a point A. When the particle is slightly displaced to its right, it starts moving and reaches the point B. The speed of the particle at B is $x\text{m/s}$. (Take, $g = 10\text{m/s}^2$)

The value of x to the nearest integer is



[18 Mar 2021 Shift 1]

Answer: 10

Solution:

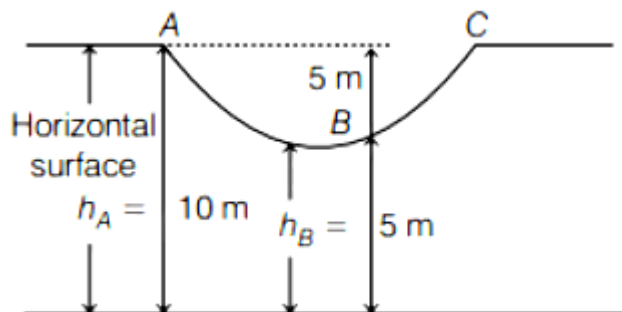
Given,

The mass of the particle, $m = 10\text{kg}$

The speed of the particle at point A, $v_A = 0\text{m/s}$

The elevation of the point A from the point B, $h_A = 5 + h_B$

Let's consider the speed of the particle at point B = v_B



Using the law of conservation of energy.

Energy at point A = Energy at point B

$$\frac{1}{2}mv_A^2 + mgh_A = \frac{1}{2}mv_B^2 + mgh_B$$

Substituting the values in the above equations, we get

$$\frac{1}{2}(10)(0)^2 + 10 \times 10 \times (5 + h_B) = \frac{1}{2}(10)v_B^2 + 10 \times 10 \times h_B$$

$$v_B = 10\text{m/s} = x\text{m/s}$$

$$\therefore x = 10$$

Question113

A constant power delivering machine has towed a box, which was initially at rest, along a horizontal straight line. The distance moved by the box in time t is proportional to
[18 Mar 2021 Shift 1]

Options:

A. $t^{\frac{2}{3}}$

B. $t^{\frac{3}{2}}$

C. t

D. $t^{\frac{1}{2}}$

Answer: B

Solution:

We know that,

Power = force $\times$ velocity [$\because$ Given, power = constant]

$\therefore$ Force $\times$ velocity = constant

$$\Rightarrow ma \times v = \text{constant}$$

$$\Rightarrow a \times v = \text{constant}$$

$$\Rightarrow \left(v \frac{dv}{dx} \right) \left(\frac{dx}{dt} \right) = C \Rightarrow \int v dv = \int C dt$$

$$\Rightarrow \frac{v^2}{2} = Ct \Rightarrow v = \sqrt{2Ct}$$

$$\Rightarrow \frac{dx}{dt} = \sqrt{2Ct} \quad \left[\because v = \frac{dx}{dt} \right]$$

$$\Rightarrow \int dx = \int \sqrt{2Ct} dt \Rightarrow x = \sqrt{2C} \left(\frac{t^{3/2}}{3/2} \right)$$

$$\Rightarrow x \propto t^{3/2}$$

Question114

A ball of mass 4kg, moving with a velocity of 10ms^{-1} , collides with a spring of length 8m and force constant 100N m^{-1} . The length of the compressed spring is $x\text{m}$. The value of x to the

nearest integer, is
[18 Mar 2021 Shift 2]

Answer: 6

Solution:

Given, mass of the ball, $m = 4\text{kg}$

Velocity of the ball, $v = 10\text{m/s}$

Force constant, $k = 100\text{N/m}$

The length of the spring, $x = 8\text{m}$

Let x is the compressed length of the spring.

Using the work-energy theorem,

"It states that kinetic energy of the ball is converted into the stored energy of spring".

$$\frac{mv^2}{2} = \frac{kx^2}{2}$$

$$\Rightarrow \frac{4(10)^2}{2} = \frac{(100)x^2}{2}$$

$$\Rightarrow x = 2\text{m}$$

The final length (compressed) of the spring

$$= 8 - 2 = 6\text{m}$$

Hence, the value of x to the nearest integer is 6 .

Question115

An object of mass m_1 collides with another object of mass m_2 , which is at rest. After the collision, the objects move with equal speeds in opposite direction. The ratio of the masses $m_2 : m_1$ is
[18 Mar 2021 Shift 2]

Options:

A. 3 : 1

B. 2 : 1

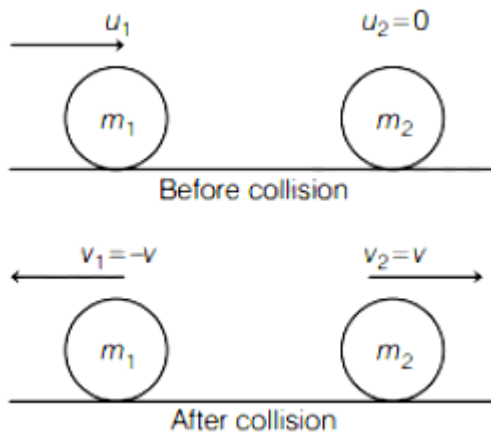
C. 1 : 2

D. 1 : 1

Answer: A

Solution:

The mass m_1 is moving with speed u_1 initially and mass m_2 is at rest. After the collision, the mass m_1 and m_2 move with speed v in opposite directions.



Using the law of conservation of linear momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\Rightarrow m_1 u_1 + m_2(0) = m_1(-v) + m_2 v$$

$$\Rightarrow m_1 u_1 = (-m_1 + m_2)v \dots (i)$$

Since, the collision is elastic because they move with same speed after the collision. Hence, coefficient of restitution, $e = 1$

$$\therefore e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow 1 = \frac{v - (-v)}{u_1 - 0}$$

$$u_1 = 2v \dots (ii)$$

Putting the above value in Eq. (i), we get

$$m_1(2v) = (-m_1 + m_2)v$$

$$\Rightarrow 3m_1 = m_2 \Rightarrow \frac{m_2}{m_1} = \frac{3}{1}$$

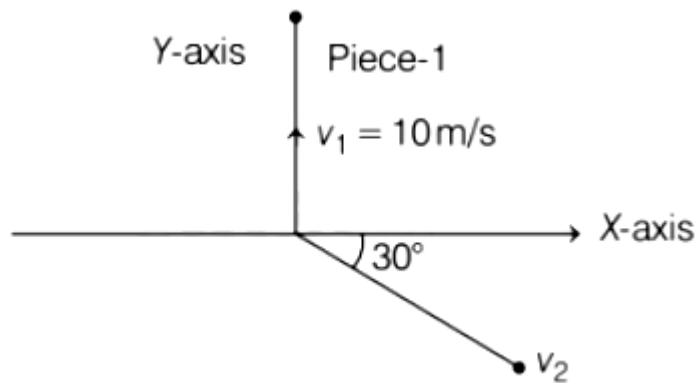
$$\Rightarrow m_2 : m_1 = 3 : 1$$

Question116

A ball of mass 10kg moving with a velocity $10\sqrt{3}\text{m/s}$ along the X-axis, hits another ball of mass 20kg which is at rest. After the collision, first ball comes to rest while the second ball disintegrates into two equal pieces. One piece starts moving along Y-axis with a speed of 10m/s.

The second piece starts moving at an angle of 30° with respect to the X-axis. The velocity of the ball moving at 30° with X-axis is $x\text{m/s}$. The configuration of pieces after collision is shown in the

figure below. The value of x to the nearest integer is



[18 Mar 2021 Shift 1]

Answer: None

Solution:

Given,

The mass of the first ball, $m_1 = 10\text{kg}$ The mass of the second ball, $m_2 = 20\text{kg}$

The initial velocity of the first ball, $u_1 = 10\sqrt{3}\text{m/s}$

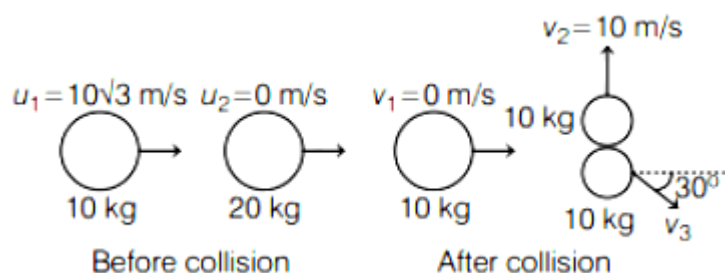
The initial velocity of the second ball, $u_2 = 0\text{m/s}$

The final velocity of the first ball, $v_1 = 0\text{m/s}$

The final velocity of the first piece of the second ball, $v_2 = 10\text{m/s}$

Let's consider the final velocity of the second piece of the second ball $= v_3$

As shown in the figure,



The net external force on the system is to be zero. Hence, we can use the law of conservation of linear momentum in both directions. In x-direction the linear momentum conserved,

In x-direction the linear momentum conserved,

$$m_1 u_{1x} + m_2 u_{2x} = m_1 v_{1x} + m_2 v_{2x} + m_3 v_{3x}$$

$$\Rightarrow 10(10\sqrt{3}) + 0 = 10(0) + 0 + 10v_3 \cos 30^\circ$$

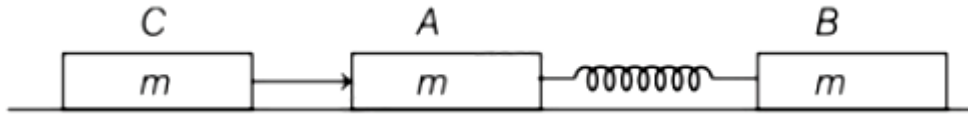
$$\Rightarrow v_3 = 20\text{m/s}$$

Hence, the velocity of the ball moving at 30° with respect to the X-axis is 20m/s .

So, the value of x to the nearest integer is 20 .

Question117

Two identical blocks A and B each of mass m resting on the smooth horizontal floor are connected by a light spring of natural length L and spring constant k . A third block C of mass m moving with a speed v along the line joining A and B collides with A. The maximum compression in the spring is



[17 Mar 2021 Shift 2]

Options:

A. $v \sqrt{\frac{m}{2k}}$

B. $\sqrt{\frac{mv}{2k}}$

C. $\sqrt{\frac{mv}{k}}$

D. $\sqrt{\frac{m}{2k}}$

Answer: A

Solution:

Let v is the speed of the third block C.

The velocity of centre of mass of A and B is

$$v_{CM} = \frac{v}{2}$$

The spring is compressed maximum by x distance.

Using the law of conservation of energy.

$$\frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{v}{2}\right)^2 + \frac{1}{2}m\left(\frac{v}{2}\right)^2 + \frac{1}{2}kx^2$$

$$\Rightarrow \frac{1}{4}mv^2 = \frac{1}{2}kx^2 \Rightarrow x = \sqrt{\frac{mv^2}{2k}}$$

$$\Rightarrow x = v \sqrt{\frac{m}{2k}}$$

Question118

A rubber ball is released from a height of 5m above the floor. It bounces back repeatedly, always rising to $\frac{81}{100}$ of the height through which it falls. Find the average speed of the ball.

(Take, $g = 10\text{ms}^{-2}$)

[17 Mar 2021 Shift 2]

Options:

A. 3.0ms^{-1}

B. 3.5ms^{-1}

C. 2.0ms^{-1}

D. 2.5ms^{-1}

Answer: D

Solution:

Velocity of rubber ball when it strikes to the ground, $v_0 = \sqrt{2gh_0}$

Using the formula of coefficient of restitution,

$$e = \frac{\text{velocity after collision}}{\text{velocity before collision}}$$

$$\Rightarrow e = \frac{v_1 - 0}{v_0 - 0} \Rightarrow e = \frac{v_1}{\sqrt{2gh_0}}$$

$$\Rightarrow v_1 = e\sqrt{2gh_0} \dots (i)$$

As, initial height of the ball = h_0 .

$$\therefore \text{The first height of the rebound, } h_1 = \frac{v_1^2}{2g}$$

$$\Rightarrow h_1 = e^2 h_0 \quad [\text{Using Eq. (i)}]$$

The nth height of the ball to the rebound,

$$h_n = \frac{v_n^2}{2g} \Rightarrow h_n = e^{2n} h_0$$

The velocity of the ball after nth rebound, $v_n = e^n v_0$

Now, the total distance travelled by the ball after nth rebound is

$$H = h_0 + 2h_1 + 2h_2 + 2h_3 \dots$$

$$H = h_0 + 2e^2 h_0 + 2e^4 h_0 + 2e^6 h_0 \dots$$

$$H = h_0 [1 + 2e^2 (1 + e^2 + e^4 + e^6 \dots)]$$

Using the formula,

$$1 + e^2 + e^4 + \dots = \frac{1}{1 - e^2}$$

$$\Rightarrow H = h_0 \left[1 + 2e^2 \left(\frac{1}{1 - e^2} \right) \right]$$

$$\Rightarrow H = h_0 \left(\frac{1 + e^2}{1 - e^2} \right)$$

$$\Rightarrow H = 5 \left(\frac{1 + (0.81)}{1 - (0.81)} \right) \quad \left(\because e^2 = \frac{h_1}{h_0} = \frac{81}{100} \right)$$

$$H = 47.6 \text{ m}$$

Now, the total time taken by the ball to come to rest

$$T = t_0 + 2t_1 + 2t_2 + \dots$$

$$T = \sqrt{\frac{2h_0}{g}} + 2\sqrt{\frac{2h_1}{g}} + 2\sqrt{\frac{2h_2}{g}} + \dots$$

$$T = \sqrt{\frac{2h_0}{g}} [1 + 2e + 2e^2 + \dots]$$

$$T = \sqrt{\frac{2h_0}{g}} [1 + 2e(1 + e + e^2 + \dots)]$$

$$T = \sqrt{\frac{2h_0}{g}} \left(\frac{1 + e}{1 - e} \right) \Rightarrow T = \sqrt{\frac{2(5)}{10}} \left(\frac{1 + \sqrt{0.81}}{1 - \sqrt{0.81}} \right)$$

$$T = 19 \text{ s}$$

The average velocity,

$$v_{\text{avg}} = \frac{H}{T} = \frac{47.6}{19}$$

$$= 2.5 \text{ m/s}$$

Hence, the average velocity is 2.5m/s.

Question119

- A boy is rolling a 0.5kg ball on the frictionless floor with the speed of 20ms^{-1} . The ball gets deflected by an obstacle on the way. After deflection it moves with 5% of its initial kinetic energy. What is the speed of the ball now ?
[17 Mar 2021 Shift 1]

Options:

A. 19.0ms^{-1}

B. 4.47ms^{-1}

C. 14.41ms^{-1}

D. 1.00ms^{-1}

Answer: B

Solution:

Given, mass of rolling ball, $m = 0.5\text{kg}$

Speed of ball, $u = 20\text{ms}^{-1}$

Before deflection,

Initial kinetic energy, $\text{KE}_i = \frac{1}{2}mu^2$

$$= \frac{1}{2} \times 0.5 \times (20)^2$$

$$= \frac{1}{2} \times 0.5 \times 400 = \frac{1}{2} \times \frac{1}{2} \times 400 = 100\text{J}$$

It is given in the question that after deflection the ball moves with 5% of its initial kinetic energy

$$\text{KE}_f = 5\% \text{ of } \text{KE}_i \Rightarrow \text{KE}_f = \frac{5}{100} \times 100 = 5\text{J}$$

If the final speed of the ball is $v\text{ms}^{-1}$, then

$$\text{KE}_f = \frac{1}{2}mv^2$$

$$\Rightarrow 5 = \frac{1}{2} \times 0.5 \times v^2 \Rightarrow 10 = 0.5 \times v^2$$

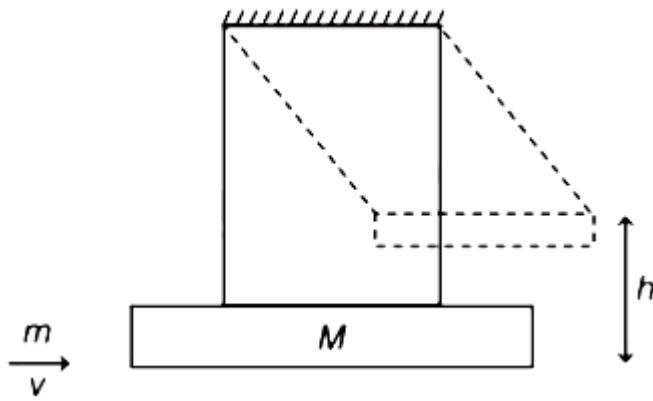
$$\Rightarrow v^2 = \frac{10}{0.5} = \frac{100}{5} \Rightarrow v^2 = 20$$

$$\Rightarrow v = \sqrt{20} = 4.47\text{ms}^{-1}$$

$$\Rightarrow v = 4.47\text{ms}^{-1}$$

Question120

A large block of wood of mass $M = 5.99\text{kg}$ is hanging from two long massless cords. A bullet of mass $m = 10\text{g}$ is fired into the block and gets embedded in it. The system (block + bullet) then swing upwards, their centre of mass rising a vertical distance $h = 9.8\text{cm}$ before the (block + bullet) pendulum comes momentarily to rest at the end of its arc. The speed of the bullet just before collision is (Take $g = 9.8\text{ms}^{-2}$)



[16 Mar 2021 Shift 2]

Options:

- A. 841.4m/s
- B. 811.4m/s
- C. 831.4m/s
- D. 821.4m/s

Answer: C

Solution:

Given

Mass of large block of wood, $M = 5.99\text{kg}$

Mass of bullet, $m = 10\text{g}$

Height at which their centre of mass rise, $h = 9.8\text{cm}$

From the law of conservation of energy,

Energy of the system when bullet gets embedded = Energy of the system till it momentarily comes to rest.

$$\Rightarrow \frac{1}{2}(M + m)v_1^2 = (M + m)gh$$

where, v_1 = velocity of bullet + block system

$$\Rightarrow v_1 = \sqrt{2gh} \dots (i)$$

According to law of conservation of momentum,

Momentum before collision = Momentum after collision.

$$\Rightarrow mv = (M + m)v_1$$

[where, v = velocity of bullet before collision]

$$\Rightarrow mv = (M + m)\sqrt{2gh} \text{ [using Eq. (i)]}$$

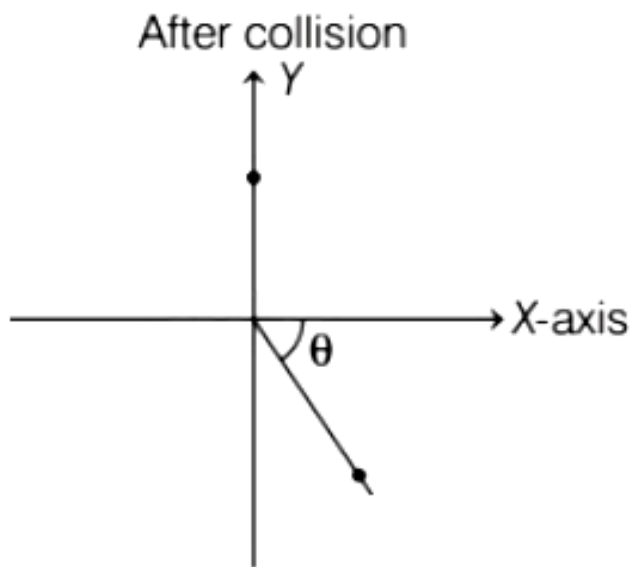
$$\Rightarrow v = \left(\frac{M + m}{m} \right) \sqrt{2gh}$$

$$\Rightarrow v = \frac{(5.99 + 0.01)}{10 \times 10^{-3}} \times \sqrt{2 \times 9.8 \times 9.8 \times 10^{-2}}$$

$$\Rightarrow v = 831.55\text{ms}^{-1}$$

Question121

A ball of mass 10kg moving with a velocity $10\sqrt{3}\text{ms}^{-1}$ along X - axis, hits another ball of mass 20kg , which is at rest. After collision, the first ball comes to rest and the second one disintegrates into two equal pieces. One of the pieces starts moving along Y -axis at a speed of 10m/s . The second piece starts moving at a speed of 20m/s at an angle θ (degree) with respect to the X -axis. The configuration of pieces after collision is shown in the figure. The value of θ to the nearest integer is

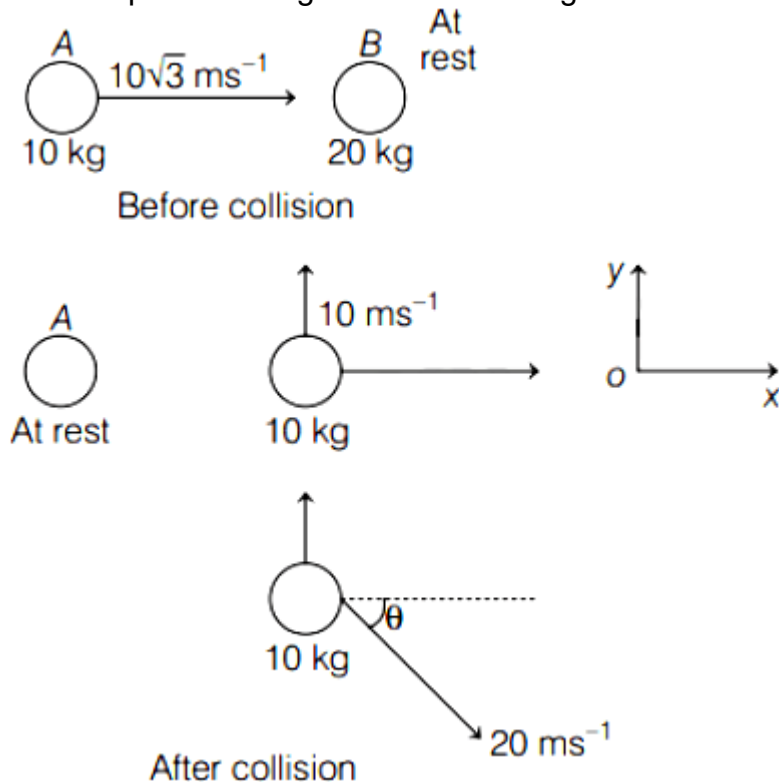


[16 Mar 2021 Shift 1]

Answer: 30

Solution:

We can represent the given situation in figure as



It means linear momentum is conserved along X -axis. According to the law of conservation of linear momentum,

$$p_i = p_f$$

$$m_A u_A + m_B u_B = m_A v_A + (m_B v_B \cos \theta)$$

$$= m_A v_A + m_{B_1} v_{B_1} \cos 90^\circ + m_{B_2} v_{B_2} \cos \theta$$

$$\Rightarrow 10 \times 10\sqrt{3} + 20 \times 0 = 10 \times 0 + 10 \times 10 \times 0 + 10 \times 20 \cos \theta$$

$$\Rightarrow 10 \times 10\sqrt{3} = 200 \cos \theta$$

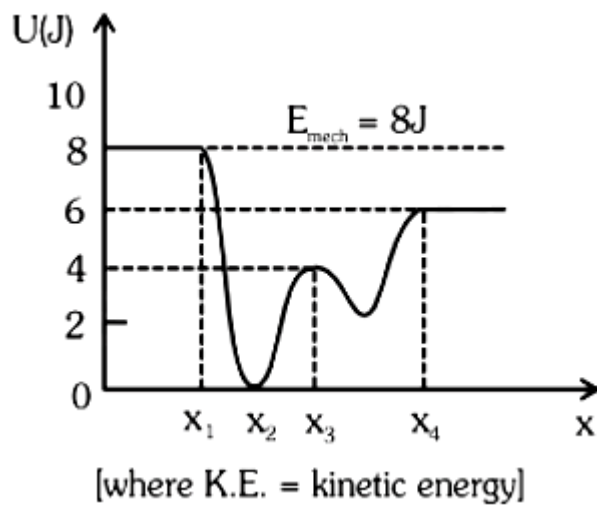
where, $\cos \theta$ being the horizontal component i.e., along X -axis

$$\Rightarrow \cos \theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = 30^\circ [\because \cos 30^\circ = \frac{\sqrt{3}}{2}]$$

Question122

Given below is the plot of a potential energy function $U(x)$ for a system, in which a particle is in one dimensional motion, while a conservative force $F(x)$ acts on it. Suppose that $E_{\text{mech}} = 8\text{J}$, the incorrect statement for this system is:



[27 Jul 2021 Shift 2]

Options:

- A. at $x > x_4$, K . E . is constant throughout the region.
- B. at $x < x_1$, K . E . is smallest and the particle is moving at the slowest speed.
- C. at $x = x_2$, K . E . is greatest and the particle is moving at the fastest speed.
- D. at $x = x_3$, K . E . = 4J

Answer: B

Solution:

$$E_{\text{mech}} = 8\text{J}$$

$$\text{(A) at } x > x_4, \quad U = \text{constant} = 6\text{J}$$

$$K = E_{\text{mech.}} - U = 2\text{J} = \text{constant}$$

$$\text{(B) at } x < x_1, \quad U = \text{constant} = 8\text{J}$$

$$K = E_{\text{mech.}} - U = 8 - 8 = 0\text{J}$$

Particle is at rest.

$$\text{(C) At } x = x_2, \quad U = 0 \Rightarrow E_{\text{mech.}} = K = 8\text{J}$$

KE is greatest, and particle is moving at fastest speed.

$$\text{(D) At } x = x_3, \quad U = 4\text{J}$$

$$U + K = 8\text{J}$$

$$K = 4\text{J}$$

Question123

A force of $F = (5y + 20)\hat{j}$ N acts on a particle. The workdone by this force when the particle is moved from $y = 0$ m to $y = 10$ m is _____ J.

[25 Jul 2021 Shift 2]

Answer: 450

Solution:

$$\begin{aligned} F &= (5y + 20)\hat{j} \\ W &= \int F \, dy = \int_0^{10} (5y + 20) \, dy \\ &= \left(\frac{5y^2}{2} + 20y \right)_0^{10} \\ &= \frac{5}{2} \times 100 + 20 \times 10 \\ &= 250 + 200 = 450 \text{ J} \end{aligned}$$

Question 124

A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the workdone by the porter in lowering the suitcase.

(take $g = 9.8 \text{ ms}^{-2}$)

[22 Jul 2021 Shift 2]

Options:

A. -62720.0 J

B. -627.2 J

C. $+627.2 \text{ J}$

D. 784.0 J

Answer: B

Solution:

$$\begin{aligned}W_{\text{Porter}} + W_{\text{mg}} &= \Delta K.E. = 0 \\W_{\text{Porter}} &= -W_{\text{mg}} = -mgh \\&= -80 \times 9.8 \times .8 = -627.2\text{J}\end{aligned}$$

Question125

**If the Kinetic energy of a moving body becomes four times its initial Kinetic energy, then the percentage change in its momentum will be :
[20 Jul 2021 Shift 2]**

Options:

- A. 100%
- B. 200%
- C. 300%
- D. 400%

Answer: A

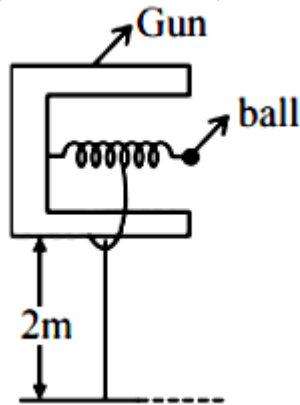
Solution:

$$\begin{aligned}K_2 &= 4K_1 \\ \frac{1}{2}mv_2^2 &= 4 \frac{1}{2}mv_1^2 \\ v_2 &= 2v_1 \\ P &= mv \\ P_2 &= mv_2 = 2mv_1 \\ P_1 &= mv_1 \\ \% \text{ change} &= \frac{\Delta P}{P_1} \times 100 = \frac{2mv_1 - mv_1}{mv_1} \times 100 = 100\%\end{aligned}$$

Question126

In a spring gun having spring constant 100 N/m a small ball 'B' of mass 100 g is put in its barrel (as shown in figure) by compressing the spring through 0.05 m . There should be a box placed at a distance ' d ' on the ground so that the ball falls in it. If the ball leaves the gun horizontally at a height of 2 m above the ground. The value of d is m.

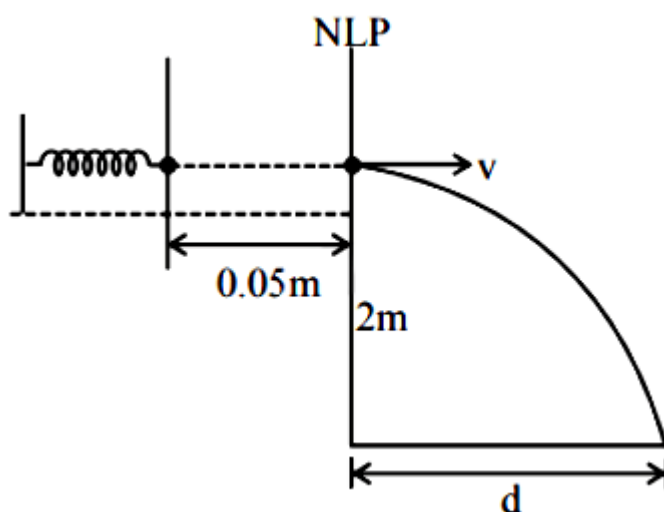
($g = 10\text{ m/s}^2$)



[20 Jul 2021 Shift 1]

Answer: 1

Solution:



$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$$

$$Kx^2 = mv^2$$

$$v = x \sqrt{\frac{k}{m}} = 0.05 \sqrt{\frac{100}{0.1}} = 0.05 \times 10\sqrt{10}$$

$$v = 0.5\sqrt{10}$$

$$\text{From } h = \frac{1}{2}gt^2$$

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 2}{10}} = \frac{2}{\sqrt{10}}$$

$$\therefore d = vt = 0.5\sqrt{10} \times \frac{2}{\sqrt{10}} = 1\text{m}$$

Question127

An automobile of mass 'm' accelerates starting from origin and initially at rest, while the engine supplies constant power P. The position is given as a function of time by:

[27 Jul 2021 Shift 2]

Options:

A. $\left(\frac{9P}{8m}\right)^{\frac{1}{2}}t^{\frac{3}{2}}$

B. $\left(\frac{8P}{9m}\right)^{\frac{1}{2}}t^{\frac{2}{3}}$

C. $\left(\frac{9m}{8P}\right)^{\frac{1}{2}}t^{\frac{3}{2}}$

D. $\left(\frac{8P}{9m}\right)^{\frac{1}{2}}t^{\frac{3}{2}}$

Answer: D

Solution:

$$P = \text{const.}$$

$$P = F v = \frac{mv^2 dv}{dx}$$

$$\int_0^x \frac{P}{m} dx = \int_0^v v^2 dv$$

$$\frac{Px}{m} = \frac{v^3}{3}$$

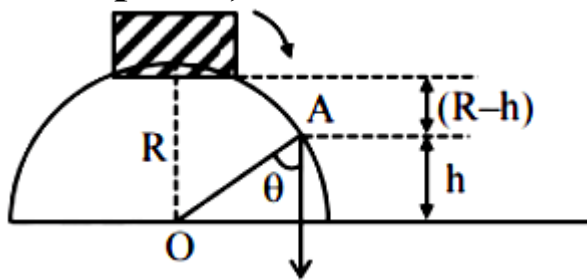
$$\left(\frac{3Px}{m}\right)^{1/3} = v = \frac{dx}{dt}$$

$$\left(\frac{3P}{m}\right)^{1/3} \int_0^t dt = \int_0^x x^{-1/3} dx$$

$$\Rightarrow x = \left(\frac{8P}{9m}\right)^{1/2} t^{3/2}$$

Question128

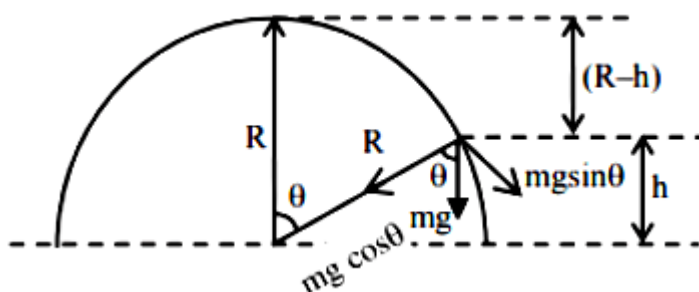
A small block slides down from the top of hemisphere of radius $R = 3$ m as shown in the figure. The height 'h' at which the block will lose contact with the surface of the sphere is ____m. (Assume there is no friction between the block and the hemisphere)



[27 Jul 2021 Shift 2]

Answer: 2

Solution:



$$mg \cos \theta = \frac{mv^2}{R} \dots\dots(1)$$

$$\cos \theta = \frac{h}{R}$$

Energy conservation

$$mg\{R - h\} = \frac{1}{2}mv^2 \dots\dots(2)$$

from (1) & (2)

$$\Rightarrow mg \left\{ \frac{h}{R} \right\} = \frac{2mg\{R-h\}}{R}$$

$$h = \frac{2R}{3} = 2m$$

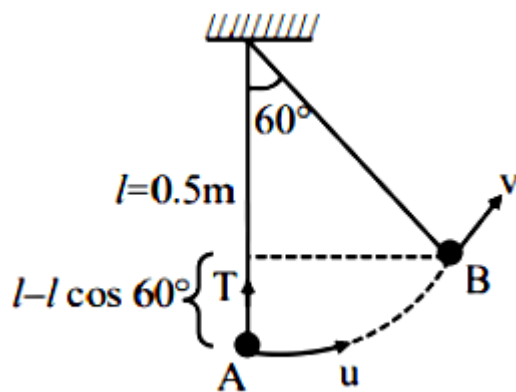
Question129

A pendulum bob has a speed of 3 m/s at its lowest position. The pendulum is 50 cm long. The speed of bob, when the length makes an angle of 60° to the vertical will be ($g = 10 \text{ m/s}^2$) _____ m/s.

[25 Jul 2021 Shift 1]

Answer: 2

Solution:



Applying work energy theorem:

$$W_g + W_T = \Delta K$$

$$-mgl(1 - \cos 60^\circ) = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$v^2 = u^2 - 2gl(1 - \cos 60^\circ)$$

$$v^2 = 9 - 2 \times 10 \times 0.5 \left(\frac{1}{2} \right)$$

$$v^2 = 4$$

$$v = 2\text{m/s}$$

Question130

A body at rest is moved along a horizontal straight line by a machine delivering a constant power. The distance moved by the body in time 't' is proportional to:
[20 Jul 2021 Shift 2]

Options:

A. $t^{\frac{3}{2}}$

B. $t^{\frac{1}{2}}$

C. $t^{\frac{1}{4}}$

D. $t^{\frac{3}{4}}$

Answer: A

Solution:

$$P = \text{constant}$$

$$\frac{1}{2}mv^2 = Pt$$

$$\Rightarrow v \propto \sqrt{t}$$

$$\frac{dx}{dt} = C\sqrt{t} \quad C = \text{constant}$$

by integration.

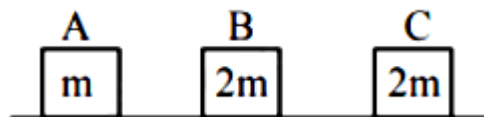
$$x = C \frac{t^{\frac{1}{2}+1}}{\frac{1}{2}+1}$$

$$x \propto t^{3/2}$$

Question131

Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. The masses of A, B and C are m, 2 m and 2 m respectively. A moves towards B with a speed of 9 m/s and makes an elastic collision with it. Thereafter B makes a completely inelastic collision with C. All motions occur along

same straight line. The final speed of C is :



[27 Jul 2021 Shift 1]

Options:

A. 6 m/s

B. 9 m/s

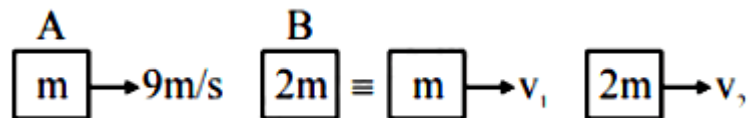
C. 4 m/s

D. 3 m/s

Answer: D

Solution:

Collision between A and B

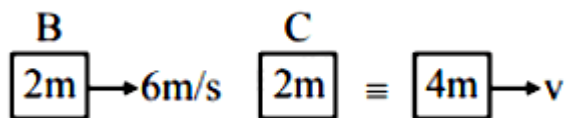


$$m \times 9 = mv_1 + 2mv_2 \text{ (from momentum conservation)}$$

$$e = 1 = \frac{v_2 - v_1}{9}$$

$$\Rightarrow v_2 = 6\text{m/sec}, v_1 = -3\text{m/sec}$$

collision between B and C



$$2m \times 6 = 4mv \text{ (from momentum conservation)}$$

$$v = 3 \text{ m/s}$$

Question132

A body of mass M moving at speed v_0 collides elastically with a mass m at rest. After the collision, the two masses move at angles θ_1 and θ_2 with respect to the initial direction of motion of the body of mass M . The largest possible value of the ratio $\frac{M}{m}$, for

which the angles θ_1 and θ_2 will be equal, is

[31 Aug 2021 Shift 1]

Options:

A. 4

B. 1

C. 3

D. 2

Answer: C

Solution:

Given, mass of body 1 = M

Mass of body 2 = m

Initial speed of body 1, $u_1 = v_0$

Initial speed of body 2, $u_2 = 0$

Final speed of body 1 and 2 = v_1 and v_2

Angle made by body 1 and 2 after collision with respect to initial direction = θ_1, θ_2

By using law of conservation of momentum, Along X-axis

$$Mv_0 + m \cdot 0 = Mv_1 \cos \theta_1 + m v_2 \cos \theta_2$$

If $\theta_1 = \theta_2 = \theta$

$$\therefore Mv_0 = Mv_1 \cos \theta + mv_2 \cos \theta \dots (i)$$

Along Y-axis,

$$Mv_1 \sin \theta_1 = mv_2 \sin \theta_2 \text{ [since, } \theta_1 = \theta_2 = \theta]$$

$$v_2 = \frac{Mv_1}{m} \dots (ii)$$

Substituting the value of v_2 in Eq. (i), we get

$$\begin{aligned} Mv_0 &= Mv_1 \cos \theta + m \left(\frac{Mv_1}{m} \right) \cos \theta \\ &= 2Mv_1 \cos \theta \end{aligned}$$

$$v_1 = \frac{v_0}{2 \cos \theta} \dots (iii)$$

By using law of conservation of energy along X-axis

$$\frac{1}{2}Mv_0^2 = \frac{1}{2}Mv_1^2 + \frac{1}{2}mv_2^2$$

$$\Rightarrow Mv_0^2 = Mv_1^2 + m \left(\frac{M}{m}v_1 \right)^2 \text{ [from Eq. (ii)]}$$

$$\Rightarrow Mv_0^2 = Mv_1^2 + \frac{M^2v_1^2}{m} = Mv_1^2 m(m+M)$$

$$\Rightarrow v_0^2 = \left(\frac{v_0}{2 \cos \theta} \right)^2 \left(\frac{m+M}{m} \right) \text{ [From Eq. (iii)]}$$

$$\Rightarrow v_0^2 = \frac{v_0^2}{4\cos^2\theta} \left(1 + \frac{M}{m} \right)$$

$$\Rightarrow 4\cos^2\theta = 1 + \frac{M}{m}$$

$$\Rightarrow \frac{M}{m} = 4\cos^2\theta - 1 \quad \left[\text{For largest possible value of } \frac{M}{m}, \theta = 0 \right]$$

$$= 4\cos^2 0^\circ - 1 = 4 - 1 = 3$$

Question133

A block moving horizontally on a smooth surface with a speed of 40m/s splits into two parts with masses in the ratio of 1 : 2. If the smaller part moves at 60m/s in the same direction, then the fractional change in kinetic energy is
[31 Aug 2021 Shift 2]

Options:

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{1}{8}$

D. $\frac{1}{4}$

Answer: C

Solution:

Let a block of mass M splits into two masses m_1 and m_2 in ratio 1: 2.

Then, mass of smaller part, $m_1 = \frac{M}{1+2} = \frac{M}{3}$

Mass of bigger part, $m_2 = 2\frac{M}{3}$

Given, speed of mass M , $v = 40$ m/s

Speed of mass m_1 , $v_1 = 60$ m/s

Let speed of mass m_2 , $v_2 = v$

Using conservation of linear momentum,

$$p_i = p_f$$

$$Mv = m_1v_1 + m_2v_2$$

$$M \times 40 = \frac{M}{3} \times 60 + \frac{2M}{3} \times v$$

$$\Rightarrow = 30 \text{ m/s}$$

∴ The fractional change in kinetic energy,

$$\begin{aligned} \frac{\Delta K}{K} &= \frac{K_f - K_i}{K_i} = \frac{K_f}{K_i} - 1 \\ &= \frac{\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2}{\frac{1}{2}Mv^2} - 1 \\ &= \frac{\frac{1}{2} \left[\frac{M}{3} \times (60)^2 + \frac{2M}{3} \times (30)^2 \right]}{\frac{1}{2}M \times (40)^2} - 1 = \frac{9}{8} - 1 = \frac{1}{8} \end{aligned}$$

Question 134

A block moving horizontally on a smooth surface with a speed of 40ms^{-1} splits into two equal parts. If one of the parts moves at 60ms^{-1} in the same direction, then the fractional change in the kinetic energy will be $x:4$, where x is [31 Aug 2021 Shift 1]

Answer: 5

Solution:

Given, initial speed of block, $u = 40\text{ms}^{-1}$

Let total mass of block = m

∴ Broken masses, $m_1 = m_2 = \frac{m}{2}$

Final speed of m_1 , $v_1 = 60\text{ms}^{-1}$

As we know that, kinetic energy, $KE = \frac{1}{2}mv^2$

∴ Initial kinetic energy $KE_i = \frac{1}{2}mu^2$

$$= \frac{1}{2}m(40)^2 = 800m \dots(i)$$

By using law of conservation of momentum,

$$mu = m_1v_1 + m_2v_2$$

$$\therefore v_2 = \frac{m \times 40 - \frac{m}{2} \times 60}{\frac{m}{2}}$$

$$= \frac{10m}{\frac{m}{2}} = 20 \text{ms}^{-1}$$

So, final kinetic energy $KE_f = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2$

$$= \frac{1}{2} \frac{m}{2} (v_1^2 + v_2^2) = \frac{m}{4} (60^2 + 20^2)$$

$$= \frac{m}{4} (3600 + 400)$$

$$= 1000m \dots (ii)$$

$\therefore$ Divide Eq (ii) by Eq. (i), we get

$$\frac{KE_f}{KE_i} = \frac{1000m}{800} = \frac{5}{4}m$$

So, $x = 5$

Question135

Two persons A and B perform same amount of work in moving a body through a certain distance d with application of forces acting at angle 45° and 60° with the direction of displacement respectively. The ratio of force applied by person A to the force applied by person B is $\frac{1}{\sqrt{x}}$. The value of x is

[27 Aug 2021 Shift 1]

Answer: 2

Solution:

Given, work done by both person is same,

$$W_A = W_B$$

Direction of first force with displacement, $\theta_1 = 45^\circ$

Direction of first force with displacement, $\theta_2 = 60^\circ$

Ratio of force applied by person A to force applied by person B is

$$F_A : F_B = 1 : \sqrt{x}.$$

Work done by both person is same,

$$W_A = W_B =$$

$$\Rightarrow F_A d \cos \theta_1 = F_B d \cos \theta_2$$

($\because$ distance is same for both = d)

$$\frac{F_A}{F_B} = \frac{\cos \theta_1}{\cos \theta_2} = \frac{\cos 60^\circ}{\cos 45^\circ}$$

$$\Rightarrow \frac{F_A}{F_B} = \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{F_A}{F_B} = \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = 2$$

Thus, the value of x is 2.

Question136

A uniform chain of length 3m and mass 3 kg overhangs a smooth table with 2m laying on the table. If k is the kinetic energy of the chain in joule as it completely slips off the table, then the value of k is

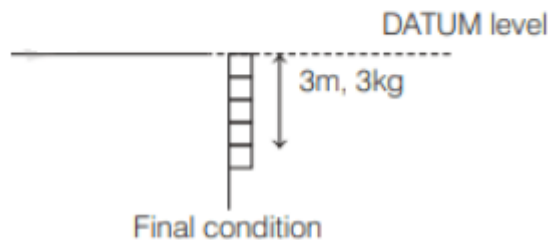
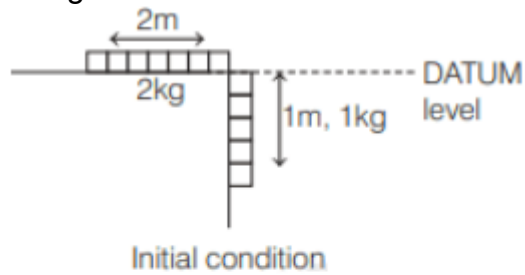
(Take, $g = 10\text{m/s}^2$)

[26 Aug 2021 Shift 1]

Answer: 40

Solution:

The given situation is shown below as



Applying law of conservation of energy, we get

$$K_{\text{initial}} + U_{\text{initial}} = K_{\text{final}} + U_{\text{final}}$$

$$K_{\text{initial}} = 0$$

$$\Rightarrow U_{\text{initial}} = -1 \times 10 \times \frac{1}{2} \text{ (1m lie below DATUM level, so negative)}$$

$$K_{\text{final}} = ?$$

$$U_{\text{final}} = -3 \times 10 \times \frac{3}{2} \text{ (3m lie below DATUM level, so negative)}$$

$$0 - 1 \times 10 \times \frac{1}{2} = K_{\text{final}} - 3 \times 10 \times \frac{3}{2}$$

$$\Rightarrow -5 = K_{\text{final}} - 45$$

$$\Rightarrow K_{\text{final}} = 45 - 5$$

$$= 40 \text{ J}$$

Question137

A body of mass m dropped from a height h reaches the ground with a speed of $0.8\sqrt{gh}$. The value of workdone by the air-friction is

[1 Sep 2021 Shift 2]

Options:

A. $-0.68 mgh$

B. mgh

C. $1.64 mgh$

D. $0.64 mgh$

Answer: A

Solution:

Given, the mass of the body = m

The height from which the body dropped = h

The speed of the body when reached the ground, $v_f = 0.8\sqrt{gh}$

Initial velocity of the body, $v = 0 \text{ m/s}$

Using the work-energy theorem,

Work done by gravity + Work done by air-friction = Final kinetic energy - Initial kinetic energy

$$W_{mg} + W_{\text{air-friction}} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

Here, work done by gravity = mgh

$$\Rightarrow mgh + W_{\text{air-friction}} = \frac{1}{2}m(0.8\sqrt{gh})^2 - \frac{1}{2}m(0)^2$$

$$\Rightarrow W_{\text{air-friction}} = \frac{0.64mgh}{2} - mgh$$

$$= 0.32 mgh - mgh = -0.68 mgh$$

The value of the work done by the air friction is -0.68 mgh.

Question138

An engine is attached to a wagon through a shock absorber of length 1.5m. The system with a total mass of 40000 kg is moving with a speed of 72kmh^{-1} , when the brakes are applied to bring it to rest. In the process of the system being brought to rest, the spring of the shock absorber gets compressed by 1.0m. If 90% of energy of the wagon is lost due to friction, the spring constant is

$$\times 10^5 \text{N/m.}$$

[1 Sep 2021 Shift 2]

Answer: 16

Solution:

Given, the length of the shock absorber, $l = 1.5 \text{ m}$

The total mass of the system, $M = 40000 \text{ kg}$

The speed of the wagon, $v = 72 \text{ km/h}$

When brakes are applied, the final velocity, $v_f = 0$

The compressed spring of the shock absorber, $x = 1 \text{ m}$ Applying the work-energy theorem,

Work done by the system = Change in kinetic energy

$$W = \Delta KE$$

$$W_{\text{friction}} + W_{\text{spring}} = \frac{1}{2}mv_f^2 + \frac{1}{2}mv_i^2$$

$$-\frac{90}{100} \left(\frac{1}{2}mv^2 \right) + W_{\text{spring}} = 0 - \frac{1}{2}mv_i^2 \quad (\because 90\% \text{ energy lost due to friction})$$

$$W_{\text{spring}} = -\frac{10}{100} \times \frac{1}{2}mv^2$$

$$-\frac{1}{2}kx^2 = \frac{1}{20}mv^2$$

$$k = \frac{mv^2}{10 \times x^2}$$

Substituting the values in the above equation, we get

$$k = \frac{40000 \times \left(72 \times \frac{5}{18} \right)^2}{10(1)^2}$$

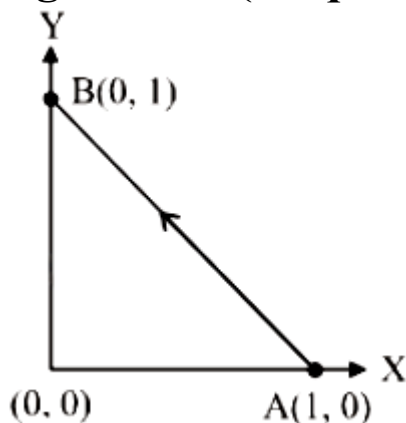
$$= 16 \times 10^5 \text{ N / m}$$

Comparing the spring constant, $k = x \times 10^5$

The value of the $x = 16$.

Question139

Consider a force $\vec{F} = -x\hat{i} + y\hat{j}$. The work done by this force in moving a particle from point A(1, 0) to B(0, 1) along the line segment is: (all quantities are in SI units)



[9 Jan. 2020 I]

Options:

A. 2J

B. $\frac{1}{2}\text{J}$

C. 1J

D. $\frac{3}{2}\text{J}$

Answer: C

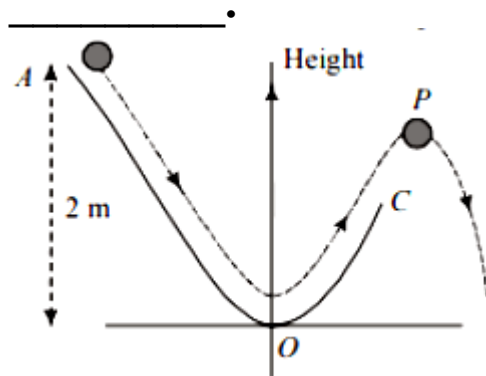
Solution:

$$\begin{aligned}\text{Work done, } W &= \int \vec{F} \cdot d\vec{s} \\ &= (-x\hat{i} + y\hat{j}) \cdot (dx\hat{i} + dy\hat{j}) \\ \Rightarrow W &= -\int_1^0 x dx + \int_0^1 y dy \\ &= \left(0 + \frac{1}{2}\right) + \frac{1}{2} = 1\text{J}\end{aligned}$$

Question140

A particle ($m = 1\text{kg}$) slides down a frictionless track (AOC) starting from rest at a point A (height 2m). After reaching C, the particle continues to move freely in air as a projectile. When it reaches its highest point P (height 1m), the kinetic energy of the particle (in J) is:

(Figure drawn is schematic and not to scale; take $g = 10\text{ ms}^{-2}$)



[NA 7 Jan. 2020 I]

Answer: 10

Solution:

Kinetic energy = change in potential energy of the particle,

$$K E = mg\Delta h$$

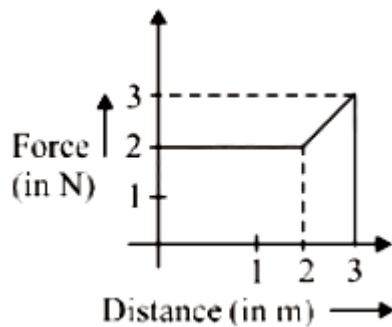
Given, $m = 1\text{kg}$

$$\Delta h = h_2 - h_1 = 2 - 1 = 1\text{m}$$

$$\therefore K E = 1 \times 10 \times 1 = 10\text{J}$$

Question141

A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3m is :



[7 Jan. 2020 II]

Options:

A. 4J

B. 2.5J

C. 6.5J

D. 5J

Answer: C

Solution:

We know area under F-x graph gives the work done by the body

$$\therefore W = \frac{1}{2} \times (3 + 2) \times (3 - 2) + 2 \times 2 = 2.5 + 4 = 6.5\text{J}$$

Using work energy theorem,
 $\Delta K.E = \text{work done}$
 $\therefore \Delta K.E = 6.5J$

Question142

A 60 HP electric motor lifts an elevator having a maximum total load capacity of 2000 kg. If the frictional force on the elevator is 4000 N, the speed of the elevator at full load is close to: (1 HP = 746 W, $g = 10 \text{ ms}^{-2}$)
[7 Jan. 2020 I]

Options:

- A. 1.7 ms^{-1}
- B. 1.9 ms^{-1}
- C. 1.5 ms^{-1}
- D. 2.0 m^{-1}

Answer: B

Solution:

Total force required to lift maximum load capacity against frictional force = 400 N

$F_{\text{total}} = M g + \text{friction}$

$$= 2000 \times 10 + 4000$$

$$= 20,000 + 4000 = 24000N$$

Using power, $P = F \times v$

$$60 \times 746 = 24000 \times v$$

$$\Rightarrow v = 1.86 \text{ m/s} \approx 1.9 \text{ m/s}$$

Hence speed of the elevator at full load is close to 1.9 ms^{-1}

Question143

Two particles of equal mass m have respective initial velocities $\hat{u}_i$ and $\hat{u} \left(\frac{\hat{i} + \hat{j}}{2} \right)$. They collide completely inelastically. The energy lost in the process is:
[9 Jan. 2020 I]

Options:

A. $\frac{1}{3}mu^2$

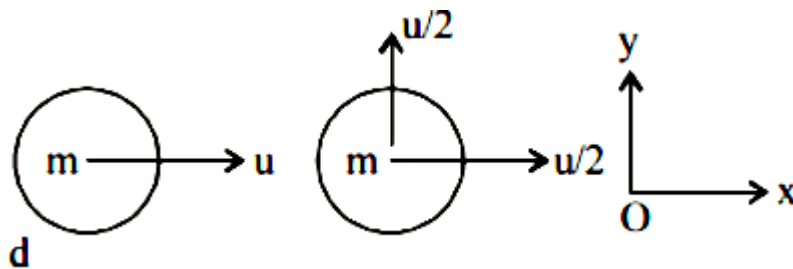
B. $\frac{1}{8}mu^2$

C. $\frac{3}{4}mu^2$

D. $\sqrt{\frac{2}{3}}mu^2$

Answer: B

Solution:



x-direction

$$mu + \frac{mu}{2} = 2mv'_x \Rightarrow v'_x = \frac{3u}{4}$$

y-direction $0 + \frac{mu}{2} = 2mv'_y \Rightarrow v'_y = \frac{u}{4}$

$$K.E._i = \frac{1}{2}mu^2 + \frac{1}{2}m \left[\left(\frac{u}{2} \right)^2 + \left(\frac{u}{2} \right)^2 \right]$$

$$= \frac{1}{2}mu^2 + \frac{mu^2}{4} = \frac{3mu^2}{4}$$

$$K.E._f = \frac{1}{2}(2m)(v'_x)^2 + \frac{1}{2}(2m)(v'_y)^2$$

$$= \frac{1}{2}2m \left[\left(\frac{3u}{4} \right)^2 + \left(\frac{u}{4} \right)^2 \right] = \frac{5}{8}mu^2$$

$\therefore$ Loss in K E = $K E_f - K E_i$

$$= mu^2 \left(\frac{6}{8} - \frac{5}{8} \right) = \frac{mu^2}{8}$$

Question144

A body A, of mass $m = 0.1\text{kg}$ has an initial velocity of $3\hat{i}\text{ms}^{-1}$. It collides elastically with another body, B of the same mass which has an initial velocity of $5\hat{j}\text{ms}^{-1}$. After collision, A moves with a velocity $\vec{v} = 4(\hat{i} + \hat{j})$. The energy of B after collision is written as $\frac{x}{10}\text{J}$. The value of x is _____.
[NA 8 Jan. 2020 I]

Answer: 1

Solution:

For elastic collision $\text{KE}_i = \text{KE}_f$

$$\frac{1}{2}m \times 25 + \frac{1}{2} \times m \times 9 = \frac{1}{2}m \times 32 + \frac{1}{2}mv_B^2$$

$$34 = 32 + V_B^2 \Rightarrow V_B = \sqrt{2}$$

$$\text{KE}_B = \frac{1}{2}mv_B^2 = \frac{1}{2} \times 0.1 \times 2 = 0.1\text{J} = \frac{1}{10}\text{J}$$

$$\therefore x = 1$$

Question145

A particle of mass m is dropped from a height h above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely inelastically, the time taken for the combined mass to reach the ground, in units of $\sqrt{\frac{h}{g}}$ is:
[8 Jan. 2020 II]

Options:

A. $\sqrt{\frac{1}{2}}$

B. $\sqrt{\frac{3}{4}}$

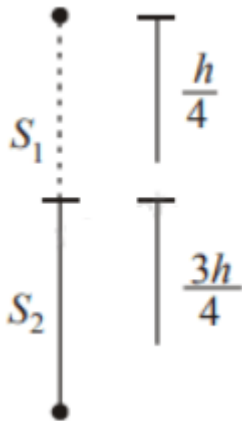
C. $\frac{1}{2}$

D. $\sqrt{\frac{3}{2}}$

Answer: D**Solution:**

Let t be the time taken by the particle dropped from height h to collide with particle thrown upward.

Using,



$$v^2 - u^2 = 2gh$$

$$\Rightarrow v^2 - 0^2 = 2gh$$

$$\Rightarrow v = \sqrt{2gh}$$

Downward distance travelled

$$s_1 = \frac{1}{2}gt^2 = \frac{1}{2}g \cdot \frac{h}{2g} = \frac{h}{4}$$

Distance of collision point from ground

$$s_2 = h - \frac{h}{4} = \frac{3h}{4}$$

Speed of (A) just before collision

$$v_1 = gt = \sqrt{\frac{gh}{2}}$$

And speed of (B) just before collision

$$v_2 = \sqrt{2gh} - \sqrt{\frac{gh}{2}}$$

Using principle of conservation of linear momentum

$$mv_1 + mv_2 = 2mv_f$$

$$\Rightarrow \frac{v_f = m \left(\sqrt{2gh} - \sqrt{\frac{gh}{2}} \right) - m \sqrt{\frac{gh}{2}} = 0}{2m}$$

After collision, time taken (t_1) for combined mass to reach the ground is

$$\Rightarrow \frac{3h}{4} = \frac{1}{2}gt_1^2$$

$$\Rightarrow t_1 = \sqrt{\frac{3h}{2g}}$$

Question146

A person pushes a box on a rough horizontal platform surface. He applies a force of 200N over a distance of 15m . Thereafter, he gets progressively tired and his applied force reduces linearly with distance to 100N . The total distance through which the box has been moved is 30m . What is the work done by the person during the total movement of the box?

[4 Sep. 2020 (II)]

Options:

A. 3280J

B. 2780J

C. 5690J

D. 5250J

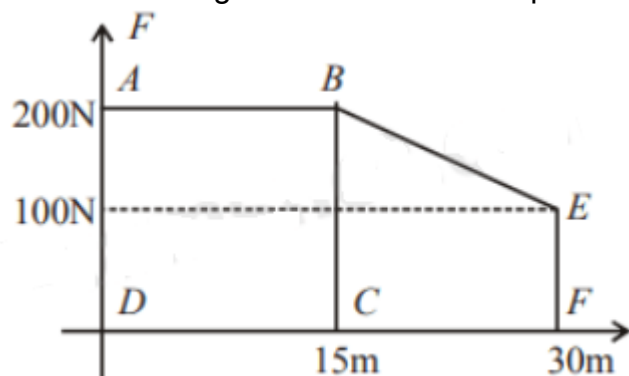
Answer: D

Solution:

The given situation can be drawn graphically as shown in figure.

Work done = Area under $F - x$ graph

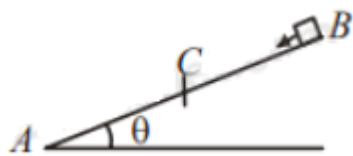
= Area of rectangle ABCD + Area of trapezium BCFE



$$W = (200 \times 15) + \frac{1}{2}(100 + 200) \times 15 = 3000 + 2250$$

$$\Rightarrow W = 5250\text{J}$$

Question147



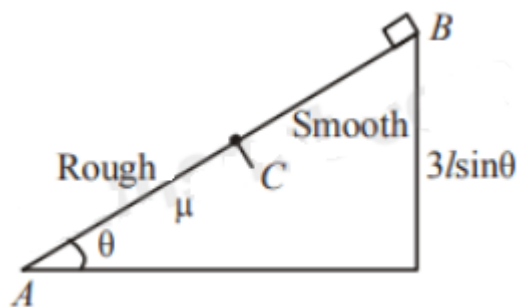
A small block starts slipping down from a point B on an inclined plane AB, which is making an angle θ with the horizontal section BC is smooth and the remaining section CA is rough with a coefficient of friction μ . It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If $BC = 2AC$, the coefficient of friction is given by $\mu = k \tan \theta$. The value of k is _____.

[NA 2 Sep. 2020 (I)]

Answer: 3

Solution:

If $AC = 1$ then according to question, $BC = 2l$ and $AB = 3l$.



Here, work done by all the forces is zero.

$$W_{\text{friction}} + W_{\text{mg}} = 0$$

$$mg(3l) \sin \theta - \mu mg \cos \theta (l) = 0$$

$$\Rightarrow \mu mg \cos \theta l = 3mgl \sin \theta$$

$$\Rightarrow \mu = 3 \tan \theta = k \tan \theta$$

$$\therefore k = 3$$

Question148

A cricket ball of mass 0.15kg is thrown vertically up by a bowling machine so that it rises to a maximum height of 20 m after leaving the machine. If the part pushing the ball applies a constant force F on the ball and moves horizontally a distance of 0.2m while launching the ball, the value of F (in N) is

($g = 10\text{ms}^{-2}$) _____.

[NA 3 Sep. 2020 (I)]

Answer: 150

Solution:

From work energy theorem,

$$W = F \cdot s = \Delta K E = \frac{1}{2}mv^2$$

$$\text{Here } v^2 = 2gh$$

$$\therefore F \cdot s = F \times \frac{2}{10} = \frac{1}{2} \times \frac{15}{100} \times 2 \times 10 \times 20$$

$$\therefore F = 150\text{N}$$

Question149

A body of mass 2 kg is driven by an engine delivering a constant power of 1 J/s. The body starts from rest and moves in a straight line. After 9 seconds, the body has moved a distance (in m)

[5 Sep. 2020 (II)]

Answer: 18

Solution:

Given, Mass of the body, $m = 2 \text{ kg}$
 Power delivered by engine, $P = 1 \text{ J/s}$
 Time, $t = 9 \text{ seconds}$
 Power, $P = Fv$

$$\Rightarrow P = mav \quad [\because F = ma]$$

$$\Rightarrow m \frac{dv}{dt} v = P \quad \left(\because a = \frac{dv}{dt} \right)$$

$$\Rightarrow v dv = \frac{P}{m} dt$$

Integrating both sides we get

$$\Rightarrow \int_0^v v dv = \frac{P}{m} \int_0^t dt$$

$$\Rightarrow \frac{v^2}{2} = Ptm \Rightarrow v = \left(\frac{2Pt}{m} \right)^{1/2}$$

$$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{2P}{m}} t^{1/2} \quad \left(\because v = \frac{dx}{dt} \right)$$

$$\Rightarrow \int_0^x dx = \sqrt{\frac{2P}{m}} \int_0^t t^{1/2} dt$$

$$\therefore \text{Distance, } x = \sqrt{\frac{2P}{m}} \frac{t^{3/2}}{3/2} = \sqrt{\frac{2P}{m}} \times \frac{2}{3} t^{3/2}$$

$$\Rightarrow x = \sqrt{\frac{2 \times 1}{2}} \times \frac{2}{3} \times 9^{3/2} = \frac{2}{3} \times 27 = 18$$

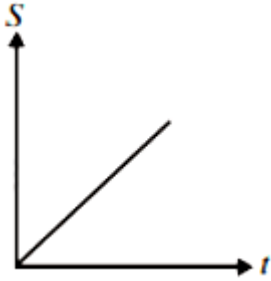
Question 150

A particle is moving unidirectionally on a horizontal plane under the action of a constant power supplying energy source. The displacement (s) - time (t) graph that describes the motion of the particle is (graphs are drawn schematically and are not to scale)

∴
[3 Sep. 2020 (II)]

Options:

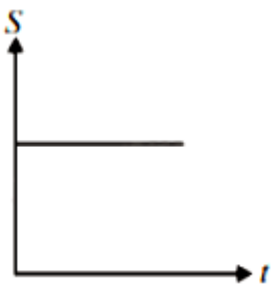
A.



B.



C.



D.



Answer: B

Solution:

We know that

Power, $P = Fv$

But $F = mav = m \frac{dv}{dt}$

$$\therefore P = mv \frac{dv}{dt} \Rightarrow P dt = mvdv$$

Integrating both sides $\int_0^t P dt = m \int_0^v v dv$

$$P \cdot t = \frac{1}{2}mv^2 \Rightarrow v = \left(\sqrt{\frac{2P}{m}} \right) t^{1/2}$$

$$\text{Distance, } s = \int_0^t v dt = \sqrt{\frac{2P}{m}} \int_0^t t^{1/2} dt = \sqrt{\frac{2P}{m}} \cdot \frac{t^{3/2}}{3/2}$$

$$\Rightarrow s = \sqrt{\frac{8P}{9m}} \cdot t^{3/2} \Rightarrow s \propto t^{3/2}$$

So, graph (b) is correct.

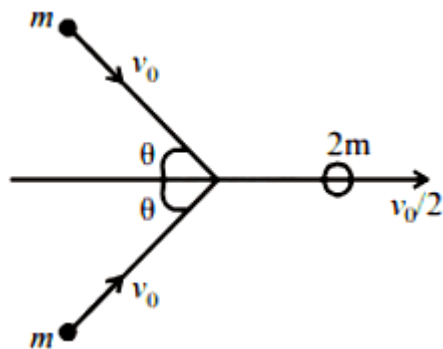
Question 151

Two bodies of the same mass are moving with the same speed, but in different directions in a plane. They have a completely inelastic collision and move together thereafter with a final speed which is half of their initial speed. The angle between the initial velocities of the two bodies (in degree) is _____.

[NA 6 Sep. 2020 (I)]

Answer: 120

Solution:



Momentum conservation along x direction,

$$2mv_0 \cos \theta = 2m \frac{v_0}{2} \Rightarrow \cos \theta = \frac{1}{2} \text{ or } \theta = 60^\circ$$

Hence angle between the initial velocities of the two bodies $= \theta + \theta = 60^\circ + 60^\circ = 120^\circ$

Question152

Particle A of mass m_1 moving with velocity $(\sqrt{3}\hat{i} + \hat{j})\text{ms}^{-1}$ collides with another particle B of mass m_2 which is at rest initially. Let

$\vec{v}_1$ and $\vec{v}_2$ be the velocities of particles A and B after collision

respectively. If $m_1 = 2m_2$ and after collision $\vec{v}_1 = (\hat{i} + \sqrt{3}\hat{j})\text{ms}^{-1}$, the angle between $\vec{v}_1$ and $\vec{v}_2$ is:

[6 Sep. 2020 (II)]

Options:

- A. 15°
- B. 60°
- C. -45°
- D. 105°

Answer: D

Solution:

Before collision,

Velocity of particle A, $u_1 = (\sqrt{3}\hat{i} + \hat{j})\text{m/s}$

Velocity of particle B, $u_2 = 0$

After collision,

Velocity of particle A, $v_1 = (\hat{i} + \sqrt{3}\hat{j})$

Velocity of particle B, $v_2 = 0$

Using principal of conservation of angular momentum

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

$$\Rightarrow 2m_2 (\sqrt{3}\hat{i} + \hat{j}) + m_2 \times 0 = 2m_2 (\hat{i} + \sqrt{3}\hat{j}) + m_2 \times \vec{v}_2$$

$$\Rightarrow 2\sqrt{3}\hat{i} + 2\hat{j} = 2\hat{i} + 2\sqrt{3}\hat{j} + \vec{v}_2$$

$$\Rightarrow \vec{v}_2 = (\sqrt{3} - 1)\hat{i} - (\sqrt{3} - 1)\hat{j}$$

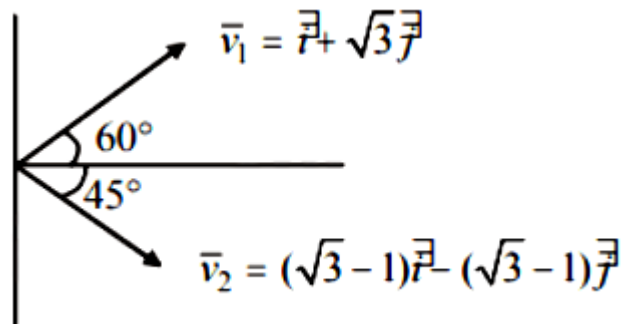
$$\Rightarrow \vec{v}_1 = \hat{i} + \sqrt{3}\hat{j}$$

For angle between $\vec{v}_1$ and $\vec{v}_2$,

$$\cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{|\vec{v}_1| |\vec{v}_2|} = \frac{2(\sqrt{3} - 1)(1 - \sqrt{3})}{2 \times 2\sqrt{2}(\sqrt{3} - 1)} = \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

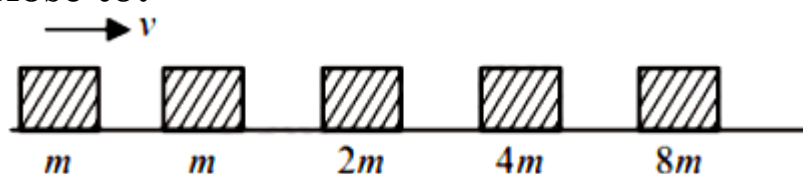
$$\Rightarrow \theta = 105^\circ$$

Angle between $\vec{v}_1$ and $\vec{v}_2$ is 105°



Question153

Blocks of masses m , $2m$, $4m$ and $8m$ are arranged in a line on a frictionless floor. Another block of mass m , moving with speed v along the same line (see figure) collides with mass m in perfectly inelastic manner. All the subsequent collisions are also perfectly inelastic. By the time the last block of mass $8m$ starts moving the total energy loss is $p\%$ of the original energy. Value of 'p' is close to:



[4 Sep. 2020 (I)]

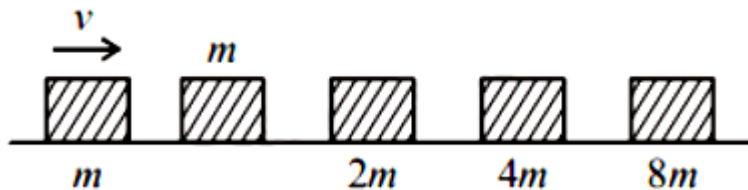
Options:

- A. 77
- B. 94
- C. 37
- D. 87

Answer: B

Solution:

According to the question, all collisions are perfectly inelastic, so after the final collision, all blocks are moving together.



Let the final velocity be v' , using momentum conservation

$$mv = 16mv' \Rightarrow v' = \frac{v}{16}$$

Now initial energy $E_i = \frac{1}{2}mv^2$

Final energy: $E_f = \frac{1}{2} \times 16m \times \left(\frac{v}{16}\right)^2 = \frac{1}{2} \frac{mv^2}{16}$

Energy loss : $E_i - E_f = \frac{1}{2}mv^2 - \frac{1}{2}m \frac{v^2}{16}$

$$\Rightarrow \frac{1}{2}mv^2 \left[1 - \frac{1}{16}\right] \Rightarrow \frac{1}{2}mv^2 \left[\frac{15}{16}\right]$$

The total energy loss is P% of the original energy.

$$\therefore \%P = \frac{\text{Energy loss}}{\text{Original energy}} \times 100$$

$$= \frac{\frac{1}{2}mv^2 \left[\frac{15}{16}\right]}{\frac{1}{2}mv^2} \times 100 = 93.75\%$$

Question154

A block of mass 1.9kg is at rest at the edge of a table, of height 1m. A bullet of mass 0.1kg collides with the block and sticks to it. If the velocity of the bullet is 20m/s in the horizontal direction just before the collision then the kinetic energy just before the

combined system strikes the floor, is [Take $g = 10\text{m/s}^2$. Assume there is no rotational motion and loss of energy after the collision is negligible.
[3 Sep. 2020 (II)]

Options:

A. 20J

B. 21J

C. 19J

D. 23J

Answer: B

Solution:

Given,

Mass of block, $m_1 = 1.9\text{ kg}$

Mass of bullet, $m_2 = 0.1\text{ kg}$

Velocity of bullet, $v_2 = 20\text{ m/s}$

Let v be the velocity of the combined system. It is an inelastic collision.

Using conservation of linear momentum

$$m_1 \times 0 + m_2 \times v_2 = (m_1 + m_2)v$$

$$\Rightarrow 0.1 \times 20 = (0.1 + 1.9) \times v$$

$$\Rightarrow v = 1\text{m/s}$$

Using work energy theorem

Work done = Change in Kinetic energy

Let K be the Kinetic energy of combined system.

$$(m_1 + m_2)gh$$

$$= K - \frac{1}{2}(m_1 + m_2)V^2$$

$$\Rightarrow 2 \times g \times 1 = K - \frac{1}{2} \times 2 \times 1^2 \Rightarrow K = 21\text{J}$$

Question155

A particle of mass m with an initial velocity $u\hat{i}$ collides perfectly elastically with a mass $3m$ at rest. It moves with a velocity $v\hat{j}$

after collision, then, v is given by :
[2 Sep. 2020 (I)]

Options:

A. $v = \sqrt{\frac{2}{3}}u$

B. $v = \frac{u}{\sqrt{3}}$

C. $v = \frac{u}{\sqrt{2}}$

D. $v = \frac{1}{\sqrt{6}}u$

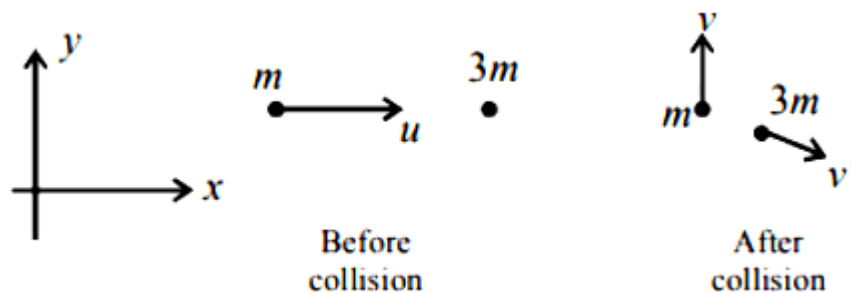
Answer: C

Solution:

From conservation of linear momentum

$$mu\hat{i} + 0 = mv\hat{j} + 3m\vec{v}'$$

$$\vec{v}' = \frac{u}{3}\hat{i} - \frac{v}{3}\hat{j}$$



From kinetic energy conservation,

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{1}{2}(3m) \left(\left(\frac{u}{3}\right)^2 + \left(\frac{v}{3}\right)^2 \right)$$

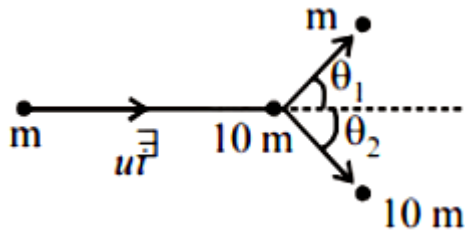
$$\text{or, } mu^2 = mv^2 + \frac{mu^2}{3} + \frac{mv^2}{3}$$

$$\therefore v = \frac{u}{\sqrt{2}}$$

Question156

A particle of mass m is moving along the x -axis with initial velocity $u\hat{i}$. It collides elastically with a particle of mass $10m$ at

rest and then moves with half its initial kinetic energy (see figure). If $\sin \theta_1 = \sqrt{n} \sin \theta_2$, then value of n is _____.



[NA 2 Sep. 2020 (II)]

Answer: 10

Solution:

From momentum conservation in perpendicular direction of initial motion.

$$mu_1 \sin \theta_1 = 10mv_1 \sin \theta_2 \dots (i)$$

It is given that energy of m reduced by half. If u_1 be velocity of m after collision, then

$$\left(\frac{1}{2}mu^2\right)\frac{1}{2} = \frac{1}{2}mu_1^2$$

$$\Rightarrow u_1 = \frac{u}{\sqrt{2}}$$

If v_1 be the velocity of mass $10m$ after collision, then

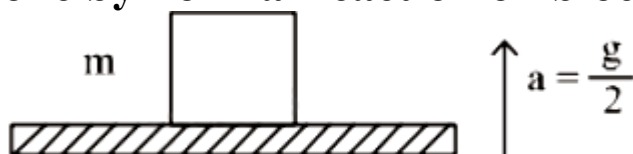
$$\frac{1}{2} \times 10m \times v_1^2 = \frac{1}{2} \frac{u^2}{2} \Rightarrow v_1 = \frac{u}{\sqrt{20}}$$

From equation (i), we have

$$\sin \theta_1 = \sqrt{10} \sin \theta_2$$

Question157

A block of mass m is kept on a platform which starts from rest with constant acceleration $g/2$ upward, as shown in fig. work done by normal reaction on block in time t is:



[10 Jan. 2019 I]

Options:

A. $-\frac{mg^2t^2}{8}$

B. $\frac{mg^2t^2}{8}$

C. 0

D. $\frac{3mg^2t^2}{8}$

Answer: D

Solution:

$$\text{Here, } N - mg = ma = \frac{mg}{2} \Rightarrow N = \frac{3mg}{2}$$

N = normal reaction Now, work done by normal reaction 'N' on block in time

$$t, W = \vec{N} \cdot \vec{S} = \left(\frac{3mg}{2} \right) \left(\frac{1}{2}gt^2 \right)$$

$$\text{or, } W = \frac{3mg^2t^2}{8}$$

Question158

A particle which is experiencing a force, given by $\vec{F} = 3\vec{i} - 12\vec{j}$, undergoes a displacement of $\vec{d} = 4\vec{i}$. If the particle had a kinetic energy of 3J at the beginning of the displacement, what is its kinetic energy at the end of the displacement?

[10 Jan. 2019 II]

Options:

A. 9J

B. 12J

C. 10J

D. 15J

Answer: D

Solution:

$$\text{Work done} = \vec{F} \cdot \vec{d} = (3\vec{i} - 12\vec{j}) \cdot (4\vec{i}) = 12\text{J}$$

From work energy theorem,

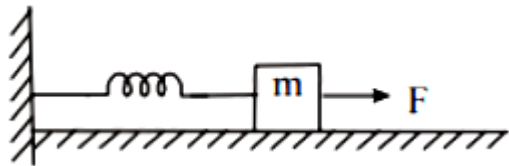
$$W_{\text{net}} = \Delta K.E. = K_f - K_i$$

$$\Rightarrow 12 = K_f - 3$$

$$\therefore K_f = 15\text{J}$$

Question159

A block of mass m , lying on a smooth horizontal surface, is attached to a spring (of negligible mass) of spring constant k . The other end of the spring is fixed, as shown in the figure. The block is initially at rest in its equilibrium position. If now the block is pulled with a constant force F , the maximum speed of the block is:



[9 Jan. 2019 I]

Options:

A. $\frac{2F}{\sqrt{mk}}$

B. $\frac{F}{\pi\sqrt{mk}}$

C. $\frac{\pi F}{\sqrt{mk}}$

D. $\frac{F}{\sqrt{mk}}$

Answer: D

Solution:

Maximum speed is at mean position or equilibrium

At equilibrium Position

$$F = kx \Rightarrow x = \frac{F}{k}$$

From work-energy theorem,

$$W_F + W_{sp} = \Delta K E$$

$$F(x) - \frac{1}{2}kx^2 = \frac{1}{2}mv^2 - 0$$

$$F\left(\frac{F}{k}\right) - \frac{1}{2}k\left(\frac{F}{k}\right)^2 = \frac{1}{2}mv^2$$

$$\Rightarrow \frac{1}{2} \frac{F^2}{k} = \frac{1}{2}mv^2$$

$$\text{or, } v_{\max} = \frac{F}{\sqrt{mk}}$$

Question160

A force acts on a 2kg object so that its position is given as a function of time as $x = 3t^2 + 5$. What is the work done by this force in first 5 seconds?

[9 Jan. 2019 II]

Options:

A. 850J

B. 950J

C. 875J

D. 900J

Answer: D

Solution:

Position, $x = 3t^2 + 5$

$$\therefore \text{Velocity, } v = \frac{dx}{dt} \Rightarrow v = \frac{d(3t^2 + 5)}{dt}$$

$$\Rightarrow v = 6t + 0$$

At $t = 0$

And, at $t = 5$ sec

$$v = 0$$

$$v = 30\text{m/s}$$

According to work-energy theorem, $w = \Delta KE$

$$\text{or } W = \frac{1}{2}mv^2 - 0 = \frac{1}{2}(2)(30)^2 = 900\text{J}$$

Question161

An alpha-particle of mass m suffers 1 -dimensional elastic collision with a nucleus at rest of unknown mass. It is scattered directly backwards losing, 64% of its initial kinetic energy. The mass of the nucleus is :

[12 Jan. 2019 II]

Options:

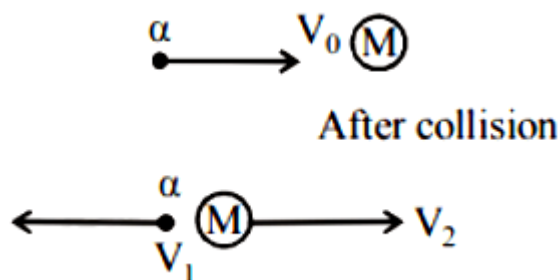
- A. $2m$
- B. $3.5m$
- C. $1.5m$
- D. $4m$

Answer: D

Solution:

Using conservation of momentum,

$$mv_0 = mv_2 - mv_1$$



$$\frac{1}{2}mv_1^2 = 0.36 \times \frac{1}{2}mv_0^2$$

$$\Rightarrow v_1 = 0.6v_0$$

The collision is elastic. So,

$$\frac{1}{2}M V_2^2 = 0.64 \times \frac{1}{2}mv_0^2 [\because M = \text{mass of nucleus}]$$

$$\Rightarrow V_2 = \sqrt{\frac{m}{M}} \times 0.8V_0$$

$$mV_0 = \sqrt{mM} \times 0.8V_0 - m \times 0.6V_0$$

$$\Rightarrow 1.6m = 0.8\sqrt{mM}$$

$$\Rightarrow 4m^2 = mM$$

$$\therefore M = 4m$$

Question 162

A piece of wood of mass 0.03kg is dropped from the top of a 100m height building. At the same time, a bullet of mass 0.02kg is fired vertically upward, with a velocity 100ms^{-1} , from the ground. The bullet gets embedded in the wood. Then the maximum height to which the combined system reaches above the top of the building before falling below is: ($g = 10\text{ms}^{-2}$)
[10 Jan. 2019 I]

Options:

A. 20m

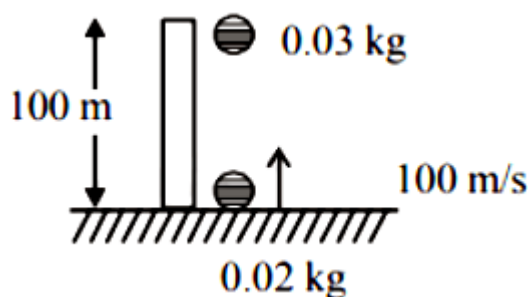
B. 30m

C. 40m

D. 10m

Answer: C

Solution:



Time taken for the particles to collide,

$$t = \frac{d}{V_{\text{rel}}} = \frac{100}{100} = 1\text{sec}$$

Speed of wood just before collision = $gt = 10\text{ m/s}$ and speed of bullet just before collision = $v - gt$

$$= 100 - 10 = 90\text{ m/s}$$

$$S = 100 \times 1 - \frac{1}{2} \times 10 \times 1 = 95\text{m}$$

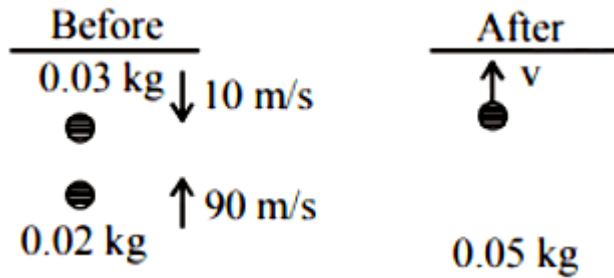
Now, using conservation of linear momentum just before and after the collision

$$-(0.03)(10) + (0.02)(90) = (0.05)v$$

$$\Rightarrow 150 = 5v \therefore v = 30 \text{ m/s}$$

Max. height reached by body

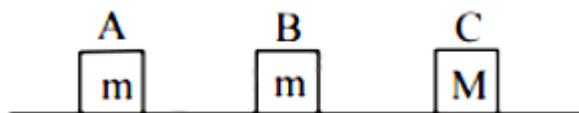
$$h = \frac{30 \times 30}{2 \times 10} = 45 \text{ m}$$



$\therefore$ Height above tower = 40 m

Question163

There block A, B and C are lying on a smooth horizontal surface, as shown in the figure. A and B have equal masses, m while C has mass M . Block A is given an initial speed v towards B due to which it collides with B perfectly inelastically. The combined mass collides with C, also perfectly inelastically $\frac{5}{6}$ th of the initial kinetic energy is lost in whole process. What is value of M/m ?



[9 Jan. 2019 I]

Options:

A. 5

B. 2

C. 4

D. 3

Answer: C

Solution:

Kinetic energy of block A

$$k_1 = \frac{1}{2}mv_0^2$$

∴ From principle of linear momentum conservation

$$mv_0 = (2m + M)v_f \Rightarrow v_f = \frac{mv_0}{2m + M}$$

According to question, of $\frac{5}{6}$ th the initial kinetic energy is lost in whole process.

$$\begin{aligned} \therefore \frac{k_i}{k_f} = 6 &\Rightarrow \frac{\frac{1}{2}mv_0^2}{\frac{1}{2}(2m + M)\left(\frac{mv_0}{2m + M}\right)^2} = 6 \\ \Rightarrow \frac{2m + M}{m} = 6 &\therefore \frac{M}{m} = 4 \end{aligned}$$

Question164

A spring whose unstretched length is l has a force constant k . The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio $\frac{k_1}{k_2}$ of the corresponding force constants, k_1 and k_2 will be:

[12 April 2019 II]

Options:

A. n

B. $\frac{1}{n^2}$

C. $\frac{1}{n}$

D. n^2

Answer: C

Solution:

$$l_1 + l_2 = l \text{ and } l_1 = nl_2$$

$$\therefore l_1 = \frac{nl}{n+1} \text{ and } l_2 = \frac{l}{n+1}$$

$$\text{As } k \propto \frac{1}{l}$$

$$\therefore \frac{k_1}{k_2} = \frac{1/(n+1)}{(nl)/(n+1)} = \frac{1}{n}$$

Question165

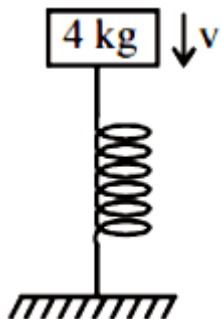
A body of mass 1kg falls freely from a height of 100m, on a platform of mass 3kg which is mounted on a spring having spring constant $k = 1.25 \times 10^6 \text{ N/m}$. The body sticks to the platform and the spring's maximum compression is found to be x . Given that $g = 10 \text{ ms}^{-2}$, the value of x will be close to :
[11 April 2019 I]

Options:

- A. 40cm
- B. 4cm
- C. 80cm
- D. 8cm

Answer: B

Solution:



Velocity of 1 kg block just before it collides with 3 kg block $= \sqrt{2gh} = \sqrt{2000} \text{ m/s}$

Using principle of conservation of linear momentum just before and just after collision, we get

$$1 \times \sqrt{2000} = 4v \Rightarrow v = \frac{\sqrt{2000}}{4} \text{ m/s}$$

Initial compression of spring

$$1.25 \times 10^6 x_0 = 30 \Rightarrow x_0 \approx 0$$

using work energy theorem,

$$\begin{aligned}
 W_g + W_{sp} &= \Delta K E \\
 \Rightarrow 40 \times x + \frac{1}{2} \times 1.25 \times 10^6 (0^2 - x^2) \\
 &= 0 - \frac{1}{2} \times 4 \times v^2 \\
 \text{solving } x &\approx 4\text{cm}
 \end{aligned}$$

Question166

A uniform cable of mass 'M' and length ' L ' is placed on a horizontal surface such that its $\left(\frac{1}{n}\right)^{\text{th}}$ part is hanging below the edge of the surface. To lift the hanging part of the cable upto the surface, the work done should be:

[9 April 2019 I]

Options:

A. $\frac{M g L}{2n^2}$

B. $\frac{M g L}{n^2}$

C. $\frac{2M g L}{n^2}$

D. $nM g L$

Answer: A

Solution:

$$\begin{aligned}
 W &= u_f - u_i \\
 &= 0 - \left(-\frac{mg}{n} \times \frac{L}{2n} \right) = \frac{M g L}{2n^2}.
 \end{aligned}$$

Question167

A wedge of mass $M = 4m$ lies on a frictionless plane. particle of mass m approaches the wedge with speed v . There is no friction

between the particle and the plane or between the particle and the wedge. The maximum height climbed by the particle on the wedge is given by:
[9 April 2019 II]

Options:

A. $\frac{v^2}{g}$

B. $\frac{2v^2}{7g}$

C. $\frac{2v^2}{5g}$

D. $\frac{v^2}{2g}$

Answer: C

Solution:

$$mv = (m + M)V'$$

$$\text{or } v = \frac{mv}{m + M} = \frac{mv}{m + 4m} = \frac{v}{5}$$

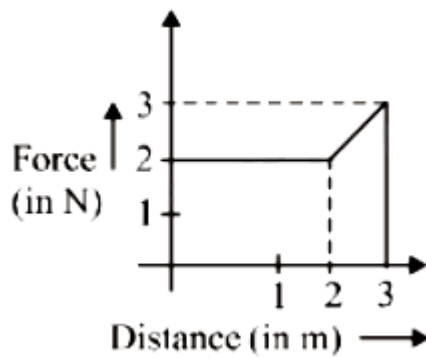
Using conservation of M E , we have

$$\frac{1}{2}mv^2 = \frac{1}{2}(m + 4m)\left(\frac{v}{5}\right)^2 + mgh$$

$$\text{or } h = \frac{2v^2}{5g}$$

Question168

A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3m is:



[8 April 2019 I]

Options:

- A. 4 J
- B. 2.5J
- C. 6.5J
- D. 5J

Answer: C

Solution:

We know area under F-x graph gives the work done by the body

$$\begin{aligned}\therefore W &= \frac{1}{2} \times (3 + 2) \times (3 - 2) + 2 \times 2 \\ &= 2.5 + 4 \\ &= 6.5\text{J}\end{aligned}$$

Using work energy theorem,

$\Delta \text{K.E} = \text{work done}$

$$\therefore \Delta \text{K.E} = 6.5\text{J}$$

Question169

A man (mass = 50kg) and his son (mass = 20kg) are standing on a frictionless surface facing each other. The man pushes his son so that he starts moving at a speed of 0.70ms^{-1} with respect to the man. The speed of the man with respect to the surface is : [12 April 2019 I]

Options:

A. 0.28ms^{-1}

B. 0.20ms^{-1}

C. 0.47ms^{-1}

D. 0.14ms^{-1}

Answer: B

Solution:

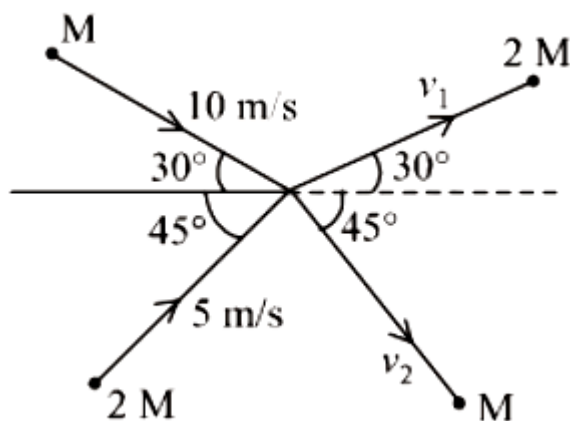
$$P_i = P_f$$

$$\text{or } 0 = 20(0.7 - v) = 50v$$

$$\text{or } v = 0.2\text{m/s}$$

Question170

Two particles, of masses M and $2M$, moving, as shown, with speeds of 10m/s and 5m/s , collide elastically at the origin. After the collision, they move along the indicated directions with speeds v_1 and v_2 , respectively. The values of v_1 and v_2 are nearly:



[10 April 2019 I]

Options:

A. 6.5m/s and 6.3m/s

B. 3.2m/s and 6.3m/s

C. 6.5m/s and 3.2m/s

D. 3.2m/s and 12.6m/s

Answer: A

Solution:

Apply conservation of linear momentum in X and Y direction for the system then

$$M(10 \cos 30^\circ) + 2M(5 \cos 45^\circ) = 2M(v_1 \cos 30^\circ) + M(v_2 \cos 45^\circ)$$

$$5\sqrt{3} + 5\sqrt{2} = \sqrt{3}v_1 + \frac{v_2}{\sqrt{2}} \dots\dots(1)$$

Also

$$2M(5 \sin 45^\circ) - M(10 \sin 30^\circ) = 2M v_1 \sin 30^\circ - M v_2 \sin 45^\circ$$

$$5\sqrt{2} - 5 = v_1 - \frac{v_2}{\sqrt{2}} \dots\dots(2)$$

Solving equation (1) and (2)

$$(\sqrt{3} + 1)v_1 = 5\sqrt{3} + 10\sqrt{2} - 5 \Rightarrow v_1 = 6.5 \text{ m/s}$$

$$v_2 = 6.3 \text{ m/s}$$

Question171

A body of mass 2kg makes an elastic collision with a second body at rest and continues to move in the original direction but with one fourth of its original speed. What is the mass of the second body?

[9 April 2019 I]

Options:

A. 1.0kg

B. 1.5kg

C. 1.8kg

D. 1.2kg

Answer: B

Solution:

$$2u + 0 = 2\left(\frac{u}{4}\right) + mv_2$$

$$\text{and } \frac{1}{2} \times 2 \times u^2 + 0 = \frac{1}{2} \times 2 \times \left(\frac{u}{4}\right)^2 + \frac{1}{2}mv_2^2$$

On solving, we get $m = 1.5\text{kg}$

Question172

A particle of mass ' m ' is moving with speed ' 2v ' and collides with a mass ' 2m ' moving with speed ' v ' in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass ' m ', which move at angle 45° with respect to the original direction.

The speed of each of the moving particle will be:

[9 April 2019 II]

Options:

A. $\sqrt{2}v$

B. $2\sqrt{2}v$

C. $\frac{v}{(2\sqrt{2})}$

D. $v/\sqrt{2}$

Answer: B

Solution:

$$m(2v) + 2mv = 0 + 2mv' \cos 45^\circ$$

$$\text{or } v' = 2\sqrt{2}v$$

Question173

A body of mass m_1 moving with an unknown velocity of $v_1 \hat{i}$, undergoes a collinear collision with a body of mass m_2 moving

with a velocity $v_2^{\hat{i}}$. After collision, m_1 and m_2 move with velocities of $v_3^{\hat{i}}$ and $v_4^{\hat{i}}$, respectively. If $m_2 = 0.5m_1$ and $v_3 = 0.5v_1$, then v_1 is:

[8 April 2019 II]

Options:

A. $v_4 - \frac{v_2}{2}$

B. $v_4 - v_2$

C. $v_4 - \frac{v_2}{4}$

D. $v_4 + v_2$

Answer: B

Solution:

$$m_1 v_1 + m_2 v_2 = m_1 v_3 + m_2 v_4$$

$$\text{or } m_1 v_1 + (0.5m_1)v_2 = m_1 0.5v_1 + (0.5m_1)v_4$$

$$\text{On solving, } v_1 = v_4 - v_2$$

Question174

A body of mass starts moving from rest along x - axis so that its velocity varies as $v = a\sqrt{s}$ where a is a constant s and is the distance covered by the body. The total work done by all the forces acting on the body in the first second after the start of the motion is:

[Online April 16, 2018]

Options:

A. $\frac{1}{8}ma^4t^2$

B. $4ma^4t^2$

C. $8ma^4t^2$

D. $\frac{1}{4}ma^4t^2$

Answer: A

Solution:

From question, $v = a\sqrt{s} = \frac{ds}{dt}$

or, $2\sqrt{s} = at \Rightarrow S = \frac{a^2t^2}{4}$

$F = m \times \frac{a^2}{2}$

Work done $= \frac{ma^2}{2} \times \frac{a^2t^2}{4} = \frac{1}{8}ma^4t^2$

Question175

A particle is moving in a circular path of radius a under the action of an attractive potential $U = -\frac{k}{2r^2}$. Its total energyis:

[2018]

Options:

A. $-\frac{k}{4a^2}$

B. $\frac{k}{2a^2}$

C. zero

D. $-\frac{3}{2}\frac{k}{a^2}$

Answer: C

Solution:

$$F = -\frac{\partial u}{\partial r} \hat{r} = \frac{K}{r^3} \hat{r}$$

Since particle is moving in circular path

$$F = \frac{mv^2}{r} = \frac{K}{r^3} \Rightarrow mv^2 = \frac{K}{r^2}$$

$$\therefore K.E. = \frac{1}{2}mv^2 = \frac{K}{2r^2}$$

$$\text{Total energy} = P.E. + K.E.$$

$$= -\frac{K}{2r^2} + \frac{K}{2r^2} = \text{Zero} \left(\because P.E. = -\frac{K}{2r^2} \text{ given} \right)$$

Question 176

Two particles of the same mass m are moving in circular orbits because of force, given by $F(r) = \frac{-16}{r} - r^3$. The first particle is at a distance $r = 1$, and the second, at $r = 4$. The best estimate for the ratio of kinetic energies of the first and the second particle is closest to

[Online April 16, 2018]

Options:

A. 10^{-1}

B. 6×10^{-2}

C. 6×10^2

D. 3×10^{-3}

Answer: B

Solution:

As the particles moving in circular orbits, So

$$\frac{mv^2}{r} = \frac{16}{r} + r^2$$

$$\text{Kinetic energy, } K.E. = \frac{1}{2}mv^2 = \frac{1}{2}[16 + r^4]$$

$$\text{For first particle, } r = 1, K_1 = \frac{1}{2}m(16 + 1)$$

Similarly, for second particle, $r = 4$, $K_2 = \frac{1}{2}m(16 + 256)$

$$\therefore \frac{K_1}{K_2} = \frac{\frac{16+1}{2}}{\frac{16+256}{2}} = \frac{17}{272} \approx 6 \times 10^{-2}$$

Question177

In a collinear collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. If the final total kinetic energy is 50% greater than the original kinetic energy, the magnitude of the relative velocity between the two particles, after collision, is:
[2018]

Options:

A. $\frac{v_0}{4}$

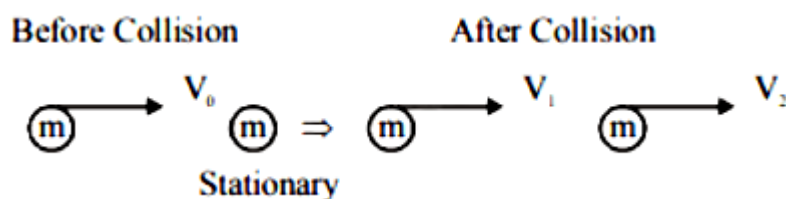
B. $\sqrt{2}v_0$

C. $\frac{v_0}{2}$

D. $\frac{v_0}{\sqrt{2}}$

Answer: B

Solution:



$$\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{3}{2}\left(\frac{1}{2}mv_0^2\right)$$

$$\Rightarrow v_1^2 + v_2^2 = \frac{3}{2}v_0^2 \dots\dots(i)$$

From momentum conservation

$$mv_0 = m(v_1 + v_2) \dots\dots(ii)$$

Squaring both sides,

$$(v_1 + v_2)^2 = v_0^2$$

$$\Rightarrow v_1^2 + v_2^2 + 2v_1v_2 = v_0^2$$

$$2v_1v_2 = -\frac{v_0^2}{2}$$

$$(v_1 - v_2)^2 = v_1^2 + v_2^2 - 2v_1v_2 = \frac{3}{2}v_0^2 + \frac{v_0^2}{2}$$

Solving we get relative velocity between the two particles

$$v_1 - v_2 = \sqrt{2}v_0$$

Question178

The mass of a hydrogen molecule is 3.32×10^{-27} kg. If 10^{23} hydrogen molecules strike, per second, a fixed wall of area 2cm^2 at an angle of 45° to the normal, and rebound elastically with a speed of 10^3m/s , then the pressure on the wall is nearly:
[2018]

Options:

A. $2.35 \times 10^3\text{N/m}^2$

B. $4.70 \times 10^3\text{N/m}^2$

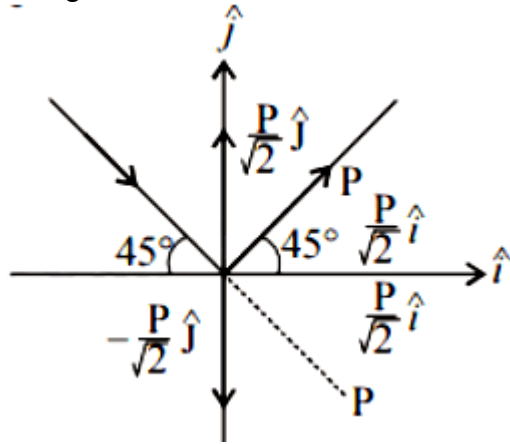
C. $2.35 \times 10^2\text{N/m}^2$

D. $4.70 \times 10^2\text{N/m}^2$

Answer: A

Solution:

Change in momentum



$$\Delta P = \frac{P}{\sqrt{2}}\hat{j} + \frac{P}{\sqrt{2}}\hat{j} + \frac{P}{\sqrt{2}}\hat{i} - \frac{P}{\sqrt{2}}\hat{i}$$

$$\Delta P = \frac{2P}{\sqrt{2}}\hat{j} = I_H \text{ molecule}$$

$$\Rightarrow I_{\text{wall}} = -\frac{2P}{\sqrt{2}}\hat{j}$$

Pressure, P

$$= \frac{F}{A} = \frac{\sqrt{2}P}{A}n (\because n = \text{no. of particles})$$

$$= \frac{\sqrt{2} \times 3.32 \times 10^{-27} \times 10^3 \times 10^{23}}{2 \times 10^{-4}} = 2.35 \times 10^3 \text{ N/m}^2$$

Question 179

It is found that if a neutron suffers an elastic collinear collision with deuterium at rest, fractional loss of its energy is p_d ; while for its similar collision with carbon nucleus at rest, fractional loss of energy is P_c . The values of P_d and P_c are respectively: [2018]

Options:

A. (.89, .28)

B. (.28, .89)

C. (0,0)

D. (0,1)

Answer: A

Solution:

For collision of neutron with deuterium:



Applying conservation of momentum :

$$mv + 0 = mv_1 + 2mv_2 \dots\dots(i)$$

$$v_2 - v_1 = v \dots\dots(ii)$$

∴ Collision is elastic, $e = 1$

From eqn (i) and eqn (ii) $v_1 = -\frac{v}{3}$

$$P_d = \frac{\frac{1}{2}mv^2 - \frac{1}{2}mv_1^2}{\frac{1}{2}mv^2} = \frac{8}{9} = 0.89$$

Now, For collision of neutron with carbon nucleus



Applying Conservation of momentum

$$mv + 0 = mv_1 + 12mv_2 \dots\dots(iii)$$

$$v = v_2 - v_1 \dots\dots(iv)$$

From eqn (iii) and eqn (iv) $v_1 = -\frac{11}{13}v$

$$P_c = \frac{\frac{1}{2}mv^2 - \frac{1}{2}m\left(\frac{11}{13}v\right)^2}{\frac{1}{2}mv^2} = \frac{48}{169} \approx 0.28$$

Question180

A proton of mass m collides elastically with a particle of unknown mass at rest. After the collision, the proton and the unknown particle are seen moving at an angle of 90° with respect to each other. The mass of unknown particle is:
[Online April 15, 2018]

Options:

A. $\frac{m}{\sqrt{3}}$

B. $\frac{m}{2}$

C. $2m$

D. m

Answer: D

Solution:

Apply principle of conservation of momentum along x-direction,

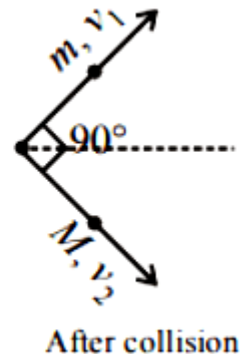
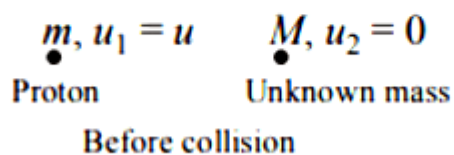
$$mu = mv_1 \cos 45^\circ + M v_2 \cos 45^\circ$$

$$mu = \frac{1}{\sqrt{2}}(mv_1 + M v_2) \dots (i)$$

Along y -direction,

$$0 = mv_1 \sin 45^\circ - M v_2 \sin 45^\circ$$

$$0 = (mv_1 - Mv_2) \frac{1}{\sqrt{2}} \dots (ii)$$



$$\text{Coefficient of restitution } e = 1 = \frac{v_2 - v_1 \cos 90^\circ}{u \cos 45^\circ} \quad (\because \text{Collision is elastic})$$

$$\Rightarrow \frac{v_2}{u\sqrt{2}} = 1$$

$$\Rightarrow u = \sqrt{2}v_2 \dots (iii)$$

Solving eqs (i), (ii), & (iii), we get mass of unknown particle, $M = m$

Question181

A body of mass $m = 10^{-2}\text{kg}$ is moving in a medium and experiences a frictional force $F = -kv^2$. Its initial speed is $v_0 = 10\text{ms}^{-1}$. If, after 10s, its energy is $\frac{1}{8}mv_0^2$, the value of k will be: [2017]

Options:

A. 10^{-4}kgm^{-1}

B. $10^{-1}\text{kgm}^{-1}\text{s}^{-1}$

C. 10^{-3}kgm^{-1}

D. 10^{-3}kgs^{-1}

Answer: A

Solution:

Let V_f is the final speed of the body.

From questions,

$$\frac{1}{2}mV_f^2 = \frac{1}{8}mV_0^2 \Rightarrow V_f = \frac{V_0}{2} = 5\text{m/s}$$

$$F = m \left(\frac{dV}{dt} \right) = -kV^2 \therefore (10^{-2}) \frac{dV}{dt} = -kV^2$$

$$\int_{10}^5 \frac{dV}{V^2} = -100K \int_0^{10} dt$$

$$\frac{1}{5} - \frac{1}{10} = 100K(10) \text{ or } K = 10^{-4} \text{kgm}^{-1}$$

Question182

An object is dropped from a height h from the ground. Every time it hits the ground it loses 50% of its kinetic energy. The total distance covered as $t \rightarrow \infty$ is
[Online April 8, 2017]

Options:

A. $3h$

B. ∞

C. $\frac{5}{3}h$

D. $\frac{8}{3}h$

Answer: A

Solution:

$(\text{K.E.})' = 50\%$ of K.E. after hit i.e.,

$$\frac{1}{2}mv^2 = \frac{50}{100} \times \frac{1}{2}mv^2 \Rightarrow v' = \frac{v}{\sqrt{2}}$$

$$\text{Coefficient of restitution} = \frac{1}{\sqrt{2}}$$

Now, total distance travelled by object is

$$H = h \left(\frac{1+e^2}{1-e^2} \right) = h \left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}} \right) = 3h$$

Question183

A time dependent force $F = 6t$ acts on a particle of mass 1kg. If the particle starts from rest, the work done by the force during the first 1 second will be [2017]

Options:

- A. 9J
- B. 18J
- C. 4.5J
- D. 22J

Answer: C

Solution:

Using, $F = ma = m \frac{dV}{dt}$

$$6t = 1 \cdot \frac{dV}{dt} \quad [\because m = 1\text{kg given }]$$

$$\int_0^v dV = \int 6t dt = 6 \left[\frac{t^2}{2} \right]_0^1 = 3\text{ms}^{-1} \quad [\because t = 1 \text{ sec given }]$$

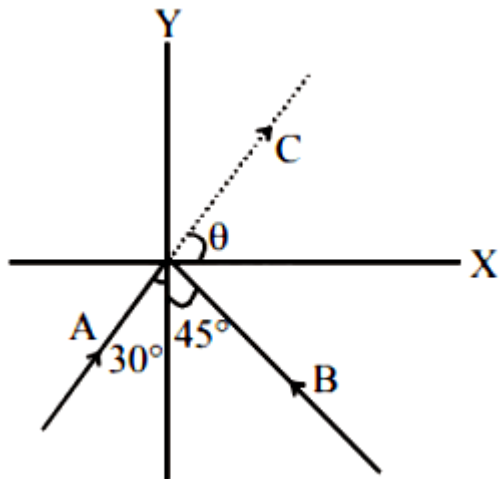
From work-energy theorem,

$$W = \Delta K E = \frac{1}{2}m(V^2 - u^2) = \frac{1}{2} \times 1 \times 9 = 4.5\text{J}$$

Question184

Two particles A and B of equal mass M are moving with the same speed v as shown in the figure. They collide completely inelastically and move as a single particle C. The angle θ that the

path of C makes with the X -axis is given by:



[Online April 9, 2017]

Options:

A. $\tan \theta = \frac{\sqrt{3} + \sqrt{2}}{1 - \sqrt{2}}$

B. $\tan \theta = \frac{\sqrt{3} - \sqrt{2}}{1 - \sqrt{2}}$

C. $\tan \theta = \frac{1 - \sqrt{2}}{\sqrt{2}(1 + \sqrt{3})}$

D. $\tan \theta = 1 - \sqrt{3} + \sqrt{2}$

Answer: A

Solution:

For particle C,

According to law of conservation of linear momentum, vertical component,

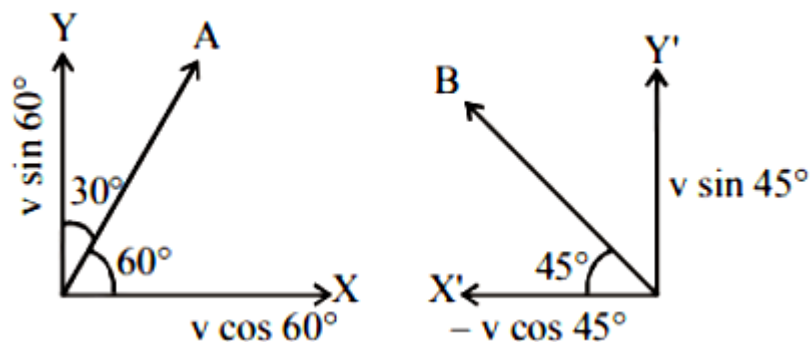
$$2 mv' \sin \theta = mv \sin 60^\circ + mv \sin 45^\circ$$

$$2mv' \sin \theta = \frac{mv}{\sqrt{2}} + \frac{mv\sqrt{3}}{2} \dots\dots (i)$$

Horizontal component,

$$2 mv' \cos \theta = mv \sin 60^\circ - mv \cos 45^\circ$$

$$2mv' \cos \theta = \frac{mv}{2} + \frac{mv}{\sqrt{2}} \dots\dots (ii)$$

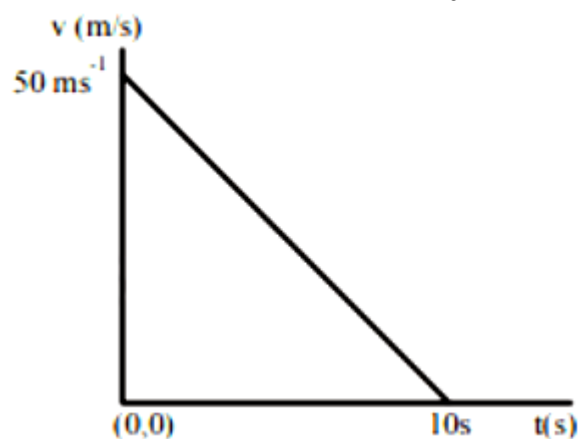


Dividing eqⁿ (i) by eqⁿ (ii),

$$\tan \theta = \frac{\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}}{\frac{1}{2} - \frac{1}{\sqrt{2}}} = \frac{\sqrt{2} + \sqrt{3}}{1 - \sqrt{2}}$$

Question185

Velocity-time graph for a body of mass 10kg is shown in figure. Work-done on the body in first two seconds of the motion is :



[Online April 10, 2016]

Options:

- A. -9300J
- B. 12000J
- C. -4500J
- D. -12000J

Answer: C

Solution:

$$\text{Acceleration (a)} = \frac{v - u}{t} = \frac{(0 - 50)}{(10 - 0)} = -5 \text{ m/s}^2$$

$$u = 50 \text{ m/s}$$

$$\therefore v = u + at = 50 - 5t$$

Velocity in first two seconds $t = 2$

$$v_{(at=2)} = 40 \text{ m/s}$$

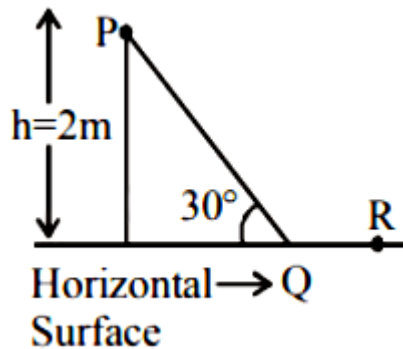
From work-energy theorem,

$$\Delta K.E. = W = \frac{1}{2}(40^2 - 50^2) \times 10 = -4500 \text{ J}$$

Question 186

A point particle of mass m , moves along the uniformly rough track PQR as shown in the figure. The coefficient of friction, between the particle and the rough track equals μ . The particle is released, from rest from the point P and it comes to rest at a point R. The energies, lost by the ball, over the parts, PQ and QR, of the track, are equal to each other, and no energy is lost when particle changes direction from PQ to QR.

The value of the coefficient of friction μ and the distance $x (= QR)$, are, respectively close to :



[2016]

Options:

- A. 0.29 and 3.5m
- B. 0.29 and 6.5m
- C. 0.2 and 6.5m
- D. 0.2 and 3.5m

Answer: A

Solution:

Work done by friction at QR = μmgx

In triangle, $\sin 30^\circ = \frac{1}{2} = \frac{2}{PQ}$

$\Rightarrow PQ = 4m$

Work done by friction at PQ = $\mu mg \times \cos 30^\circ \times 4$

$$= \mu mg \times \frac{\sqrt{3}}{2} \times 4 = 2\sqrt{3}\mu mg$$

Since work done by friction on parts PQ and QR are equal, $\mu mgx = 2\sqrt{3}\mu mg$

$$\Rightarrow x = 2\sqrt{3} \cong 3.5m$$

Using work energy theorem $mg \sin 30^\circ \times 4 = 2\sqrt{3}\mu mg + \mu mgx$

$$\Rightarrow 2 = 4\sqrt{3}\mu$$

$$\Rightarrow \mu = 0.29$$

Question 187

A person trying to lose weight by burning fat lifts a mass of 10kg upto a height of 1m 1000 times. Assume that the potential energy lost each time he lowers the mass is dissipated. How much fat will he use up considering the work done only when the weight is lifted up? Fat supplies 3.8×10^7 J of energy per kg which is converted to mechanical energy with a 20% efficiency rate. Take $g = 9.8 \text{ ms}^{-2}$:

[2016]

Options:

A. $9.89 \times 10^{-3} \text{ kg}$

B. $12.89 \times 10^{-3} \text{ kg}$

C. $2.45 \times 10^{-3} \text{ kg}$

D. $6.45 \times 10^{-3} \text{ kg}$

Answer: B

Solution:

$$n = \frac{W}{\text{input}} = \frac{mgh \times 1000}{\text{input}} = \frac{10 \times 9.8 \times 1 \times 1000}{\text{input}}$$

$$\text{Input} = \frac{98000}{0.2} = 49 \times 10^4 \text{J}$$

$$\text{Fat used} = \frac{49 \times 10^4}{3.8 \times 10^7} = 12.89 \times 10^{-3} \text{kg.}$$

Question188

A particle of mass M is moving in a circle of fixed radius R in such a way that its centripetal acceleration at time t is given by $n^2 R t^2$ where n is a constant. The power delivered to the particle by the force acting on it, is:
[Online April 10, 2016]

Options:

A. $\frac{1}{2} M n^2 R^2 t^2$

B. $M n^2 R^2 t$

C. $M n R^2 t^2$

D. $M n R^2 t$

Answer: B

Solution:

Centripetal acceleration

$$a_c = n^2 R t^2$$

$$a_c = \frac{v^2}{R} = n^2 R t^2$$

$$v^2 = n^2 R^2 t^2$$

$$v = n R t$$

$$a_c = \frac{dv}{dt} = n R$$

$$\text{Power} = m a_t v = m n R n R t = M n^2 R^2 t$$

Question189

A car of weight W is on an inclined road that rises by 100 m over a distance of 1K m and applies a constant frictional force $\frac{W}{20}$ on the car. While moving uphill on the road at a speed of 10ms^{-1} , the car needs power P . If it needs power $\frac{P}{2}$ while moving downhill at speed v then value of v is:
[Online April 9, 2016]

Options:

A. 20ms^{-1}

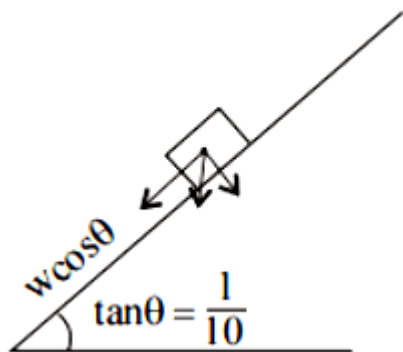
B. 5ms^{-1}

C. 15ms^{-1}

D. 10ms^{-1}

Answer: C

Solution:



While moving downhill power

$$P = \left(W \sin \theta + \frac{W}{20} \right) 10$$

$$P = \left(\frac{W}{10} + \frac{W}{20} \right) 10 = \frac{3W}{2}$$

$$\frac{P}{2} = \frac{3W}{4} = \left(\frac{W}{10} - \frac{W}{20} \right) v$$

$$\frac{3}{4} = \frac{v}{20} \Rightarrow v = 15\text{m/s}$$

∴ Speed of car while moving downhill $v = 15\text{ m/s}$.

Question190

**A neutron moving with a speed 'v' makes a head on collision with a stationary hydrogen atom in ground state. The minimum kinetic energy of the neutron for which inelastic collision will take place is :
[Online April 10, 2016]**

Options:

- A. 20.4 eV
- B. 10.2 eV
- C. 12.1 eV
- D. 16.8 eV

Answer: A

Solution:

For inelastic collision $v' = \frac{m_1}{(m_1 + m_2)}v$

$$= \frac{1}{(1 + 1)}v = \frac{v}{2}$$

n → v(H) Before

(n)(H) → $\frac{v}{2}$ After

$$\text{Loss in K.E.} = \frac{1}{2}mv^2 - \frac{1}{2}(2m)\left(\frac{v}{2}\right)^2 = \frac{1}{4}mv^2$$

K.E. lost is used to jump from 1 st orbit to 2 nd orbit

$$\Delta K.E. = 10.2\text{eV}$$

Minimum K.E. of neutron for inelastic collision

$$\frac{1}{2}mv^2 = 2 \times 10.2 = 20.4\text{eV}$$

Question191

**A particle is moving in a circle of radius r under the action of a force $F = ar^2$ which is directed towards centre of the circle. Total mechanical energy (kinetic energy + potential energy) of the particle is (take potential energy = 0 for r = 0) :
[Online April 11, 2015]**

Options:

A. $\frac{1}{2}\alpha r^3$

B. $\frac{5}{6}\alpha r^3$

C. $\frac{4}{3}\alpha r^3$

D. αr^3

Answer: B

Solution:

As we know, $dU = F.dr$

$$U = \int_0^r \alpha r^2 dr = \frac{\alpha r^3}{3} \dots\dots(i)$$

$$\text{As, } \frac{mv^2}{2} = \alpha r^2$$

$$m^2 v^2 = m \alpha r^3$$

$$\text{or, } 2m(K.E.) = \frac{1}{2}\alpha r^3 \dots\dots(ii)$$

Total energy = Potential energy + kinetic energy Now, from eqn (i) and (ii)

Total energy = K.E. + P.E.

$$= \frac{\alpha r^3}{3} + \frac{\alpha r^3}{2} = \frac{5}{6}\alpha r^3$$

Question192

A block of mass $m = 0.1\text{kg}$ is connected to a spring of unknown spring constant k . It is compressed to a distance x from its equilibrium position and released from rest. After approaching half the distance $\left(\frac{x}{2}\right)$ from equilibrium position, it hits another block and comes to rest momentarily, while the other block moves with a velocity 3ms^{-1} .

The total initial energy of the spring is:

[Online April 10, 2015]

Options:

A. 0.3J

B. 0.6J

C. 0.8J

D. 1.5J

Answer: B

Solution:

Applying momentum conservation

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$0.1u + m(0) = 0.1(0) + m(3)$$

$$0.1u = 3m$$

$$\frac{1}{2}0.1u^2 = \frac{1}{2}m(3)^2$$

Solving we get, $u = 3$

$$\frac{1}{2}kx^2 = \frac{1}{2}K \left(\frac{x}{2} \right)^2 + \frac{1}{2}(0.1)3^2$$

$$\Rightarrow \frac{3}{4}kx^2 = 0.9$$

$$\Rightarrow \frac{3}{2} \times \frac{1}{2}kx^2 = 0.9$$

$$\therefore \frac{1}{2}K x^2 = 0.6J \text{ (total initial energy of the spring)}$$

Question193

A particle of mass m moving in the x direction with speed $2v$ is hit by another particle of mass $2m$ moving in the y direction with speed v . If the collision is perfectly inelastic, the percentage loss in the energy during the collision is close to :

[2015]

Options:

A. 56%

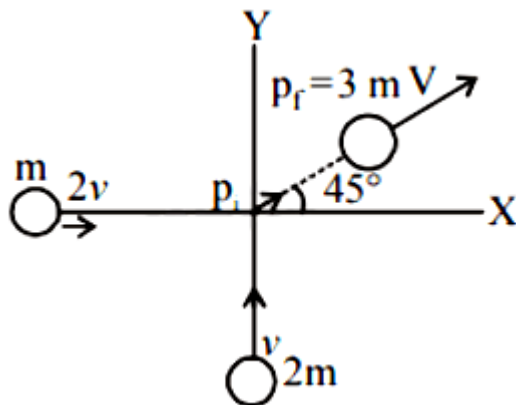
B. 62%

C. 44%

D. 50%

Answer: A

Solution:



Initial momentum of the system

$$p_i = \sqrt{[m(2V)^2 + 2m(2V)^2]} \\ = \sqrt{2}m \times 2V$$

Final momentum of the system $= 3mV$

By the law of conservation of momentum

$$2\sqrt{2}mv = 3mV \Rightarrow \frac{2\sqrt{2}v}{3} = V_{\text{combined}}$$

Loss in energy

$$\Delta E = \frac{1}{2}m_1 V_1^2 + \frac{1}{2}m_2 V_2^2 - \frac{1}{2}(m_1 + m_2)V_{\text{combined}}^2$$

$$DE = 3mv^2 - \frac{4}{3}mv^2 = \frac{5}{3}mv^2 = 55.55\%$$

Percentage loss in energy during the collision $\approx 56\%$

Question194

When a rubber-band is stretched by a distance x , it exerts restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is:

[2014]

Options:

A. $aL^2 + bL^3$

B. $\frac{1}{2}(aL^2 + bL^3)$

C. $\frac{aL^2}{2} + \frac{bL^3}{3}$

$$D. \frac{1}{2} \left(\frac{aL^2}{2} + \frac{bL^3}{3} \right)$$

Answer: C

Solution:

Work done in stretching the rubber-band by a distance dx is

$$dW = F dx = (ax + bx^2)dx$$

Integrating both sides,

$$W = \int_0^L ax dx + \int_0^L bx^2 dx = \frac{aL^2}{2} + \frac{bL^3}{3}$$

Question195

A bullet loses $\left(\frac{1}{n}\right)^{\text{th}}$ of its velocity passing through one plank.

The number of such planks that are required to stop the bullet can be:

[Online April 19, 2014]

Options:

A. $\frac{n^2}{2n-1}$

B. $\frac{2n^2}{n-1}$

C. infinite

D. n

Answer: A

Solution:

Let u be the initial velocity of the bullet of mass m .

After passing through a plank of width x , its velocity decreases to v .

$$\therefore u - v = \frac{4}{n} \text{ or, } v = u - \frac{4}{n} = \frac{u(n-1)}{n}$$

If F be the retarding force applied by each plank, then using work – energy theorem,

$$F_x = \frac{1}{2}mu^2 - \frac{1}{2}mv^2 = \frac{1}{2}mu^2 - \frac{1}{2}mu^2 \frac{(n-1)^2}{n^2}$$

$$= \frac{1}{2}mu^2 \left[\frac{1 - (n-1)^2}{n^2} \right]$$

$$F_x = \frac{1}{2}mu^2 \left(\frac{2n-1}{n^2} \right)$$

Let P be the number of planks required to stop the bullet.

Total distance travelled by the bullet before coming to rest = Px

Using work-energy theorem again,

$$F(Px) = \frac{1}{2}mu^2 - 0$$

$$\text{or, } P(F_x) = P \left[\frac{1}{2}mu^2 \frac{(2n-1)}{n^2} \right] = \frac{1}{2}mu^2$$

$$\therefore P = \frac{n^2}{2n-1}$$

Question196

A spring of unstretched length 1 has a mass m with one end fixed to a rigid support. Assuming spring to be made of a uniform wire, the kinetic energy possessed by it if its free end is pulled with uniform velocity v is:

[Online April 12, 2014]

Options:

A. $\frac{1}{2}mv^2$

B. mv^2

C. $\frac{1}{3}mv^2$

D. $\frac{1}{6}mv^2$

Answer: D

Question197

A bullet of mass 4g is fired horizontally with a speed of 300 m/s into 0.8 kg block of wood at rest on a table. If the coefficient of friction between the block and the table is 0.3, how far will the block slide approximately?

[Online April 12, 2014]

Options:

- A. 0.19 m
- B. 0.379 m
- C. 0.569 m
- D. 0.758 m

Answer: B

Solution:

Given, $m_1 = 4\text{g}$, $u_1 = 300\text{m/s}$

$m_2 = 0.8\text{kg} = 800$, $u_2 = 0\text{m/s}$

From law of conservation of momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Let the velocity of combined system

$$= v \text{ m/s}$$

then,

$$4 \times 300 + 800 \times 0 = (800 + 4) \times v$$

$$v = \frac{1200}{804} = 1.49\text{m/s}$$

Now, $\mu = 0.3$ (given)

$$a = \mu g = 0.3 \times 10 \text{ (take } g = 10\text{m/s}^2)$$

$$= 3\text{m/s}^2$$

then, from $v^2 = u^2 + 2as$ as

$$(1.49)^2 = 0 + 2 \times 3 \times s$$

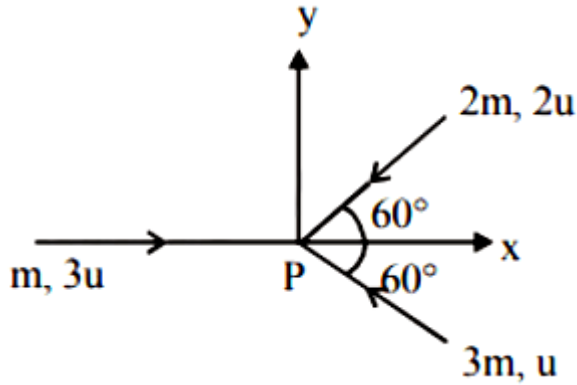
$$s = \frac{(1.49)^2}{6}$$

$$s = \frac{2.22}{6}$$

$$= 0.379\text{m}$$

Question198

Three masses m , $2m$ and $3m$ are moving in x - y plane with speed $3u$, $2u$ and u respectively as shown in figure. The three masses collide at the same point at P and stick together. The velocity of resulting mass will be:



[Online April 12, 2014]

Options:

- A. $\frac{u}{12}(\hat{i} + \sqrt{3}\hat{j})$
- B. $\frac{u}{12}(\hat{i} - \sqrt{3}\hat{j})$
- C. $\frac{u}{12}(-\hat{i} + \sqrt{3}\hat{j})$
- D. $\frac{u}{12}(-\hat{i} - \sqrt{3}\hat{j})$

Answer: D

Solution:

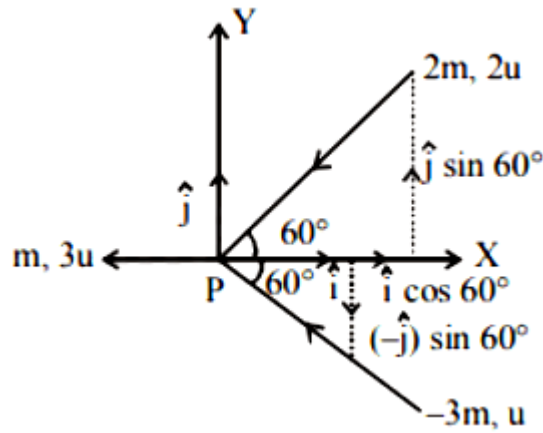
From the law of conservation of momentum we know that,
 $m_1 u_1 + m_2 u_2 + \dots = m_1 v_1 + m_2 v_2 + \dots$

Given $m_1 = m$, $m_2 = 2m$ and $m_3 = 3m$

and $u_1 = 3u$, $u_2 = 2u$ and $u_3 = u$

Let the velocity when they stick = $\vec{v}$

Then, according to question,



$$\begin{aligned}
 m \times 3u(\hat{i}) + 2m \times 2u(-\hat{i} \cos 60^\circ - \hat{j} \sin 60^\circ) + 3m \times u(-\hat{i} \cos 60^\circ + \hat{j} \sin 60^\circ) &= (m + 2m + 3m)\vec{v} \\
 \Rightarrow 3mu\hat{i} - 4mu\frac{\hat{i}}{2} - 4mu\left(\frac{\sqrt{3}}{2}\hat{j}\right) - 3mu\frac{\hat{i}}{2} + 3mu\left(\frac{\sqrt{3}}{2}\hat{j}\right) &= 6m\vec{v} \\
 \Rightarrow mu\hat{i} - \frac{3}{2}mu\hat{i} - \frac{\sqrt{3}}{2}mu\hat{j} &= 6m\vec{v} \\
 \Rightarrow -\frac{1}{2}mu\hat{i} - \frac{\sqrt{3}}{2}mu\hat{j} &= 6m\vec{v} \\
 \Rightarrow \vec{v} = \frac{u}{12}(-\hat{i} - \sqrt{3}\hat{j})
 \end{aligned}$$

Question199

Two springs of force constants 300N/m (Spring A) and 400N/m (Spring B) are joined together in series. The combination is compressed by 8.75cm . The ratio of energy stored in A and B is $\frac{E_A}{E_B}$. Then $\frac{E_A}{E_B}$ is equal to:

[Online April 9, 2013]

Options:

- A. $\frac{4}{3}$
- B. $\frac{16}{9}$
- C. $\frac{3}{4}$
- D. $\frac{9}{16}$

Answer: A

Solution:

Given : $k_A = 300\text{N /m}$, $k_B = 400\text{N /m}$

Let when the combination of springs is compressed by force F . Spring A is compressed by x .
Therefore compression in spring B

$$x_B = (8.75 - x)\text{cm}$$

$$F = 300 \times x = 400(8.75 - x)$$

Solving we get, $x = 5\text{cm}$

$$x_B = 8.75 - 5 = 3.75\text{cm}$$

$$\frac{E_A}{E_B} = \frac{\frac{1}{2}k_A(x_A)^2}{\frac{1}{2}k_B(x_B)^2} = \frac{300 \times (5)^2}{400 \times (3.75)^2} = \frac{4}{3}$$

Question200

A wind-powered generator converts wind energy into electrical energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed n , the electrical power output will be most likely proportional to
[Online April 25, 2013]

Options:

A. v^4

B. v^2

C. v

D. v

Answer: D

Question201

A 70kg man leaps vertically into the air from a crouching position. To take the leap the man pushes the ground with a constant force F to raise himself. The center of gravity rises by 0.5m before he leaps. After the leap the c.g. rises by another 1m. The maximum power delivered by the muscles is: (Take $g = 10\text{ms}^{-2}$)

[Online April 23, 2013]

Options:

A. 6.26×10^3 Watts at the start

B. 6.26×10^3 Watts at take off

C. 6.26×10^4 Watts at the start

D. 6.26×10^4 Watts at take off

Answer: B

Question202

This question has statement I and statement II. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement - I: A point particle of mass m moving with speed v collides with stationary point particle of mass M . If the

maximum energy loss possible is given as $f \left(\frac{1}{2}mv^2 \right)$ then

$$f = \left(\frac{m}{M + m} \right)$$

Statement - II: Maximum energy loss occurs when the particles get stuck together as a result of the collision.

[2013]

Options:

- A. Statement - I is true, Statment - II is true, Statement - II is the correct explanation of Statement - I.
- B. Statement-I is true, Statment - II is true, Statement - II is not the correct explanation of Statement - II.
- C. Statement - I is true, Statment - II is false.
- D. Statement - I is false, Statment - II is true.

Answer: D

Solution:

$$\begin{aligned}\text{Maximum energy loss} &= \frac{P^2}{2m} - \frac{P^2}{2(m+M)} \left[\because K.E. = \frac{P^2}{2m} = \frac{1}{2}mv^2 \right] \\ &= \frac{P^2}{2m} \left[\frac{M}{(m+M)} \right] = \frac{1}{2}mv^2 \left\{ \frac{M}{m+M} \right\}\end{aligned}$$

Statement II is a case of perfectly inelastic collision.

By comparing the equation given in statement I with above equation, we get\

$$f = \left(\frac{M}{m+M} \right) \text{ instead of } \left(\frac{m}{M+m} \right)$$

Hence statement I is wrong and statement II is correct.

Question203

**A projectile of mass M is fired so that the horizontal range is 4 km. At the highest point the projectile explodes in two parts of masses M/4 and 3M/4 respectively and the heavier part starts falling down vertically with zero initial speed. The horizontal range (distance from point of firing) of the lighter part is :
[Online April 23, 2013]**

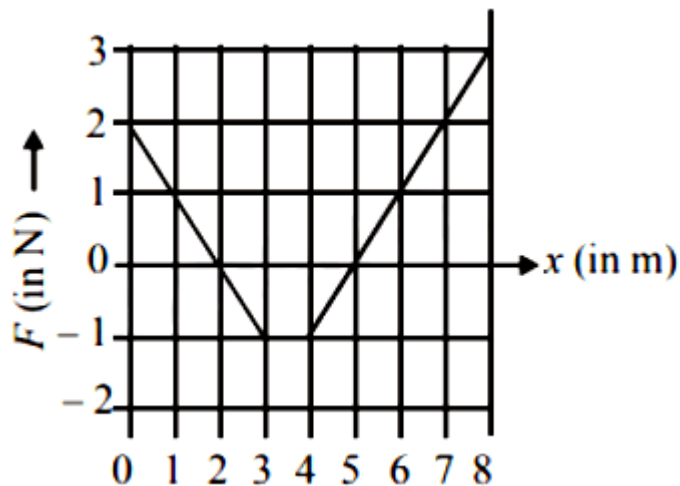
Options:

- A. 16 km
- B. 1 km
- C. 10 km
- D. 2 km

Answer: C

Question204

The force $\vec{F} = F \hat{i}$ on a particle of mass 2kg, moving along the x - axis is given in the figure as a function of its position x. The particle is moving with a velocity of 5m/s along the x -axis at $x = 0$. What is the kinetic energy of the particle at $x = 8\text{m}$?



[Online May 26, 2012]

Options:

- A. 34J
- B. 34.5J
- C. 4.5J
- D. 29.4J

Answer: D

Question205

A particle gets displaced by $\Delta \vec{r} = (2\hat{i} + 3\hat{j} + 4\hat{k})\text{m}$ under the action of a force $\vec{F} = (7\hat{i} + 4\hat{j} + 3\hat{k})$. The change in its kinetic energy is

[Online May 7, 2012]

Options:

A. 38J

B. 70J

C. 52.5J

D. 126J

Answer: A

Solution:

According to work-energy theorem,

Change in kinetic energy = work done

$$\begin{aligned} &= \vec{F} \cdot \Delta \vec{r} = (7\hat{i} + 4\hat{j} + 3\hat{k}) \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) \\ &= 14 + 12 + 12 = 38\text{J} \end{aligned}$$

Question206

A moving particle of mass m , makes a head on elastic collision with another particle of mass $2m$, which is initially at rest. The percentage loss in energy of the colliding particle on collision, is close to

[Online May 19, 2012]

Options:

A. 33%

B. 67%

C. 90%

D. 10%

Answer: C

Solution:

Fractional decrease in kinetic energy of mass m

$$= 1 - \left(\frac{m_2 - m_1}{m_2 + m_1} \right)^2 = 1 - \left(\frac{2 - 1}{2 + 1} \right)^2$$

$$= 1 - \left(\frac{1}{3} \right)^2 = 1 - \frac{1}{9} = \frac{8}{9}$$

Percentage loss in energy

$$= \frac{8}{9} \times 100 \approx 90\%$$

Question207

Two bodies A and B of mass m and 2m respectively are placed on a smooth floor. They are connected by a spring of negligible mass. A third body C of mass m is placed on the floor. The body C moves with a velocity v_0 along the line joining A and B and collides elastically with A. At a certain time after the collision it is found that the instantaneous velocities of A and B are same and the compression of the spring is x_0 . The spring constant k will be

[Online May 12, 2012]

Options:

A. $m \frac{v_0^2}{x_0}$

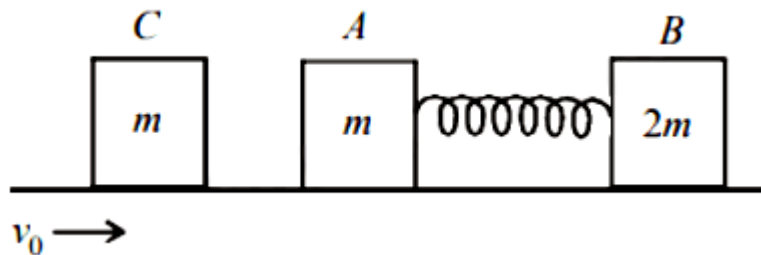
B. $m \frac{v_0}{2x_0}$

C. $2m \frac{v_0}{x_0}$

D. $\frac{2}{3}m\left(\frac{v_0}{x_0}\right)^2$

Answer: D

Solution:



Initial momentum of the system block (C) = mv_0 . After striking with A, the block C comes to rest and now both block A and B moves with velocity v when compression in spring is x_0 .

By the law of conservation of linear momentum

$$mv_0 = (m + 2m)v \Rightarrow v = \frac{v_0}{3}$$

By the law of conservation of energy

K.E. of block C = K.E. of system + P.E. of system

$$\frac{1}{2}mv_0^2 = \frac{1}{2}(3m)\left(\frac{v_0}{3}\right)^2 + \frac{1}{2}kx_0^2$$

$$\Rightarrow \frac{1}{2}mv_0^2 = \frac{1}{6}mv_0^2 + \frac{1}{2}kx_0^2$$

$$\Rightarrow \frac{1}{2}kx_0^2 = \frac{1}{2}mv_0^2 - \frac{1}{6}mv_0^2 = \frac{mv_0^2}{3}$$

$$\therefore k = \frac{2}{3}m\left(\frac{v_0}{x_0}\right)^2$$

Question208

A projectile moving vertically upwards with a velocity of 200ms^{-1} breaks into two equal parts at a height of 490m. One part starts moving vertically upwards with a velocity of 400ms^{-1} . How much time it will take, after the break up with the other part to hit the ground?

[Online May 12, 2012]

Options:

A. $2\sqrt{10}\text{s}$

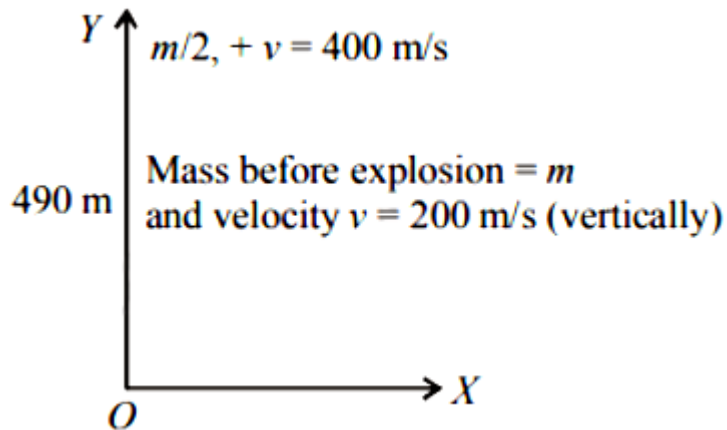
B. 5s

C. 10s

D. $\sqrt{10}$ s

Answer: C

Solution:



Momentum before explosion = Momentum after explosion

$$m \times 200\hat{j} = \frac{m}{2} \times 400\hat{j} + \frac{m}{2}v$$

$$= \frac{m}{2}(400\hat{j} + v)$$

$$\Rightarrow 400\hat{j} - 400\hat{j} = v$$

$$\therefore v = 0$$

i.e., the velocity of the other part of the mass, $v = 0$

Let time taken to reach the earth by this part be t

Applying formula, $h = ut + \frac{1}{2}gt^2$

$$490 = 0 + \frac{1}{2} \times 9.8 \times t^2$$

$$\Rightarrow t^2 = \frac{980}{9.8} = 100$$

$$\therefore t = \sqrt{100} = 10 \text{ sec}$$

Question 209

At time $t = 0$ a particle starts moving along the x -axis. If its kinetic energy increases uniformly with time ' t ', the net force acting on it must be proportional to
[2011 RS]

Options:

A. constant

B. t

C. $\frac{1}{\sqrt{t}}$

D. $\sqrt{t}$

Answer: C

Solution:

$$K.E. \propto t$$

$$K.E. = ct \quad [\text{Here } c = \text{constant}]$$

$$\Rightarrow \frac{1}{2}mv^2 = ct$$

$$\Rightarrow \frac{(mv)^2}{2m} = ct$$

$$\Rightarrow \frac{p^2}{2m} = ct \quad (\because p = mv)$$

$$\Rightarrow p = \sqrt{2ctm}$$

$$\Rightarrow F = \frac{dp}{dt} = \frac{d(\sqrt{2ctm})}{dt}$$

$$\Rightarrow F = \sqrt{2cm} \times \frac{1}{2\sqrt{t}}$$

$$\Rightarrow F \propto \frac{1}{\sqrt{t}}$$

Question210

The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$,

where a and b are constants and x is the distance between the atoms. If the dissociation energy of the molecule is

$D = [U(x = \infty) - U_{\text{at equilibrium}}]$, D is

[2010]

Options:

A. $\frac{b^2}{2a}$

B. $\frac{b^2}{12a}$

C. $\frac{b^2}{4a}$

D. $\frac{b^2}{6a}$

Answer: D

Solution:

At equilibrium : $F = \frac{-d U(x)}{d x}$

$$\Rightarrow F = \frac{-d}{d x} \left[\frac{a}{x^{12}} - \frac{b}{x^6} \right]$$

$$\Rightarrow F = - \left[\frac{12a}{x^{13}} + \frac{6b}{x^7} \right]$$

$$\Rightarrow \frac{12a}{x^{13}} = \frac{6b}{x^7} \Rightarrow x = \left(\frac{2a}{b} \right)^{\frac{1}{6}}$$

$$\therefore U_{\text{at equilibrium}} = \frac{a}{\left(\frac{2a}{b} \right)^2} - \frac{b}{\left(\frac{2a}{b} \right)} = -\frac{b^2}{4a} \text{ and } U_{(x=\infty)} = 0$$

$$\therefore D = 0 - \left(-\frac{b^2}{4a} \right) = \frac{b^2}{4a}$$

Question211

Statement -1: Two particles moving in the same direction do not lose all their energy in a completely inelastic collision.

Statement -2 : Principle of conservation of momentum holds true for all kinds of collisions.

[2010]

Options:

A. Statement -1 is true, Statement -2 is true ; Statement -2 is the correct explanation of Statement -1.

B. Statement -1 is true, Statement -2 is true; Statement -2 is not the correct explanation of Statement -1

C. Statement -1 is false, Statement -2 is true.

D. Statement -1 is true, Statement -2 is false.

Answer: A

Solution:

In completely inelastic collision, all initial kinetic energy is not lost but loss in kinetic energy is as large as it can be. Linear momentum remains conserved in all types of collision. Statement -2 explains statement -1 correctly because applying the principle of conservation of momentum, we can get the common velocity and hence the kinetic energy of the combined body.

Question 212

An athlete in the olympic games covers a distance of 100 m in 10s. His kinetic energy can be estimated to be in the range [2008]

Options:

A. 200J – 500J

B. $2 \times 10^5\text{J} - 3 \times 10^5\text{J}$

C. 20, 000J – 50, 000J

D. 2, 000J – 5, 000J

Answer: D

Solution:

The average speed of the athlete

$$v = \frac{s}{t} = \frac{100}{10} = 10\text{m/s}$$

$$\therefore \text{K.E.} = \frac{1}{2}mv^2$$

Assuming the mass of athlete to 40 kg his average K.E would be

$$\text{K.E.} = \frac{1}{2} \times 40 \times (10)^2 = 2000\text{J}$$

Assuming mass to 100kg average kinetic energy

$$K.E. = \frac{1}{2} \times 100 \times (10)^2 = 5000J$$

Question213

A block of mass 0.50 kg is moving with a speed of 2.00 ms^{-1} on a smooth surface. It strikes another mass of 1.00 kg and then they move together as a single body. The energy loss during the collision is [2008]

Options:

A. 0.16 J

B. 1.00 J

C. 0.67 J

D. 0.34 J

Answer: C

Solution:

Initial kinetic energy of the system

$$K.E._i = \frac{1}{2}mu^2 + \frac{1}{2}M(0)^2$$

$$= \frac{1}{2} \times 0.5 \times 2 \times 2 + 0 = 1J$$

Momentum before collision = Momentum after collision

$$m_1u_1 + m_2u_2 = (m + M) \times v$$

$$\therefore 0.5 \times 2 + 1 \times 0 = (0.5 + 1) \times v \Rightarrow v = \frac{2}{3} \text{m/s}$$

Final kinetic energy of the system is

$$K.E._f = \frac{1}{2}(m + M)v^2$$

$$= \frac{1}{2}(0.5 + 1) \times \frac{2}{3} \times \frac{2}{3} = \frac{1}{3}J$$

$$\therefore \text{Energy loss during collision} = \left(1 - \frac{1}{3}\right)J = 0.67J$$

Question214

A 2 kg block slides on a horizontal floor with a speed of 4m/s. It strikes a uncompressed spring, and compresses it till the block is motionless. The kinetic friction force is 15N and spring constant is 10, 000N /m. The spring compresses by [2007]

Options:

- A. 8.5cm
- B. 5.5cm
- C. 2.5cm
- D. 11.0cm

Answer: B

Solution:

Suppose the spring gets compressed by x before stopping.

kinetic energy of the block = P.E. stored in the spring + work done against friction.

$$\Rightarrow \frac{1}{2} \times 2 \times (4)^2 = \frac{1}{2} \times 10,000 \times x^2 + 15 \times x$$

$$\Rightarrow 10,000x^2 + 30x - 32 = 0$$

$$\Rightarrow 5000x^2 + 15x - 16 = 0$$

$$\therefore x = \frac{-15 \pm \sqrt{(15)^2 - 4 \times (5000)(-16)}}{2 \times 5000}$$

$$= 0.055\text{m} = 5.5\text{cm}.$$

Question215

A particle is projected at 60° to the horizontal with a kinetic energy K. The kinetic energy at the highest point is [2007]

Options:

- A. K /2

B. K

C. Zero

D. $K/4$

Answer: D

Solution:

Let u be the velocity with which the particle is thrown and m be the mass of the particle. Then

$$K = \frac{1}{2}mu^2 \dots\dots(1)$$

At the highest point the velocity is $u \cos 60^\circ$ (only the horizontal component remains, the vertical component being zero at the top-most point). Therefore kinetic energy at the highest point.

$$K' = \frac{1}{2}m(u \cos 60^\circ)^2 = \frac{1}{2}mu^2 \cos^2 60^\circ = \frac{K}{4} \text{ [From 1]}$$

Question216

A particle of mass 100g is thrown vertically upwards with a speed of 5m/s. The work done by the force of gravity during the time the particle goes up is [2006]

Options:

A. $-0.5J$

B. $-1.25J$

C. $1.25J$

D. $0.5J$

Answer: B

Solution:

Given, Mass of the particle, $m = 100g$

Initial speed of the particle, $u = 5 \text{ m/s}$

Final speed of the particle, $v = 0$

Work done by the force of gravity

= Loss in kinetic energy of the body.

$$= \frac{1}{2}m(v^2 - u^2) = \frac{1}{2} \times \frac{100}{1000}(0^2 - 5^2)$$

$$= -1.25\text{J}$$

Question217

The potential energy of a 1kg particle free to move along the x - axis is given by $V(x) = \left(\frac{x^4}{4} - \frac{x^2}{2} \right) \text{J}$.

The total mechanical energy of the particle is 2J . Then, the maximum speed (in m/s) is
[2006]

Options:

A. $\frac{3}{\sqrt{2}}$

B. $\sqrt{2}$

C. $\frac{1}{\sqrt{2}}$

D. 2

Answer: A

Solution:

Potential energy

$$V(x) = \frac{x^4}{4} - \frac{x^2}{2} \text{ joule}$$

For maxima of minima

$$\frac{dV}{dx} = 0 \Rightarrow x^3 - x = 0 \Rightarrow x = \pm 1$$

$$\Rightarrow \text{Min. P.E.} = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}\text{J}$$

$$\text{K.E.}_{(\text{max.})} + \text{P.E.}_{(\text{min.})} = 2 \text{ (Given)}$$

$$\therefore \text{K.E.}_{(\text{max.})} = 2 + \frac{1}{4} = \frac{9}{4}$$

$$\text{K.E.}_{\text{max}} = \frac{1}{2}mv_{\text{max}}^2$$

$$\Rightarrow \frac{1}{2} \times 1 \times v_{\text{max}}^2 = \frac{9}{4} \Rightarrow v_{\text{max}} = \frac{3}{\sqrt{2}}$$

Question 218

A mass of M kg is suspended by a weightless string. The horizontal force that is required to displace it until the string makes an angle of 45° with the initial vertical direction is [2006]

Options:

A. $Mg(\sqrt{2} + 1)$

B. $Mg\sqrt{2}$

C. $\frac{Mg}{\sqrt{2}}$

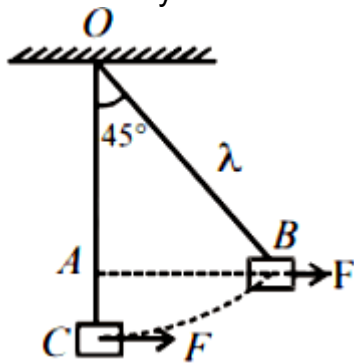
D. $Mg(\sqrt{2} - 1)$

Answer: D

Solution:

Work done by tension + Work done by force (applied) + Work done by gravitational force = change in kinetic energy

Work done by tension is zero



$$\Rightarrow 0 + F \times AB - Mg \times AC = 0$$

$$\Rightarrow F = Mg \left(\frac{AC}{AB} \right) = Mg \left[\frac{1 - \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \right]$$

$$[\because AB = l \sin 45^\circ = \frac{l}{\sqrt{2}} \text{ and } AC = OC - OA = l - l \cos 45^\circ = l \left(1 - \frac{1}{\sqrt{2}} \right)]$$

where l = length of the string.]

$$\Rightarrow F = Mg(\sqrt{2} - 1)$$

Question219

A bomb of mass 16kg at rest explodes into two pieces of masses 4kg and 12kg. The velocity of the 12kg mass is 4ms^{-1} . The kinetic energy of the other mass is [2006]

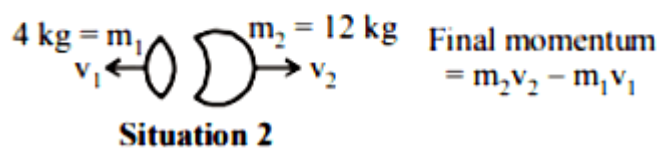
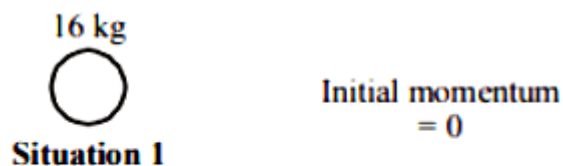
Options:

- A. 144J
- B. 288J
- C. 192J
- D. 96J

Answer: B

Solution:

Let the velocity and mass of 4kg piece be v_1 and m_1 and that of 12kg piece be v_2 and m_2 .



Applying conservation of linear momentum

$$16 \times 0 = 4 \times v_1 + 12 \times 4$$

$$\Rightarrow v_1 = -\frac{12 \times 4}{4} = -12\text{ms}^{-1}$$

Kinetic energy of 4kg mass

$$\therefore \text{K.E.} = \frac{1}{2}m_1v_1^2 = \frac{1}{2} \times 4 \times 144 = 288\text{J}$$

Question220

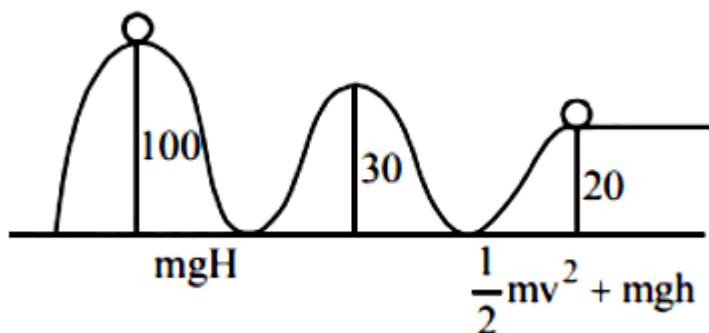
A spherical ball of mass 20 kg is stationary at the top of a hill of height 100 m. It rolls down a smooth surface to the ground, then climbs up another hill of height 30 m and finally rolls down to a horizontal base at a height of 20 m above the ground. The velocity attained by the ball is [2005]

Options:

- A. 20 m/s
- B. 40 m/s
- C. $10\sqrt{30}$ m/s
- D. 10 m/s

Answer: B

Solution:



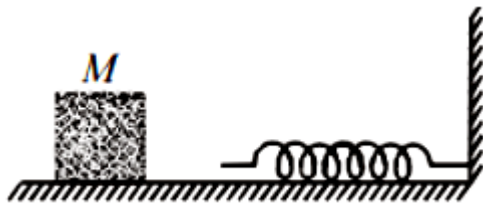
Using conservation of energy,
Total energy at 100 m height
= Total energy at 20m height

$$m(10 \times 100) = m\left(\frac{1}{2}v^2 + 10 \times 20\right)$$

$$\text{or } \frac{1}{2}v^2 = 800 \text{ or } v = \sqrt{1600} = 40\text{m/s}$$

Question221

The block of mass M moving on the frictionless horizontal surface collides with the spring of spring constant k and compresses it by length L . The maximum momentum of the block after collision is



[2005]

Options:

A. $\frac{kL^2}{2M}$

B. $\sqrt{Mk}L$

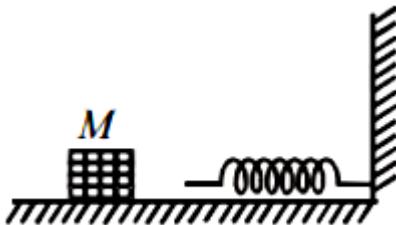
C. $\frac{ML^2}{k}$

D. zero

Answer: B

Solution:

When the spring gets compressed by length L .
K.E. lost by mass M = P.E. stored in the compressed spring.



$$\frac{1}{2}Mv^2 = \frac{1}{2}kL^2$$

$$\Rightarrow v = \sqrt{\frac{k}{M}} \cdot L$$

Momentum of the block, $= M \times v$

$$= M \times \sqrt{\frac{k}{M}} \cdot L = \sqrt{kM} \cdot L$$

Question222

A mass ' m ' moves with a velocity ' v ' and collides inelastically with another identical mass. After collision the 1st mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial

direction of motion. Find the speed of the 2nd mass after collision.

[2005]

Options:

A. $\sqrt{3}v$

B. v

C. $\frac{v}{\sqrt{3}}$

D. $\frac{2}{\sqrt{3}}v$

Answer: D

Solution:

Considering conservation of momentum along x-direction,

$$mv = mv_1 \cos \theta \dots\dots(1)$$

where v_1 is the velocity of second mass

In y-direction,

$$0 = \frac{mv}{\sqrt{3}} - mv_1 \sin \theta$$

$$\text{or } m_1 v_1 \sin \theta = \frac{mv}{\sqrt{3}} \dots(2)$$

Squaring and adding eqns. (1) and (2) we get

$$v_1^2 = v^2 + \frac{v^2}{3} \Rightarrow v_1 = \frac{2}{\sqrt{3}}v$$

Question223

A uniform chain of length 2m is kept on a table such that a length of 60cm hangs freely from the edge of the table. The total mass of the chain is 4kg. What is the work done in pulling the entire chain on the table?

[2004]

Options:

A. 12J

B. 3.6J

C. 7.2J

D. 1200J

Answer: B

Solution:

Mass of over hanging part of the chain

$$m' = \frac{4}{2} \times (0.6)\text{kg} = 1.2\text{kg}$$

Weight of hanging part of the chain

$$= 1.2 \times 10 = 12\text{N}$$

C.M. of hanging part = 0.3m below the table

$$\text{Workdone in putting the entire chain on the table} = 12 \times 0.30 = 3.6\text{J}$$

Question224

A force $\vec{F} = (5\vec{i} + 3\vec{j} + 2\vec{k})\text{N}$ is applied over a particle which displaces it from its origin to the point $\vec{r} = (2\vec{i} - \vec{j})\text{m}$. The work done on the particle in joules is [2004]

Options:

A. +10

B. +7

C. -7

D. +13

Answer: B

Solution:

Given, Force, $\vec{F} = (5\hat{i} + 3\hat{j} + 2\hat{k})$

Displacement, $\vec{x} = (2\hat{i} - \hat{j})$

Work done,

$$W = \vec{F} \cdot \vec{x} = (5\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (2\hat{i} - \hat{j}) \\ = 10 - 3 = 7 \text{ joules}$$

Question 225

A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement x is proportional to [2004]

Options:

A. x

B. e^x

C. x^2

D. $\log_e x$

Answer: C

Solution:

Given : retardation proportional to displacement
i.e., $a = -kx$ [Here, $k = \text{constant}$]

$$\text{But } a = v \frac{dv}{dx}$$

$$\therefore \frac{v dv}{dx} = -kx \Rightarrow \int_{v_1}^{v_2} v dv = - \int_0^x kx dx$$

$$\Rightarrow \left[\frac{v^2}{2} \right]_{v_1}^{v_2} = -k \left[\frac{x^2}{2} \right]_0^x$$

$$\Rightarrow (v_2^2 - v_1^2) = -\frac{kx^2}{2}$$

$$\Rightarrow \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2}m \left(-\frac{kx^2}{2} \right)$$

$\therefore$ Loss in kinetic energy, $\therefore \Delta K \propto x^2$

Question 226

A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle, the motion of the particles takes place in a plane. It follows that
[2004]

Options:

- A. its kinetic energy is constant
- B. its acceleration is constant
- C. its velocity is constant
- D. it moves in a straight line

Answer: A

Solution:

Work done by such force is always zero since force is acting in a direction perpendicular to velocity.

$\therefore$ From work-energy theorem = $\Delta K = 0$

K remains constant.

Question227

A body of mass ' m ', accelerates uniformly from rest to ' v_1 ' in time ' t_1 '. The instantaneous power delivered to the body as a function of time ' t ' is
[2004]

Options:

A. $\frac{mv_1 t^2}{t_1}$

B. $\frac{mv_1^2 t}{t_1^2}$

C. $\frac{mv_1 t}{t_1}$

D. $\frac{mv_1^2 t}{t_1}$

Answer: B

Solution:

Let a be the acceleration of body

Using, $v = u + at$

$$v_1 = 0 + at_1 \Rightarrow a = \frac{v_1}{t_1}$$

Velocity of the body at instant t $v = at$

$$\Rightarrow v = \frac{v_1 t}{t_1}$$

Instantaneous power, $P = \vec{F} \cdot \vec{v} = (m \vec{a}) \cdot \vec{v}$

$$= \left(\frac{mv_1}{t_1} \right) \left(\frac{v_1 t}{t_1} \right) = m \left(\frac{v_1}{t_1} \right)^2 t$$

Question 228

A spring of spring constant $5 \times 10^3 \text{ N/m}$ is stretched initially by 5cm from the unstretched position. Then the work required to stretch it further by another 5cm is

[2003]

Options:

A. 12.50N – m

B. 18.75N – m

C. 25.00N – m

D. 6.25N – m

Answer: B

Solution:

Spring constant, $k = 5 \times 10^3 \text{ N/m}$

Let x_1 and x_2 be the initial and final stretched position of the spring, then

$$\begin{aligned}
 \text{Work done, } W &= \frac{1}{2}k(x_2^2 - x_1^2) \\
 &= \frac{1}{2} \times 5 \times 10^3 [(0.1)^2 - (0.05)^2] \\
 &= \frac{5000}{2} \times 0.15 \times 0.05 = 18.75 \text{ N m}
 \end{aligned}$$

Question229

A wire suspended vertically from one of its ends is stretched by attaching a weight of 200N to the lower end.

**The weight stretches the wire by 1 mm. Then the elastic energy stored in the wire is
[2003]**

Options:

A. 0.2 J

B. 10 J

C. 20 J

D. 0.1 J

Answer: D

Solution:

The elastic potential energy

$$\begin{aligned}
 &= \frac{1}{2} \times \text{Force} \times \text{extension} \\
 &= \frac{1}{2} \times F \times x \\
 &= \frac{1}{2} \times 200 \times 0.001 = 0.1 \text{ J}
 \end{aligned}$$

Question230

A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time ' t ' is

**proportional to
[2003]**

Options:

A. $t^{3/4}$

B. $t^{3/2}$

C. $t^{1/4}$

D. $t^{1/2}$

Answer: B

Solution:

Power, $P = F v = ma \cdot v$

$$\Rightarrow P = \frac{mdv}{dt} v = c = \text{constant} \left(\because F = ma = \frac{mdv}{dt} \right)$$

$$mv_0 v = cd t$$

Integrating both sides, we get

$$m \int_0^v v dv = c \int_0^t dt$$

$$\Rightarrow \frac{1}{2}mv^2 = ct$$

$$\Rightarrow \frac{v^2}{2} = \frac{c \cdot t}{m}$$

$$\Rightarrow v_2 = \frac{2c \cdot t}{m}$$

$$\Rightarrow v = \sqrt{\frac{2c}{m}} \times t^{1/2}$$

$$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{2c}{m}} \times t^{1/2} \text{ where } v = \frac{dx}{dt}$$

$$\Rightarrow \int_e^x dx = \sqrt{\frac{2c}{m}} \times \int_0^t t^{1/2} dt$$

$$\Rightarrow x = \sqrt{\frac{2c}{m}} \times \frac{2t^{3/2}}{3} \Rightarrow x \propto t^{3/2}$$

Question231

Consider the following two statements:

A. Linear momentum of a system of particles is zero

**B. Kinetic energy of a system of particles is zero. Then
[2003]**

Options:

- A. A does not imply B and B does not imply A
- B. A implies B but B does not imply A
- C. A does not imply B but B implies A
- D. A implies B and B implies A

Answer: C

Solution:

Kinetic energy of a system of particle is zero only when the speed of each particles is zero. This implies momentum of each particle is zero, thus linear momentum of the system of particle has to be zero.

Also if linear momentum of the system is zero it does not mean linear momentum of each particle is zero. This is because linear momentum is a vector quantity. In this case the kinetic energy of the system of particles will not be zero.

$\therefore$ A does not imply B but B implies A.

Given, force, $F = 200\text{N}$ extension of wire, $x = 1\text{mm}$.

Question232

**A spring of force constant $800\text{N} / \text{m}$ has an extension of 5cm .
The work done in extending it from 5cm to 15cm is
[2002]**

Options:

- A. 16J
- B. 8J
- C. 32J
- D. 24J

Answer: B

Solution:

Small amount of work done in extending the spring by dx is

$$dW = kx dx$$

$$\therefore W = k \int_{0.05}^{0.15} x dx$$

$$= \frac{800}{2} [(0.15)^2 - (0.05)^2]$$

$$= 400 [(0.15 + 0.05)(0.15 - 0.05)]$$

$$= 400 \times 0.2 \times 0.1 = 8J$$

Question 233

A ball whose kinetic energy is E , is projected at an angle of 45° to the horizontal. The kinetic energy of the ball at the highest point of its flight will be [2002]

Options:

A. E

B. $E / \sqrt{2}$

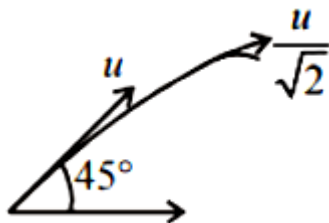
C. $E/2$

D. zero

Answer: C

Solution:

Let u be the speed with which the ball of mass m is projected. Then the kinetic energy (E) at the point of projection is



$$E = \frac{1}{2}mu^2 \dots\dots(i)$$

When the ball is at the highest point of its flight, the speed of the ball is $\frac{u}{\sqrt{2}}$ (Remember that

the horizontal component of velocity does not change during a projectile motion).

∴ The kinetic energy at the highest point

$$= \frac{1}{2}m \left(\frac{u}{\sqrt{2}} \right)^2 = \frac{1}{2} \frac{mu^2}{2} = \frac{E}{2} \text{ [From (i)]}$$
